

بِسْمِ اللَّهِ الرَّحْمَنِ الرَّحِيمِ



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Design and Development of a Unified Digital Platform for Travel Agencies and Flight Offers in Sudan (AirBridgeSD)

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الآية

قال تعالى:

﴿وَقَالُوا الْحَمْدُ لِلَّهِ الَّذِي هَدَانَا لِهَذَا وَمَا كُنَّا لِنَهْتَدِيَ لَوْلَا أَنَّ هَدَانَا اللَّهُ﴾

[سورة الأعراف: الآية 43]

الإهداء

﴿رَبِّ أَوْزِعْنِي أَنْ أَشْكُرَ نِعْمَتَكَ الَّتِي أَنْعَمْتَ عَلَيَّ وَعَلَىٰ وَالِدَيَّ وَأَنْ أَعْمَلَ صَالِحًا تَرْضَاهُ﴾

[سورة النمل: 19]

يُهدى هذا العمل إلى والدينا الكرام لدعمهم المستمر وتشجيعهم لنا طوال مسيرتنا الأكاديمية، وإلى إخوتنا وأخواتنا كما يُهدى إلى أصدقائنا الأوفياء لرفقتهم القيمة وجهودهم المشتركة خلال السنوات الجامعية. علاوة على ذلك، نتوجه بهذا الإهداء إلى مشرفنا الأكاديمي وأساتذتنا الأفاضل لتوجيهاتهم القيمة، ورؤاهم التقنية، وملاحظاتهم البناءة التي ساهمت في تشكيل هذا المشروع، وأخيراً، إلى جامعة السودان للعلوم والتكنولوجيا لتوفيرها الأساس الأكاديمي لتطوير هذا العمل.

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Abstract

The travel and aviation sector in Sudan heavily relies on traditional offline coordination and manual communication via social media platforms, leading to total exclusion of local travel data where global search engines fail to index the Sudanese domestic market due to the absolute lack of local travel APIs. To address this structural infrastructure gap, this research introduces **AirBridgeSD**, a unified, multi-agency digital flight comparison and reservation platform tailored specifically for the Sudanese aviation market. The significance of this study lies in bridging the regional digital divide, organizing a fragmented market, and providing pricing transparency for consumers while boosting the operational efficiency of local businesses. Methodologically, the study follows an **Design Science Research** (DSR) Method combined with a **Case Study** approach, gathering critical workflow and pricing requirements through direct discussions with local travel agencies. For the software development lifecycle, the Agile **Feature-Driven Development** (FDD) methodology was adopted to systematically manage the platform's domain-focused requirements and diverse data sources.

From a technical perspective, the platform utilizes a fully responsive web front-end and a backend powered by the Laravel MVC framework and a MySQL database. The system handles data using a dual-layered hybrid routing strategy: it interfaces with the modern **Duffel API** sandbox to securely fetch live international flight schedules, while deploying specialized administrative dashboards that empower local Sudanese agencies to manually input and manage their offline domestic flights, local tariffs, and commissions. Functionally, users can filter multi-agency options under one clean interface and initiate a practical booking request workflow tailored to regional payment habits. The project successfully delivered a functional multi-agency web marketplace, and the study concludes that combining **hybrid web architectures** with decentralized data entry offers a highly practical, secure, and cost-effective solution to overcome severe infrastructure deficits in emerging digital markets.

المستخلص

يعتمد قطاع السفر والطيران في السودان بشكل كبير على الطرق التقليدية والتنسيق اليدوي عبر تطبيقات التواصل الاجتماعي، مما تسبب في غياب واستبعاد كامل للبيانات الرقمية محلياً جعل محركات البحث العالمية عاجزة عن فهرسة سوق السفر الداخلي لعدم وجود واجهات برمجة تطبيقات (APIs) تربط الوكالات السودانية محلياً. يهدف هذا البحث إلى حل هذه المشكلة عبر بناء منصة الويب الموحدة **AirBridgeSD** لتمكين المسافرين من البحث والمقارنة بين عروض رحلات الطيران المختلفة لعدة وكالات سفر سودانية في مكان واحد. وتكمن أهمية الدراسة في سد فجوة الانعدام الرقمي محلياً، وتنظيم السوق المتشتت، وتوفير الشفافية للمستهلك مع رفع الكفاءة التشغيلية للوكالات الناشئة. ولتحقيق ذلك، يتبع البحث منهجية **البحث العلمي التصميمي (DSR Design Science Research)** مدعوماً بأسلوب **دراسة الحالة (Case Study)** لجمع البيانات وفهم آليات عمل الأسعار والعمليات من خلال مناقشات مباشرة مع الوكالات، بينما تم تبني **منهجية التطوير الموجه بالخصائص (FDD)** لإدارة دورة حياة تطوير البرمجيات والتحكم في منطق النظام المعقد.

من الناحية البرمجية والتنفيذية، تم بناء الواجهة الأمامية بتصميم مستجيب يناسب الهواتف والحواسيب، وتطوير الخلفية بواسطة إطار عمل **Laravel** وقاعدة بيانات **MySQL**. يوجه النظام البيانات برمجياً بمسارين: الأول يتصل بواجهة **Duffel API** الدولية لجلب الرحلات العالمية والأسعار الحية في بيئة اختبارية آمنة، والثاني يوفر لوحات تحكم مخصصة تتيح للوكالات السودانية إدخال وجدولة رحلاتها الداخلية وأسعارها وعمولاتها يدوياً، مما يحل فجوة غياب الـ **APIs** المحلية. ووظيفياً، يسهل الموقع على المستخدمين تصفية ومقارنة العروض، ثم تقديم طلبات حجز مرنة تتوافق مع عادات الدفع والتعامل الحالية في السودان. أثمر المشروع عن تصميم وتسليم منصة سوق ويب عملت بكفاءة على توحيد قنوات السفر، ويستنتج البحث أن دمج **معماريات الويب الهجينة** مع إدخال البيانات اللامركزي يمثل حلاً عملياً واقتصادياً للتغلب على عجز البنية التحتية الرقمية في الأسواق الناشئة دون تكبد تكاليف تراخيص تجارية باهظة.

Research Terms

Term	Meaning	Description
AirBridgeSD	A virtual bridge established to span operational gaps and streamline connection between fragmented service sectors in the aviation industry in Sudan.	A unified digital platform (AirBridgeSD) that aggregates Sudanese travel agencies, syncing local databases with aviation systems to display flight schedules and optimize agency workflows.
FDD	Feature Driven Development	An iterative, model-driven Agile framework designed for complex, large-scale software projects. It revolves around developing and delivering small, client-valued software features.
DSR	Design Science Research Approach	A methodology focused on creating and evaluating an artifact (the platform) to solve a practical real-world problem.
SPA	Single Page Application	A web application that interacts with the user by dynamically rewriting the current web page with new data retrieved from the server, without performing a full page reload.
IDE	Integrated Development Environment	A software application that provides comprehensive facilities to computer programmers for software development, including a code editor, debugger, and build automation tools.
API	Application Programming Interface	A set of protocols and tools for building software and applications, allowing structured data exchange between different systems over a network.
SMTP	Simple Mail Transfer Protocol	A standard communication protocol used by application servers to send, route, and deliver email messages across the internet.

TLS / SSL	Transport Layer Security / Secure Sockets Layer	Cryptographic protocols designed to provide secure and encrypted communication over a computer network, preventing data interception and ensuring data integrity in transit.
UI	User Interface	The series of screens, pages, and visual elements that enable a person to interact with a software product or digital service.
UX	User Experience	The internal experience that a person has as they interact with every aspect of a product's design, interface, and functionality.
GIT	Distributed Version Control System	A version control system for tracking changes in source code during software development, enabling collaboration and history management across distributed teams.
ERD	Entity Relationship Diagram	A graphical representation that illustrates the logical structure of a database, showing entities (tables), their attributes, and the relationships between them.
SQL	Structured Query Language	The standard language used to manage and manipulate relational databases, including querying, inserting, updating, and deleting data.
ACID	Atomicity, Consistency, Isolation, Durability	A set of properties that guarantee database transactions are processed reliably, ensuring data validity even in the event of system failures.
JSON	JavaScript Object Notation	A lightweight, human-readable data interchange format used to transmit structured data between a server and a web application. It is the primary format used by the AirBridgeSD REST API for all request and response payloads.
GDS	Global Distribution System	A centralised reservation network used by travel agencies to access and book airline tickets, hotel rooms, and other travel products. Legacy GDS providers include Amadeus, Sabre, and Travelport. AirBridgeSD integrates with the Duffel NDC API rather than a traditional GDS.

NDC	New Distribution Capability	An IATA-defined XML-based data transmission standard that enables airlines to distribute rich content and ancillary services directly to travel agents and consumers via API, bypassing legacy GDS intermediaries. AirBridgeSD uses the Duffel NDC API for international flight data.
MVC	Model-View-Controller	A software architectural pattern that separates an application into three interconnected components: the Model (data and business logic), the View (user interface), and the Controller (request handling and routing). Laravel implements this pattern on the AirBridgeSD backend.
REST / RESTful	Representational State Transfer	An architectural style for designing networked applications. A RESTful API communicates over HTTP using standard methods (GET, POST, PUT, DELETE) and returns data in a stateless, resource-oriented format. The AirBridgeSD backend exposes a RESTful API consumed by the React.js frontend.
ORM	Object-Relational Mapper	A programming technique that maps database tables to objects in application code, enabling developers to interact with the database using the programming language rather than raw SQL. Laravel's ORM is called Eloquent.
CRUD	Create, Read, Update, Delete	The four fundamental operations performed on persistent data storage. All AirBridgeSD management features (offer management, airline catalogue, commission rules) are implemented as CRUD operations via the RESTful API.
PHP	Hypertext Preprocessor	A widely-used open-source server-side scripting language especially suited for web development. AirBridgeSD's backend is built using PHP 8.2 via the Laravel framework.
HTML	HyperText Markup Language	The standard markup language used to create the structure of web pages. AirBridgeSD's frontend is rendered as HTML in the browser via React.js component rendering.

CSS	Cascading Style Sheets	A stylesheet language used to describe the visual presentation of HTML elements. AirBridgeSD uses Tailwind CSS, a utility-first CSS framework, for responsive and mobile-first styling.
IATA	International Air Transport Association	The global trade association for the world's airlines, responsible for setting international aviation standards including the two-letter airline codes and three-letter airport codes used throughout the AirBridgeSD platform.
UML	Unified Modelling Language	A standardised visual language used in software engineering to model the design of a system. AirBridgeSD documentation includes UML Use Case, Activity, Sequence, Class, and ER Diagrams.
CSR	Client-Side Rendering	A web rendering approach where the browser downloads a minimal HTML page and uses JavaScript to render content dynamically. AirBridgeSD uses CSR via React.js, with the Laravel backend serving only API data.
PK	Primary Key	A column (or set of columns) in a database table that uniquely identifies each row. Every table in the AirBridgeSD database schema has a primary key defined on the id column.
FK	Foreign Key	A column in one database table that references the primary key of another table, establishing and enforcing a referential integrity constraint between the two tables.
UUID	Universally Unique Identifier	A 128-bit identifier that is unique across all systems and time without requiring a central registry. Used in AirBridgeSD to generate unique filenames for uploaded agency licence documents and storing emails.
SDLC	Software Development Life Cycle	A structured process used for planning, creating, testing, and deploying software systems. AirBridgeSD follows the Feature-Driven Development (FDD) model as its SDLC methodology.

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CHAPTER 1

INTRODUCTION

1.1 Background:

The rapid advancement of information technology has significantly transformed the global travel and tourism industry. Modern web-based travel platforms have enabled users to search, compare, and evaluate flight options from multiple providers through a single digital interface capabilities that were previously accessible only to professional travel agents through specialised enterprise systems. As a result, digitised travel markets have seen marked improvements in price transparency, booking efficiency, and consumer accessibility.

Despite this global progression, the Sudanese travel sector has not undergone a comparable transformation. Local travel agencies continue to operate through manual and fragmented processes, relying on telephone enquiries, WhatsApp messaging, physical office visits, and social media pages to communicate offers and coordinate reservations. No centralised, consumer-facing platform currently exists to aggregate, compare, or display flight information from multiple Sudanese agencies in one place. Customers seeking flight options must contact agencies individually, receive inconsistent pricing information, and navigate a booking process that is time-consuming and largely opaque.

This disconnect is further compounded by the absence of any standardised digital infrastructure within the local travel sector. Sudanese agencies do not expose structured data feeds or digital interfaces, which means that even globally established flight comparison platforms cannot integrate local offers into their systems. As a consequence, locally negotiated packages, agency-specific pricing, and region-specific travel offers remain entirely invisible to modern digital discovery tools.

There is therefore a clear and growing need for a localised platform purpose-built for the Sudanese travel market, one that bridges the gap between global flight data and local agency operations, and that accommodates the operational realities on the ground. The proposed system, **AirBridgeSD**,

directly addresses this need by providing a unified digital environment through which customers can discover and compare flight offers from multiple local agencies alongside internal local flights from local airlines and internationally available options, all within a single accessible interface.

1.2 Problem Statement:

The Sudanese travel and flight booking sector is characterised by severe fragmentation and a near-total absence of digital infrastructure. Local travel agencies operate independently, relying on manual communication channels including telephone calls, WhatsApp messaging, and Facebook pages to publish offers and receive customer inquiries. This operational model creates two distinct and measurable deficiencies that directly affect both consumers and the agencies themselves.

From the consumer perspective, travellers seeking flight options in Sudan have no centralised platform through which to search, compare, or evaluate offers from multiple agencies simultaneously. Each agency maintains its pricing and scheduling information in isolated, non-digital formats, compelling customers to contact multiple offices individually in order to perform a basic price comparison. This process is time-consuming, error-prone, and lacks transparency.

From the market infrastructure perspective, the absence of standardised digital APIs within the Sudanese travel sector renders the country's aviation market entirely invisible to international flight aggregation platforms such as Skyscanner, Wego, and Google Flights [4, 8]. These platforms operate exclusively by querying GDS endpoints and NDC APIs; because no such endpoints exist for Sudanese agencies, locally negotiated prices, domestic travel packages, and regional flight schedules are completely excluded from global comparison engines.

Existing local attempts at digitisation such as single-carrier booking portals and agency-only B2B systems address only isolated dimensions of

this problem. They either restrict comparison to a single airline, or are inaccessible to consumers entirely. No platform currently exists that unifies multiple Sudanese travel agencies under a single consumer-facing interface while simultaneously providing those agencies with tools to manage their offers digitally.

To quantify the scope of this problem, a preliminary requirements-gathering study was conducted through structured and semi-structured interviews with six Sudanese travel agencies . The findings confirmed the fragmentation described above: all of the interviewed agencies relied exclusively on WhatsApp, Facebook, phone calls, and social-media pages to publish their offers and receive customer inquiries, and none operated a public, web-based interface through which customers could view or compare their prices. The agencies further reported that, in the absence of any online presence, customers were required to contact each office individually to request and compare fares, a manual process the agencies identified as a barrier to reaching a wider customer base. These observations establish a concrete, evidence-based gap in the Sudanese travel market that the AirBridgeSD platform is designed to address.

In summary, the core problem addressed by this research is the **absence of a unified, consumer-facing digital marketplace for Sudanese travel agencies** that enables customers to search and compare multi-agency flight offers while empowering agencies to manage and publish their services through a structured, web-based platform.

1.3 Aims:

The primary aim of this research is to design and develop **AirBridgeSD**: a unified, responsive web-based platform that aggregates flight ticket offers from multiple Sudanese travel agencies and presents them to consumers through a single, transparent interface. The platform addresses the digital infrastructure gap identified in the Sudanese aviation market by adopting a

hybrid data architecture that combines real-time international flight data sourced from the Duffel NDC API with locally managed agency offers stored in a dedicated relational database. In doing so, the research contributes to a practical, cost-effective model for digitalising fragmented travel markets in emerging economies where standardised API ecosystems are absent.

1.4 Objectives:

The following objectives define the specific, measurable outcomes required to fulfil the aim stated in Section 1.3:

- O1 - Unified Flight and Offer Aggregation:** To design and implement a responsive web interface that consolidates and displays flight ticket offers and promotional travel packages from multiple registered Sudanese travel agencies within a single, consumer-facing platform.
- O2 - Multi-Criteria Search and Filtering:** To develop dynamic search and filtering mechanisms enabling users to sort and narrow flight results by lowest price, shortest duration, preferred airline, and specific agency, applicable to both standard flights and promotional offers.
- O3 - Automated WhatsApp Routing and Request Tracking:** To implement a booking request workflow that automatically generates a pre-formatted WhatsApp message containing complete flight details, price, and agency information and routes it directly to the selected agency's verified contact, enabling off-platform booking finalisation.
- O4 - Dynamic Commission Calculation Engine:** To implement a backend commission engine that allows each agency to define custom pricing rules (fixed monetary fee or percentage markup) per airline and trip type, and automatically computes the final price displayed to the consumer.
- O5 - Agency Self-Service Management Dashboard:** To build a secure, role-based dashboard through which approved travel agencies can independently manage their profiles, configure airline partnerships, set

commission structures, publish promotional offers, and submit inquiries to the platform administrator.

O6 - Centralised Administration and Governance Panel: To develop a master administration panel that enables the platform administrator to review and approve or reject agency registration requests, manage the airline catalogue, monitor platform-wide activity metrics, and respond to customer and agency support messages.

O7 - Secure Agency Onboarding and Authentication: To implement a structured agency registration and verification workflow in which new agencies submit credentials and documentation for administrative review, and upon approval receive automated email access notifications and are granted authenticated dashboard access.

1.5 Scope:

AirBridgeSD is a responsive, hybrid web-based platform designed specifically for the Sudanese travel market. The system's scope is bounded by the functional and technical capabilities deliverable within a single development cycle and is constrained by the local infrastructure realities of the Sudanese aviation sector. To define the project boundaries clearly, the scope is categorised as follows.

1.5.1 In-Scope (Features Included)

- **Hybrid Search and Aggregation Hub:** A responsive interface that pools and displays real-time international flight data (retrieved via the Duffel NDC API) alongside customised domestic and international promotional offers managed locally in the system's MySQL database.
- **Multi-Criteria Search and Filtering:** Dynamic sorting and filtering mechanisms by lowest price, shortest flight duration, specific airlines, and individual agencies, applicable to both standard flights and promotional travel offers, supported by a keyword text-search field.

- **Integrated Communication and Feedback Channels:** An automated WhatsApp link generator that pre-fills selected flight details for direct user-to-agency booking initiation, alongside a built-in contact form for users and agencies to submit inquiries, feedback, or complaints to the platform administrator. The platform intentionally routes booking requests through WhatsApp rather than completing bookings internally. This decision is grounded in findings gathered during the Case Study phase, which indicated that direct messaging remains a common communication channel between agencies and customers. The platform therefore supports existing workflows while introducing structured digital request initiation.
- **Dynamic Pricing and Commission Engine:** Backend logic enabling agencies to define custom commission rules (fixed monetary amount or percentage) per airline and trip type, with automatic calculation and display of the final consumer-facing price.
- **Agency Onboarding and Workflow Dashboard:** A structured registration workflow with automated SMTP email notifications for application approval or rejection. Upon approval, agencies gain authenticated access to a dashboard for profile management, offer publication, airline partnership configuration, and commission management.
- **Comprehensive Administration Control Centre:** A master dashboard granting the platform administrator full system governance: agency account management (approve, reject, activate, deactivate, delete), airline catalogue management, and centralised complaint and inquiry management.

1.5.2 Out-of-Scope (Features Excluded)

- **Direct Live Seat Issuance and Ticketing:** The platform does not issue official airline tickets, generate Passenger Name Records (PNRs), or perform seat-lock reservations. All ticketing is completed off-platform by the selected travel agency.
- **Electronic Payment Processing:** Integration with local payment service providers (including BOK — Bank of Khartoum, O-CASH, MyCashi,

SyberPay, Fawry, and Maroof) or international payment gateways is excluded from the current version. Financial transactions occur directly between the traveller and the agency.

- **Customer Account System and Booking History:** Full customer account registration, login, and persistent booking history are excluded from the current version to reduce onboarding friction.
- **Transaction history functionality:** is currently limited because booking finalisation occurs outside the platform through agency communication channels.
- **Automated synchronisation with agency management systems:** is not implemented in the current version because no standardised integration interfaces were identified during requirements gathering.
- **Additional Agency Services:** The platform displays flight ticket prices and promotional offers only, hotel bookings, car rentals, tour packages, visa services, and any other agency services are outside the platform.
- **Native Mobile Application:** The current scope is limited to a responsive web platform accessible via mobile browsers. Dedicated native Android or iOS applications are not included in this version of the system.

1.6 Motivation:

The motivation for this research derives from a direct and observable inefficiency in the Sudanese travel market. In the absence of any centralized digital comparison platform, travellers seeking competitive flight fares are required to contact multiple travel agencies individually by telephone, physical visit, or social media in order to collect and compare pricing information. This process is not only time-consuming but also structurally inequitable: it disadvantages travellers with limited mobility or time, and it prevents smaller travel agencies from reaching a broader market through digital channels.

From a technical perspective, the opportunity to address this gap exists. Modern web development frameworks, cloud hosting platforms, and

developer-accessible flight APIs provide the necessary tools to construct a functional aggregation platform at a fraction of the cost previously required by legacy GDS integrations. The Duffel NDC API, in particular, offers a developer-friendly sandbox environment that enables the integration of live international flight data without the financial and legal barriers associated with enterprise-level GDS subscriptions.

From a social and economic perspective, digitalising the travel sector even incrementally has the potential to improve price transparency for consumers, increase agency visibility and operational efficiency, and contribute to the broader digital transformation of the Sudanese service economy. The research is therefore motivated not only by a technical opportunity but also by a genuine need for practical infrastructure improvement in an underserved regional market.

1.7 Research Methodology:

1.7.1 Research Type:

This study adopts the Design Science Research (DSR) approach. This methodology focuses on designing and developing a practical software solution to solve a real-world problem. It is suitable for this project because the main outcome is a functional system, the AirBridgeSD platform, which aims to address challenges in the Sudanese travel and aviation market.

In addition, a Case Study approach was used during the requirements gathering phase. Structured and semi-structured discussions were conducted with representatives from local Sudanese travel agencies to better understand their workflows, pricing methods, commission handling, and communication processes. The collected information helped define the system requirements, database structure, and user interactions.

Together, these approaches provided a suitable foundation for the project. The Case Study helped identify real market needs, while the Design

Science Research approach guided the design and development of the proposed system.

1.7.2 Development Methodology:

For the software development lifecycle, the Feature-Driven Development (FDD) methodology was adopted. FDD is an Agile development approach that focuses on building the system through small, well-defined features delivered incrementally. It was chosen for this project because it provides a balance between structured planning and iterative development.

FDD was suitable for this project for several reasons:

- **System Complexity:**

The platform contains multiple actor roles, including Customers, Agencies, and Administrators, each with different functionalities and workflows. FDD helped organize the system into manageable feature groups.

- **Incremental Development:**

The feature-based approach allowed modules such as the Agency Dashboard, Commission Management, Special Offers, and Duffel API integration to be developed and tested separately before full integration.

- **Structured Agile Process:**

Compared to other Agile methods, FDD provides clearer system structure and design planning while still supporting iterative development. This was useful for managing the backend logic and the integration between local database flights and external API data.

The project development process followed the five main stages of the FDD methodology as shown in figure 1.1:

1. Develop an Overall Model:

At this stage, the Sudanese travel and aviation market was analyzed to identify the main system actors, their roles, and the overall workflow of the platform. This helped define the system boundaries and the interactions

between customers, agencies, administrators, local flight data, and external flight APIs.

2. Build a Features List:

The system requirements were divided into smaller feature groups based on user roles and business functionalities. Features such as flight search, commission management, agency offers, authentication, and booking requests were identified and organized.

3. Plan by Feature:

The project tasks and implementation priorities were planned according to the importance and dependencies of each feature. This stage helped organize the development workflow and ensure gradual progress throughout the project lifecycle.

4. Design by Feature:

For each feature group, the required system models and technical designs were prepared, including workflows, UML diagrams, database structure, and interaction logic between system components.

5. Build by Feature:

The system was implemented incrementally, where each feature was developed, tested, and integrated into the platform separately before proceeding to the next feature. This approach supported continuous improvement and easier management of the project models.



Figure 1.1 Feature-Driven Mode

CHAPTER 2

LITERATURE REVIEW

2.1 Introduction:

The rapid adoption of web technologies has substantially transformed the operational frameworks of the global travel and aviation industry. Contemporary flight booking platforms enable users to access real-time pricing, seat availability, and multi-carrier schedules through a single digital interface capabilities that were previously restricted to professional travel agents using proprietary Global Distribution Systems (GDS).

Despite this global progression, the Sudanese travel sector remains largely disconnected from these digital advancements. Local travel agencies continue to depend on manual communication channels, including telephone enquiries, WhatsApp messaging, and physical office visits, to publish offers and coordinate reservations. This structural gap forms the contextual basis of the present research.

This section presents a systematic review of eight relevant studies encompassing both international flight aggregation platforms and local Sudanese travel systems to establish the current state of the field, identify technical and operational gaps, and justify the design decisions adopted in the AirBridgeSD platform. Each study is evaluated according to its technical architecture, functional scope, and observed limitations relative to the Sudanese travel market.

2.2 Theoretical Background:

Online travel management and flight comparison platforms rely heavily on specific technological frameworks to ensure data integrity, real-time synchronization, and global distribution. This section outlines the core components of these systems and evaluates their integration challenges within the local Sudanese ecosystem.

2.2.1 Global Travel Aggregation Systems

Aggregation systems act as centralized platforms that extract, consolidate, and standardize real-time data from disparate external sources, including global airlines, intermediate travel brokers, and central reservation engines. These architectures utilize Application Programming Interfaces (APIs) and advanced data extraction techniques to maintain a unified inventory.

Technical Infrastructure Requirements:

- **Real-Time Global APIs:** To guarantee instantaneous synchronization of volatile pricing and seat availability.
- **Distributed Caching Mechanisms:** Implemented to optimize search performance, minimize network latency, and reduce repetitive API queries.
- **Relational Database Management Systems (RDBMS):** Large-scale databases architected for transactional indexing of multi-leg flight data.
- **Advanced Query Routing and Search Algorithms:** Designed to execute complex multi-criteria filtering, sorting, and comparison matrices.
- **Local Integration Limitation:** While highly efficient globally (e.g., Skyscanner, Wego), these systems completely lack integration with the Sudanese travel market. Local agencies operate on isolated, offline structures and do not expose public APIs, creating a total data disconnect.

2.2.2 Direct Airline Booking Infrastructures

These proprietary, closed-loop systems are deployed directly by specific airline carriers (e.g., Badr Airlines, Tarco Aviation) to manage their dedicated operations.

Core System Components:

- **Central Flight Scheduling Database:** Manages operational timelines, routing matrices, and fleet distribution.
- **Real-Time Seat Inventory Systems:** Tracks specific cabin configurations and live seat availability.
- **Transactional Booking Engines:** Processes reservation lifecycles, user authentication, and secure payment gateway handshakes.

- **Operational Limitation:** These applications operate strictly in siloed environments. They function exclusively for a single carrier and inherently cannot aggregate, compare, or cross-reference options from competing airlines or regional travel agents.

2.2.3 Global Distribution Systems (GDS) and Travel Management Software

To interface with global inventories, accredited agencies utilize enterprise travel networks and high-level software abstraction layers, such as:

- **Legacy Global Distribution Systems (GDS):** Amadeus, Sabre, and Travelport.
- **Modern Developer-Centric APIs:** Duffel Infrastructure.
- **Operational Capabilities:**
 - Executing live passenger name record (PNR) generation and seat reservations.
 - Automated electronic ticket issuing and itinerary modification workflows.
 - Real-time querying of multi-airline availability and fare rules.

Access Barrier: These B2B infrastructures are strictly restricted. These systems are exclusively business-to-business in orientation and are not accessible to end consumers, demand specialized terminal training, and require expensive corporate subscription licenses, making them inaccessible to the average local traveler.

2.2.4 The Sudanese Travel Market Landscape and Gaps

The travel and aviation sector in Sudan presents unique infrastructural and technical deficits that impede digital transformation:

- **Absence of Consolidated Digital Aggregators:** The local market completely lacks a unified, consumer-facing flight comparison ecosystem.
- **API Inexistence:** Local travel agencies do not possess, expose, or maintain API endpoints, creating severe barriers to automated data aggregation.

- **Data Fragmentation:** Pricing records, scheduling, and package structures remain fully non-digitized and decentralized.
- **Manual Operational Reliance:** Consumers and agencies heavily depend on primitive, manual communication channels—such as direct WhatsApp chats, traditional Facebook postings, and physical branch visits—to verify options.

2.3 Literature Studies:

2.3.1 Study 1: Tarco Aviation Online Booking System (2024):

Tarco Aviation is one of the major Sudanese airlines and provides a web-based platform that allows customers to search for flights operated by the company and submit booking requests online[28]. The system is mainly designed for travelers looking for domestic and regional routes offered specifically by Tarco. Through the platform, users can select travel dates, view available flights, check seat availability, and submit reservation requests without visiting travel agencies. This platform represents one of the early digital solutions introduced in the Sudanese aviation sector.

The system is built using a traditional web-application structure. The front-end uses common web technologies such as HTML , CSS , and JavaScript to display the user interface. The back-end is developed using PHP or a similar content management framework, while MySQL is used to store flight schedules, seat availability, and booking information. Since the system handles only Tarco's own flights, the overall data size remains relatively small and manageable.

Functionally, the Tarco platform offers the essential features of an airline booking system. Users can browse flight schedules, check seat availability in real time, select the preferred flight, and send a booking request using an online form. The system also sends email confirmations to notify users that their request has been received. These features help reduce reliance on traditional booking methods such as phone calls and in-person visits to airline offices.

Pros:

- Provides a simple and accessible online booking option for Tarco flights.
- Allows users to view schedules, seat availability, and submit reservation requests.
- Uses a relational database to store flight and booking data in an organized way.
- Reduces manual work and improves customer convenience.

Cons:

- Limited to Tarco flights only; no comparison with other airlines or agencies.
- Does not integrate with Global Distribution Systems (GDS) such as Duffel or Sabre.
- Lacks advanced search features such as flexible dates, filters, or price recommendations.
- Customers must visit other websites to explore additional flight options.

Observed Limitations and Addressed Gaps:

Although Tarco Aviation's booking system provides a useful starting point for digital flight reservations in Sudan, it is restricted to a single airline and does not support price comparison or multi-agency integration. Users still need to check other agencies individually to find alternative flights or better prices. The proposed platform in this research aims to address these gaps by creating a **unified system** that brings together offers from multiple Sudanese travel agencies. This allows users to compare schedules, prices, and routes in one place, offering more transparency and a better booking experience than single-provider systems like Tarco's.

2.3.2 Study 2: Haboob Travel Multi-Service Booking Platform (2019):

Haboob Travel is a Sudanese digital platform that offers multiple travel services, including flight booking, hotel reservations, tour packages, and transportation [27]. It was developed to make the travel process easier for Sudanese users by allowing them to browse international flights and

accommodation options through a single system. The platform is accessible through both mobile devices and web browsers, and it represents one of the early local attempts to introduce a complete online travel booking environment in Sudan.

The mobile access strategy for the Haboob platform has not been publicly disclosed; however, based on the features described on the platform's public-facing pages, the use of a cross-platform mobile framework is consistent with its observed functionality. Regarding backend infrastructure, the platform's connectivity with airline reservation systems is not publicly documented; nonetheless, based on its described capability to integrate with external APIs and provide real-time flight data, integration with an airline host system or GDS-compatible interface is architecturally consistent with its feature set. The platform utilises cloud-based databases, combining SQL (Structured Query Language) for organised reservation data and NoSQL for large and frequently changing hotel and flight datasets. This structure helps the system manage large volumes of data and support real-time search.

Haboob Travel includes many useful features. Users can search flights with filters, explore hotels, book tours, and rent cars. The system also supports user accounts, secure login, discount codes, multiple languages, and online payments through international gateways. These features give users a modern experience and allow them to complete reservations directly through the platform.

Pros:

- Provides a wide range of travel services in one platform.
- Connects to global APIs that support hundreds of airlines and hotel options.
- Offers both mobile and web access with modern interface design.
- Supports online payments and provides automatic booking confirmations.
- Gives users a convenient and centralized booking experience.

Cons:

- Works as a single-agency system and does not include local Sudanese travel agencies.

- Does not show locally negotiated prices or domestic flight offers.
- Depends heavily on external APIs, which may cause data delays or limited accuracy.
- Smaller Sudanese agencies cannot join due to high costs of GDS connections.
- Lacks dashboards for agencies to manage their own offers independently.

Observed Limitations and Addressed Gaps:

While Haboob Travel provides a strong global travel platform, it does not solve a key issue in the Sudanese market: the absence of a local, multi-agency comparison system. Sudanese customers still need to contact agencies individually to find local prices or special offers. The proposed system in this research addresses this gap by creating a **unified local platform** that gathers flight offers from multiple Sudanese travel agencies. It provides transparent pricing, customized local search filters, and a booking request workflow that matches Sudanese travel habits. Unlike Haboob, the proposed system allows agencies to manage their offers through dashboards, creating a more organized and competitive environment.

2.3.3 Study 3: Blue Orange B2B Flight Hosting System:

Blue Orange is a Sudanese Business-to-Business (B2B) hosting system designed specifically for travel agencies[26]. It provides agencies with access to flight booking tools, airline fares, and ticketing services through a secure login. Unlike public booking platforms, Blue Orange does not offer a customer-facing interface. Instead, it focuses on providing agencies with a professional environment to manage daily booking operations in an organised and efficient manner.

The system offers each registered agency a dedicated dashboard through which they can search for flights, view available fares, issue tickets, create Passenger Name Records (PNRs), generate invoices, and manage reservations. Blue Orange's backend connectivity with airline reservation systems is not

publicly documented; however, based on its described PNR generation capability, integration with an airline host system or a GDS-compatible interface is architecturally consistent with its feature set. The system's database manages agency profiles, booking records, pricing details, and transaction logs.

Blue Orange serves as a replacement for traditional manual methods, assisting agencies in reducing operational errors and improving productivity. It centralises tasks that many Sudanese agencies typically perform through fragmented channels such as WhatsApp, phone calls, spreadsheets, or handwritten notes.

Pros:

- Provides a professional system for Sudanese agencies to manage flight bookings.
- Offers dashboards for flight search, ticketing, PNR creation, and reservation management.
- Reduces manual work and increases booking accuracy.
- Helps agencies access airline fares and generate invoices in an organized manner.

Cons:

- No public-facing interface for customers to search or compare flight prices.
- Does not include a customer booking request feature.
- Does not display offers from different Sudanese agencies in one place.
- Agencies cannot publish or share their prices directly with customers through the system.

Observed Limitations and Addressed Gaps:

Although Blue Orange improves the internal operations of Sudanese travel agencies, it remains strictly an agency-only system and does not support customer needs. There is no unified marketplace where travelers can compare ticket prices from different Sudanese agencies, view available flights, or request

bookings online. Customers must still contact agencies individually, which is time-consuming and lacks transparency.

The proposed system in this research addresses these gaps by offering a customer-facing, multi-agency platform. It gathers offers from multiple Sudanese travel agencies into one place and allows users to search, filter, compare, and submit booking requests. Agencies will also receive dashboards within the system, enabling them to publish and manage their offers directly. This creates a complete and unified solution that does not currently exist in Sudan.

2.3.4 Study 4: Wego Global Flight and Hotel Comparison Platform (2023):

Wego is an international travel metasearch platform that allows users to search and compare flights and hotels from hundreds of airlines and online travel agencies[3]. It is widely used in the Middle East and Asia and provides travelers with real-time information on prices, routes, and availability. The system works as a data-aggregation and comparison engine rather than a direct booking platform, meaning users search and filter itineraries on Wego but are routed to external checkout provider endpoints to finalize transactions. Wego is known for its fast performance, user-friendly interface, and wide coverage of global travel data.

Architecturally, Wego relies on large-scale parallel integrations with external travel data streams. Instead of relying on a single infrastructure type, it aggregates data by interfacing with legacy **Global Distribution Systems (GDS)** such as **Sabre** and **Amadeus**, alongside modern **New Distribution Capability (NDC)** API hubs like **Duffel**, and carrier-specific RESTful web services. The platform utilizes distributed, cloud-native backend microservices to compute and process millions of concurrent search queries. To optimize data handling, it employs a hybrid data layer combining highly indexed Relational Database Management Systems (RDBMS) for structural transaction logs, alongside high-throughput NoSQL databases to cache dynamic, fast-changing

flight matrices. The front-end is optimized for cross-platform responsiveness (web and mobile), delivering unified user experience layout states.

Functionally, Wego offers a comprehensive suite of search and analytics tools. Users can execute advanced filtering layers, multi-city route processing, flexible-date grid calendars, and localized fare alerts. The system processes asynchronous price-change notifications via instant background worker queues and supports seamless multi-currency and multi-language interface switching, making it an optimal global comparative utility.

Pros:

- Aggregates live flight and accommodation datasets across hundreds of international carriers and Online Travel Agencies (OTAs).
- Implements low-latency multi-criteria filtering engines and predictive price sorting.
- Optimizes client side rendering performance across modern mobile and web environments.

Cons:

- Lacks direct structural control over the booking pipeline, redirecting users to external third-party checkouts which can trigger sudden transaction failures.
- Completely excludes regional providers and local ticket consolidators who lack exposed digital endpoints.

Observed Limitations and Addressed Gaps:

Although Wego provides an exceptional framework for international travel metasearch, its architecture exhibits a total coverage blackout regarding the Sudanese domestic and regional travel ecosystem. Because Wego queries data strictly from established global GDS and online NDC endpoints, it is inherently incapable of indexing Sudanese travel agencies, local private packages, or custom-negotiated offline tariffs. Furthermore, Wego assumes a standardized digital transaction pipeline, lacking any support for localized operational habits such as manual booking confirmation requests or regional offline payment

verification workflows. Finally, independent local agencies are entirely locked out of the Wego environment because it offers no provider-facing content management consoles.

The proposed **AirBridgeSD** platform addresses these architectural gaps by designing a localized, multi-agency marketplace tailored specifically for Sudan's aviation realities. While the proposed system incorporates modern API integration logic for international lookups, it overrides the global metadata gap by deploying a dedicated Laravel-powered administrative dashboard. This empowers local Sudanese travel agencies to independently digitize, publish, and control their custom schedules, localized pricing, and flight packages. Consequently, **AirBridgeSD** successfully transitions a completely disconnected, offline local market into a unified web engine that supports local booking workflows, which global tools like Wego fail to accommodate.

2.3.5 Study 5: Online Comparison Shopping Behavior of Travel Consumers (Xiang and Gretzel):

This study by Xiang and Gretzel [2] investigates how travelers behave when searching for and comparing flight options online. Instead of assuming that customers follow a simple, linear path when choosing flights, the researchers focus on the idea of cross-site shopping, where users move between different types of travel platforms before making a final booking decision. The study provides a deeper understanding of how travelers use comparison sites and airline websites at different stages of their search journey.

According to the study, travelers usually begin their search on Online Travel Agencies (OTAs) and metasearch engines, as these platforms offer a wide range of options and allow quick price comparison. However, when they are ready to finalize their booking, many customers switch to airline websites for more accurate details, updated policies, and a higher sense of trust and security. This shows that the booking journey is not linear; instead, customers use each platform for a specific purpose depending on what information they need.

The study concludes that price is a major factor in the early stages of search, but the final booking decision depends on the platform's reliability and the quality of information it provides—such as baggage rules, seat choices, or updated flight details.

Pros:

- Provides strong empirical evidence about the real behavior of travelers when comparing flights online.
- Clearly identifies the different roles of comparison platforms (for discovery) and airline websites (for final booking).
- Offers valuable insights for designers and managers of travel systems on how to improve user experience and increase conversions.

Cons:

- The study is based on data from developed Western markets, which may not fully match the behavior of users in Sudan or similar regions.
- Focuses heavily on price and trust as the main decision factors, while other factors such as agency involvement or local booking habits may be more important in markets like Sudan.

Observed Limitations and Addressed Gaps:

The study provides an important understanding of how travelers switch between comparison sites and airline websites, highlighting the complementary roles these platforms play during the search and booking process. However, its findings are limited by the geographical and cultural scope of the sample, which may not accurately reflect the behavior of travelers in emerging markets with different digital infrastructures. The current project benefits from these insights but adapts them to the Sudanese context, where local agencies, manual communication, and non-linear booking habits are more common.

2.3.6 Study 6: Skyscanner Global Travel Metasearch Platform

Skyscanner is a leading global travel metasearch engine and aggregator designed to allow end-users to compare flights, hotels, and car rentals across hundreds of providers[7]. It operates as a data-consolidation layer rather than a direct booking engine, routing users to external airlines or Online Travel Agencies (OTAs) to finalize transactions.

The system is built on a highly scalable, distributed microservices architecture. The front-end leverages modern JavaScript frameworks (such as React) to handle complex, asynchronous state management and deliver highly responsive interface rendering. The back-end is powered by a multi-tiered infrastructure using robust programming environments like Java and Python, engineered to handle massive concurrent traffic. Data synchronization relies heavily on distributed caching mechanisms (such as Redis) alongside advanced search optimization algorithms to parse millions of flight combinations. For global distribution data, it integrates extensively with commercial RESTful APIs and real-time structured data feeds.

Functionally, Skyscanner provides an advanced suite of search capabilities. Users can execute multi-city or flexible-date queries, view comprehensive price matrices via interactive calendars, and apply intricate filtering layers (e.g., stops, duration, alliance partnerships, carbon emissions). The platform incorporates automated pricing alerts, dispatching instant notifications to users via email or webhooks when fare metrics fluctuate. Additionally, it integrates analytical recommendation systems that estimate whether a flight price is likely to rise or fall based on historical data trends.

Pros:

- Delivers a highly optimized, single-interface comparison ecosystem covering global flight inventories.
- Utilizes sophisticated caching and parallel query routing to minimize search latency across external databases.
- Maximizes customer convenience through advanced UX features like predictive pricing and flexible multi-criteria filtering.

Cons:

- Suffers from occasional data desynchronization, leading to price mismatches between the dashboard and the actual airline checkout page due to cache expiration latencies.
- Completely dependent on external digital infrastructure; it cannot index or display service providers that lack active web endpoints or public APIs.

Observed Limitations and Addressed Gaps:

Although Skyscanner represents the peak of global travel aggregation, it suffers from total exclusion regarding the Sudanese aviation market. Because domestic and regional airlines in Sudan, alongside local travel agencies, operate on offline structures, manual records, or closed networks without public APIs, Skyscanner cannot index or display flights within or out of Sudan. The proposed AirBridgeSD platform addresses this architectural gap directly. Instead of relying on non-existent local APIs, the platform provides a dedicated administrative dashboard allowing Sudanese agencies to manually upload, manage, and digitize their internal flight schedules and custom tariffs, bringing a completely disconnected market into a unified web interface.

2.3.7 Study 7: Amadeus Global Distribution System (GDS)**Framework**

Amadeus is the premier Global Distribution System (GDS) worldwide, functioning as a massive business-to-business (B2B) transactional network that connects global aviation carriers with commercial travel platforms[29]. In architectural and academic research, Amadeus represents the foundational benchmark for modern flight aggregation and reservation lifecycles.

Architecturally, the enterprise core of Amadeus is built on a high-availability hybrid cloud system that interfaces directly with legacy airline central

reservation systems (CRSs). It relies strictly on global aviation regulations, such as the International Air Transport Association (IATA) standards and EDIFACT protocols, to guarantee absolute transaction integrity and synchronization across thousands of concurrent nodes. For independent software engineers and academic researchers, the "Amadeus for Developers" portal historically served as a critical gateway. This platform offered self-service RESTful APIs that allowed client applications to query live flight search engines, parse multi-currency tariff matrices, and test complex multi-leg booking workflows using structured JSON data payloads secured under industry-standard OAuth 2.0 authentication protocols.

Functionally, the Amadeus ecosystem provides a comprehensive programmatic interface for global travel management:

- **Advanced Cryptic and Programmatic Search Matrices:** Executes parallel query routing across multiple airline flight schedules to return live fare tracking and real-time inventory availability.
- **Interactive Seat-Map Validation:** Delivers dynamic cabin layout rendering and real-time baggage allowance verification patterns.
- **Standardized Booking Lifecycles:** Processes real-time transaction states to capture extensive passenger information (APIS data) and instantly generate official Passenger Name Records (PNRs) and verification codes.

Pros:

- Delivers absolute data accuracy backed by global aviation standards and direct, low-latency mainframe connections.
- Standardizes multiple disparate airline APIs and communication rules into a single uniform data schema for modern web developers.
- Provides a highly scalable sandbox environment equipped with comprehensive technical documentation and mock response payloads.

Cons:

- Imposes heavy financial, legal, and formal accreditation barriers when attempting to transition a client system from testing sandboxes to the live production environment.
- Inherits the structural complexity of legacy GDS databases, resulting in a steep learning curve and significant engineering overhead for small-scale operations.
- Suffers from regional access constraints, geo-blocking, and unpredictable service deprecations for independent developer tiers.

Observed Limitations and Addressed Gaps:

The integration of a global enterprise system like Amadeus faces insurmountable operational and structural barriers within the local Sudanese travel context. First, the accessible "Amadeus for Developers" free tier experienced sudden service restrictions, culminating in its final discontinuation for unaccredited independent developers in March 2026, which created a total service dependency risk. Second, and more critically, Amadeus acts strictly as a global digital index; it is completely blind to the local Sudanese domestic travel market. Because Sudanese travel agencies and local regional routes operate entirely offline, on closed private servers, or through manual coordination, their customized pricing tariffs, flight schedules, and agency commission rates are completely absent from the Amadeus GDS database.

The proposed AirBridgeSD platform addresses these critical gaps through a dual-layered, self-reliant system architecture:

- **The Global Service Gap Resolution:** Rather than being halted by the decommissioning of the Amadeus free tier, the proposed platform executes a technical migration to the Duffel API to securely manage global flight patterns and international searches in a developer-friendly sandbox.
- **The Local Digital Inexistence Gap Resolution:** To solve the total absence of local agency APIs in Sudan, AirBridgeSD eliminates reliance on external GDS indexing. The system implements a dedicated local backend using Laravel and MySQL. This custom-architected subsystem provides local Sudanese travel agencies with an intuitive web dashboard to manually input,

register, and digitize their internal domestic flights and regional packages, successfully uniting global data standards and local market realities under one centralized platform.

2.4 Conclusion:

The review of local and international travel-booking systems shows that, although many platforms offer useful features like global search, real-time results, and advanced filters, they do not address the specific needs of the Sudanese travel market. Most existing systems either focus on a single airline, depend on international data sources, or lack the tools needed by Sudanese agencies that still rely heavily on manual communication and offline processes.

The studies clearly highlight that there is no platform that brings Sudanese travel agencies together or allows customers to compare locally negotiated prices in one place. Agencies also lack digital dashboards, and customers have no structured way to submit booking requests or track their status online.

From these findings, a consistent gap becomes clear: the Sudanese market needs a localized, multi-agency platform that collects flight offers, improves price transparency, and supports organized communication between customers and agencies.

The proposed system aims to fill this gap by creating the first unified Sudanese flight comparison and booking-request platform. It will allow agencies to manage their offers digitally and give customers an easier and more transparent way to search, compare, and request bookings, helping modernize the flight-booking process in Sudan.

2.5 Summary of literature Review:

The following comprehensive synthesis matrix aggregates the operational objectives, technical architectures, and core results of the reviewed systems and theoretical literature, establishing the framework for **AirBridgeSD**:

Study	Objective	Tools and Techniques	Results
Study 1: Tarco Aviation Online Booking System (2024)	To provide a direct, single-carrier web portal for domestic and regional flight search and reservation requests in Sudan.	HTML, CSS, JavaScript (Front-end), PHP/CMS Framework (Back-end), and MySQL Relational Database.	Digitized direct-carrier booking workflows; eliminated manual agency dependency for single-carrier customers; managed localized data safely within low-scale database environments.
Study 2: Haboob Travel Multi-Service Booking Platform (2019)	To deliver a multi-service client platform for cross-border flight aggregation, hotel bookings, tour packages, and car rentals.	Mobile Frameworks (React Native/Flutter), External GDS API Integration, Hybrid Cloud Storage using SQL (Reservations) and NoSQL (Dynamic Flight feeds).	Unified international travel discovery via mobile and web apps; deployed secure user sessions and international payment gateways; completely excluded localized travel agency pricing packages.
Study 3: Blue Orange B2B Flight Hosting System	To provide a secure, backend Business-to-Business portal for Sudanese travel agencies to automate internal corporate reservation processes.	Enterprise Dashboards, Airline Host Systems/GDS Integration, Database storage for Agency Profiles, PNR logs, and Financial transaction records.	Automated legacy administrative tasks (PNR generation, invoice creation, and fare lookups); completely eliminated client-facing interface options, leaving consumers dependent on manual outreach.
Study 4: Wego Global Comparison Platform (2023)	To aggregate multi-airline data feeds into an international consumer metasearch engine optimized for real-time price sorting and parallel route filtering.	Multi-source API Aggregation (Legacy GDS like Sabre/Amadeus, and Modern NDC APIs like Duffel), Distributed Cloud Infrastructure, Hybrid Database Layer (RDBMS and NoSQL mix).	Accomplished real-time price comparison and parallel routing analytics for global travel; failed to index local offline Sudanese travel agency records, customized tariffs, and regional manual payment habits.

Study 5: Online Comparison Shopping Behavior (Xiang and Gretzel)	To empirically analyze user search pathways and non-linear cross-site shopping habits of travel consumers during flight selection.	Empirical Field Research, Behavioral Data Mining, Consumer Tracking Models, Qualitative and Quantitative Analysis.	Proved that consumers switch non-linearly between aggregators (discovery phase) and trusted carrier sites (booking execution phase); emphasized the critical role of data trust and clarity over base price.
Study 6: Skyscanner Global Travel Metasearch Platform	To deliver a global, high-speed metasearch engine for instant multi-provider flight routing and dynamic calendar price comparison matrices.	Distributed Microservices, React JS, Java, Python Multi-Tiered Backend, Distributed Caching (Redis), structured RESTful API parsing algorithms.	Achieved advanced multi-criteria filtering, webhooks-based price alerting, and predictive analytics; created total market data disconnect inside offline aviation contexts like Sudan.
Study 7: Amadeus Global Distribution System Framework	To build a secure, global B2B transaction network for baseline passenger record management, flight configuration tracking, and global fare distribution.	High-availability Hybrid Cloud, Mainframe Databases, IATA standards, EDIFACT, Self-Service REST APIs, OAuth 2.0 Security, and sandbox test databases.	Standardized global aviation logic into universal JSON schemas with absolute transaction integrity; exposed vulnerabilities to regional sandbox closures (March 2026), necessitating hybrid backend designs.

Table 2.1 Summary Of Literature Review

CHAPTER 3

SYSTEM ANALYSIS AND DESIGN

3.1 Introduction:

System analysis and design involves understanding the requirements and constraints of the **AirBridgeSD** platform, a unified digital system for air travel agencies in Sudan and decomposing them into detailed components to ensure effective design and implementation.

3.2 System Analysis:

3.2.1 Functional Requirements:

3.2.1.1 Customer Functional Requirements:

- **Search for Flights:** The customer can search for available flights by entering departure/destination locations, travel dates (one-way or round-trip), number of passengers, and travel class.
- **View Available Flights:** The customer can view and browse a comprehensive list of available flights provided by different registered travel agencies.
- **View Flight Details:** The customer can view full itinerary details including airline, departure/arrival times, flight duration, airport nodes, baggage allowance, seat availability, agency name, and final price.
- **Filter Flights:** The customer can filter and sort flight results based on the lowest price, shortest duration, and preferred airline.
- **View Flight Offers:** The customer can view promotional flight packages and travel offers published by various agencies.
- **View Offer Details:** The customer can view full details of any flight offer, including airline, scheduling, baggage, seat capacity, providing agency, promotional price, and the offer's validity date.
- **Filter Flight Offers:** The customer can filter and search travel offers by price, duration, airline, and travel dates.
- **Search Flight Offers:** The customer can search within travel offers using a search field that filters by title, name, or specific travel agency.

- **Send Booking Request (via WhatsApp):** The customer can initiate a booking where the system automatically structures a detailed WhatsApp message containing the complete selected itinerary and routes it to the designated agency.
- **Contact Platform Administrator:** The customer can submit a support form to the admin by entering their name, email, phone number, selecting the message type (e.g., Suggestion, Complaint), and writing the message body.
- **View General Information:** The customer can browse platform static pages including Agency Directory, Terms of Use, About the System, Privacy Policy, and FAQs.

3.2.1.2 Agency Functional Requirements

- **Register Agency Account:** The agency can submit an onboarding request by filling out basic credentials, uploading official verification documents, and providing account details. Access to the dashboard remains restricted until Admin approval.
- **Login to Dashboard:** Upon receiving Admin approval, the authorized agency can securely log into their private dashboard using their registered email and password.
- **Logout from Dashboard:** The agency can securely terminate its active session and exit the dashboard, clearing authentication tokens to prevent unauthorized access.
- **Manage Agency Profile:** The agency can update and maintain its operational profile, including agency business name, phone number, WhatsApp contact link, and physical office address.
- **View Partnered Airlines:** The agency can browse a comprehensive list of system-supported airlines provided by the platform.
- **Add Partnered Airlines:** The agency can add specific airlines to its active operational partnership list.

- **Remove Partnered Airlines:** The agency can remove airlines from its profile that it no longer works or partners with.
- **Create Commission Rules:** The agency can define customized commission rules based on the airline and flight type (one-way/round-trip) using either a fixed monetary amount or a percentage value.
- **Update Commission Rules:** The agency can modify or edit existing commission values and pricing structures for any partnered airline.
- **Delete Commission Rules:** The agency can permanently remove active commission rules from the system, reverting the flight pricing to its base fare.
- **View Travel Offers:** The agency can view a list of all active travel offers and promotional packages currently assigned to its profile.
- **Create Travel Offers:** The agency can add and publish new customized travel packages and promotional flights to the platform.
- **Remove Travel Offers:** The agency can manually delete active travel offers that it no longer wishes to display.
- **Auto-Expire Travel Offers:** The system automatically deprecates and removes travel offers from public view once their validity or expiration date passes.
- **Receive Booking Requests:** The agency receives structured customer booking requests containing full flight metrics and the displayed price directly via automated WhatsApp routing.
- **Contact Platform Administrator:** The agency can send support messages to the system administrator by completing a dedicated dashboard form with a specified subject and message body.

3.2.1.3 Admin Functional Requirements:

- **Login to Administration Panel:** The admin can securely log into the system's management backend using authorized administrator credentials.

- **Logout from Administration Panel:** The admin can securely terminate the active session, clearing all authentication tokens and access privileges.
- **View Dashboard Analytics:** The admin can view platform metrics and general statistics, including the total number of registered travel agencies, active customers, and platform traffic.
- **Review Agency Registration Requests:** The admin can view and audit registration requests, checking uploaded official documents and details submitted by new travel agencies.
- **Approve Agency Requests:** The admin can approve pending agency requests, which triggers the system to activate the account and automatically dispatch an activation email via SMTP.
- **Reject Agency Requests:** The admin can reject invalid agency registration requests, providing a reason and locking the account from platform access.
- **Suspend Agency Account:** The admin can temporarily disable or suspend an approved agency account to freeze its operations on the platform when necessary.
- **Delete Agency Account:** The admin can permanently remove an agency account and its associated records from the database if required.
- **View System Airlines:** The admin can browse the comprehensive database of airlines currently supported by the platform.
- **Add New Airlines:** The admin can input and register new airlines into the system database to make them available for agencies.
- **Update Airline Information:** The admin can edit and update existing airline data, names, logos, or identifiers.
- **Delete Airlines:** The admin can remove airlines from the global database if they are no longer operational.
- **View Support Requests and Complaints:** The admin can review contact forms, suggestions, and complaints submitted by both customers and travel agencies.

- **Respond to Support Requests:** The admin can manage and mark support tickets or complaints as "Resolved" after reviewing them.

3.2.1.4 System Functional Requirements:

- **Authentication and Access Control:** The system shall enforce role-based authentication and authorization rules to restrict access according to Customer, Agency, and Administrator permissions.
- **Agency Verification Workflow:** The system shall validate agency onboarding requests and restrict dashboard access until administrative approval is completed.
- **Dynamic Pricing and Commission Calculation:** The system shall automatically calculate and display the final ticket price by applying agency-defined commission rules (fixed value or percentage) according to the selected airline and trip type.
- **Booking Request Generation and Routing:** The system shall automatically generate a structured booking request containing selected flight details and route the request to the selected agency through WhatsApp integration.
- **Booking Request Tracking:** The system shall record each generated booking request at the time of submission and maintain a request history visible to the corresponding agency dashboard without requiring customer account registration.

3.2.2 Use Case Diagram:

The use case diagram for A Unified Digital Platform for Air Travel Agencies in Sudan System represents the interactions between different user roles and the system's core functions. It outlines how key actors such as customer, agency, and admin engage with the system to complete specific tasks. Use case

diagrams provide a visual tool to clarify system boundaries and user behaviors, and they offer a simplified overview of functional requirements [4].

3.2.2.1 Customer use case diagram

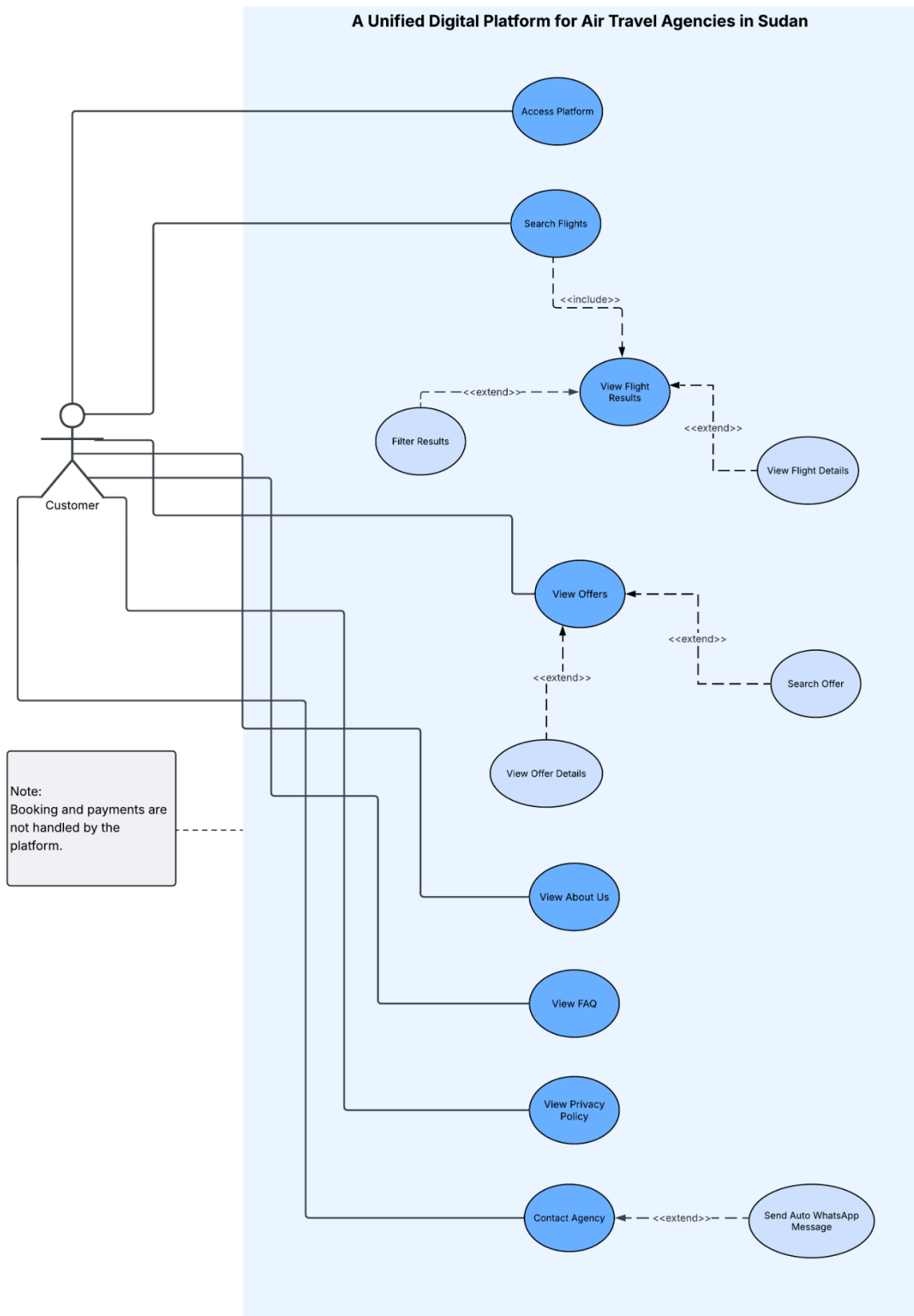


Figure 3.1 Use Case Diagram for Customer

- Customer Use Case Descriptions:

Table 3.1 Customer Access Platform Use Case Description

#	1
Name	Access Platform
Actor	Customer
Precondition	The user has a stable internet connection and a compatible web browser.
Basic Flow	1. The Customer enters the AirBridgeSD platform URL in the browser. 2. The System loads the main landing page. 3. The System displays available search fields and latest flight promotional offers.
Postcondition	The Customer successfully views the platform home screen and is ready to search for flights.

Table 3.2 Customer Search Flights Use Case Description

#	2
Name	Search Flights
Actor	Customer
Precondition	The Customer has successfully accessed the platform landing page.
Basic Flow	1. The Customer inputs travel criteria (Departure city, Destination city, Travel date, and Trip Type: One-Way or Round-Trip, and can put the numbers of passengers or the type of class). 2. The Customer clicks the "Search" button. 3. The System includes and executes the View Flight Results use case. 4. The System displays available search fields and latest flight promotional offers.
Postcondition	Flight options matching the search criteria are retrieved from the data sources.

Table 3.3 Customer View Flight Results Use Case Description

#	3
Name	View Flight Results
Actor	Customer
Precondition	<<include>> by the Search Flights use case after executing a query.
Basic Flow	1. The System fetches live flight availability and dynamically applies configured agency commissions to the pricing logic. 2. The System displays a list of available flights showing prices, airlines, and durations.
Postcondition	The Customer can see the available options with integrated markup prices with the details of trip and contact agency button.

Table 3.4 Customer Filter Results Use Case Description

#	4
Name	Filter Results
Actor	Customer
Precondition	The flight results list is currently displayed on the screen.
Basic Flow	1. The Customer selects specific filters (e.g., specific Airline, shortage duration) or chooses a sorting method (Price: Low). 2. The System updates the displayed flight list instantly based on the selected criteria.
Postcondition	The flight results are refined and rearranged according to customer preferences.

Table 3.5 Customer View Flight Details Use Case Description

#	5
Name	View Flight Details

Actor	Customer
Precondition	<<extend>> from View Flight Results when the user clicks on a specific flight card.
Basic Flow	1. The Customer clicks on a specific flight from the results list to see more information. 2. The System displays comprehensive trip details (Baggage allowance, exact transit times, and terminal info, the name of agency, contact agency button).
Postcondition	The Customer successfully views the expanded detailed specifications of the flight.

Table 3.6 Customer View Offers Use Case Description

#	6
Name	View Offers
Actor	Customer
Precondition	The Customer navigates to the promotional offers section from the landing page.
Basic Flow	1. The Customer clicks on the "Offers" tab/section. 2. The System displays special promotional flight packages published by registered travel agencies.
Postcondition	The Customer views available discount deals.

Table 3.7 Customer Search Offer Use Case Description

#	7
Name	Search Offer
Actor	Customer
Precondition	<<extend>> from View Offers when the user wants to search within the available offers.
Basic Flow	1. The Customer types specific keywords or destinations inside the search box of the offers section. 2. The System updates the promotional list to show matching deals.
Postcondition	The filtered promotional flight offers are successfully shown.

Table 3.8 Customer View Offer Details Use Case Description

#	8
Name	View Offer Details
Actor	Customer
Precondition	<<extend>> from View Offers when the Customer clicks on a specific promotional travel offer package.
Basic Flow	1. The Customer browses the listed travel offers and clicks on a specific package card. 2. The System intercepts the click and queries the database for the complete details of this promotional deal. 3. The System opens an expanded modal or detailed view showing the full itinerary, airline information, special discount rate, expiration date, the specific travel agency identity publishing the deal, the number of seats, the description of trip, and the details of contact like name of agency and the button for contact.
Postcondition	The Customer successfully views the comprehensive and full details of the chosen travel offer, ready to initiate contact for booking.

Table 3.9 Customer View About Us Use Case Description

#	9
Name	View About Us
Actor	Customer
Precondition	The Customer is browsing the platform.
Basic Flow	1. The Customer clicks on the "About Us" link in the navigation menu or footer. 2. The System displays the platform's vision, mission, how the platform works, and company information.
Postcondition	The Customer views the institutional background information of AirBridgeSD.

Table 3.10 Customer View FAQ Use Case Description

#	10
Name	View FAQ
Actor	Customer
Precondition	The Customer is browsing the platform.
Basic Flow	1. The Customer clicks on the "FAQ" (Frequently Asked Questions) section link. 2. The System loads the common questions list along with their pre-defined answers.
Postcondition	The Customer reviews informational help guidelines.

Table 3.11 Customer View Privacy Policy Use Case Description

#	11
Name	View Privacy Policy
Actor	Customer
Precondition	The Customer is browsing the platform.
Basic Flow	1. The Customer clicks on the "Privacy Policy" legal link in the footer. 2. The System displays the terms, data handling standards, and user privacy conditions.
Postcondition	The Customer views the platform legal policies document.

Table 3.12 Customer Contact Agency Use Case Description

#	12
Name	Contact Agency
Actor	Customer
Precondition	The Customer selects a preferred flight option or promotional offer to initiate booking.
Basic Flow	1. The Customer clicks the "Contact Agency" action button. 2. The System triggers the extending function Send Auto

	WhatsApp Message.
Postcondition	The platform readies the redirection pipeline for external communication.

Table 3.13 Customer Send Auto WhatsApp Message Use Case Description

#	13
Name	Send Auto WhatsApp Message
Actor	Customer
Precondition	<<extend>> triggered directly from the Contact Agency action.
Basic Flow	1. The System parses the selected flight data (Airline, Date, Price) and compiles a formatted text message. 2. The System opens the external WhatsApp Web/App interface pre-filled with the agency's phone number and the compiled text message.
Postcondition	The Customer leaves the system boundary to finalize and track negotiation details directly with the agency via WhatsApp.

3.2.2.2 Agency use case diagram

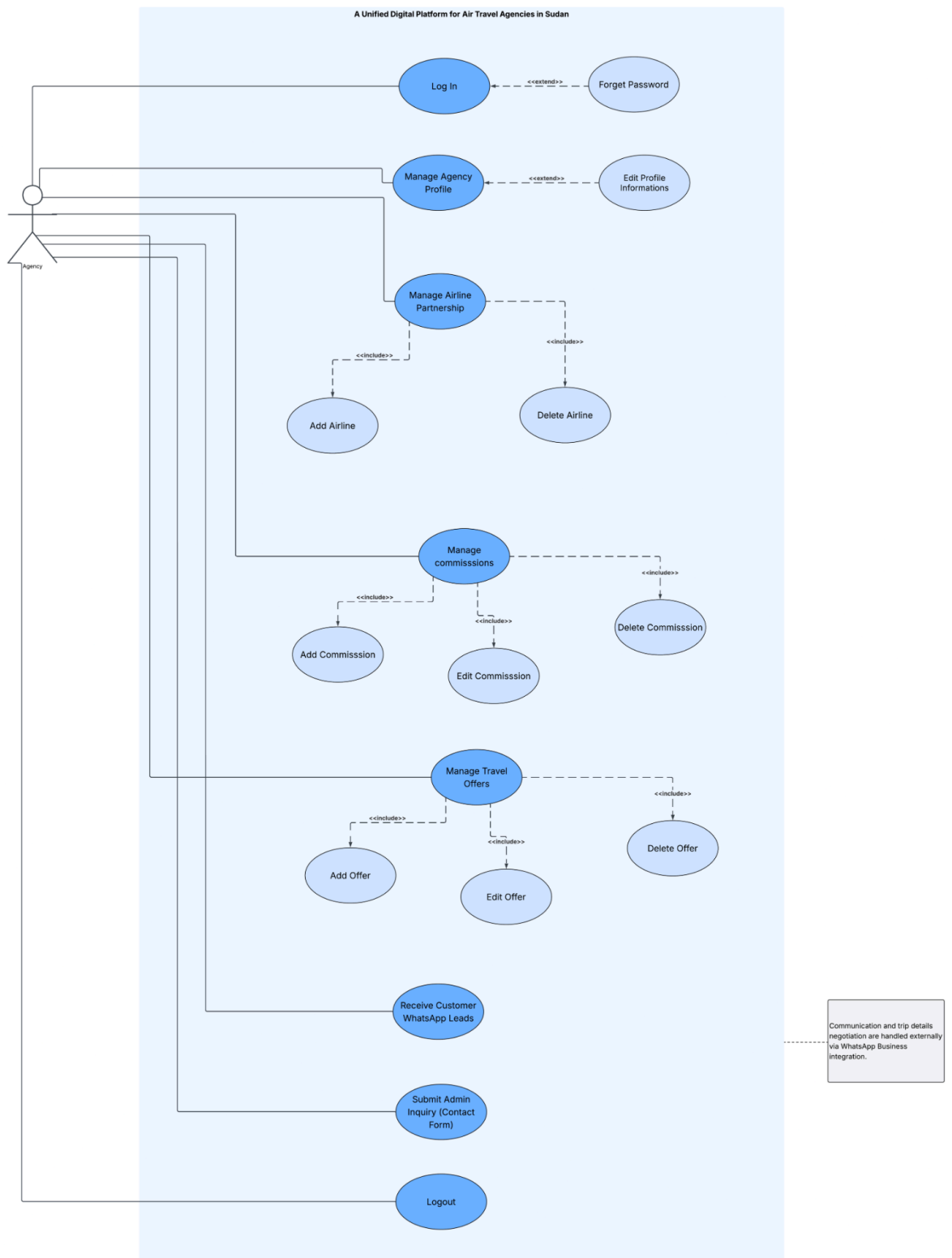


Figure 3.2 Use Case Diagram for Agency Dashboard

- Agency Use Case Descriptions:

Table 3.14 Agency Login Use Case Description

#	14
Name	Login
Actor	Agency
Precondition	The Agency has an active, registered account approved by the platform Admin.
Basic Flow	1. The Agency enters their registered Email and Password. 2. The System validates the credentials against the database. 3. The System redirects the Agency to their secure Dashboard workspace.
Postcondition	The Agency gains access to the dashboard features.

Table 3.15 Agency forgot Password Use Case Description

#	15
Name	forgot Password
Actor	Agency
Precondition	<<extend>> from Log In when the Agency cannot remember their credentials.
Basic Flow	1. The Agency clicks on the "Forget Password" link on the login screen. 2. The Agency inputs their registered email address. 3. The System verifies the email and transmits a secure password reset link to the user's inbox email.
Postcondition	A recovery email is successfully sent to the Agency.

Table 3.16 Agency Manage Agency Profile Use Case Description

#	16
Name	Manage Agency Profile

Actor	Agency
Precondition	The Agency is successfully logged in.
Basic Flow	1. The Agency navigates to the profile settings tab. 2. The System displays the current agency profile details and can update the information, (e.g., name of company, phone number, email, name of responsible account, company logos image and location).
Postcondition	The Agency successfully views their current account profile interface.

Table 3.17 Agency Edit Profile Information Use Case Description

#	17
Name	Edit Profile Informations
Actor	Agency
Precondition	<<extend>> from Manage Agency Profile when the user triggers the edit mode.
Basic Flow	1. The Agency clicks on the "Edit Profile" button. 2. The Agency modifies details (such as contact information, phone numbers). 3. The Agency clicks "Save Changes". 4. The System validates fields, updates the database, and re-renders the profile page.
Postcondition	The updated profile configuration is saved.

Table 3.18 Agency Manage Airline Partnership Use Case Description

#	18
Name	Manage Airline Partnership
Actor	Agency
Precondition	The Agency is logged in and authorized to update operational airline listings.
Basic Flow	1. The Agency accesses the Airline Partnership section from the side menu. 2. The System fetches and lists current airline entities.

	3. The System includes and manages specific structural CRUD modifications (Add Airline , Edit Airline , Delete Airline).
Postcondition	The core structural airline array configuration is prepared for data modification.

Table 3.19 Agency Add Airline Use Case Description

#	19
Name	Add Airline
Actor	Agency
Precondition	<<include>> by Manage Airline Partnership triggered from the central partnership.
Basic Flow	1. The Agency clicks "Add New Airline". 2. The System displays the popup with a select form for the Agency to choose. 3. The Agency chooses the airline from the select form. 4. The Agency clicks "Save". 5. The System appends the new entity to the data tables.
Postcondition	A new airline partner is added to the system scope.

Table 3.20 Agency Delete Airline Use Case Description

#	20
Name	Delete Airline
Actor	Agency
Precondition	<<include>> by Manage Airline Partnership triggered from the database management workspace.
Basic Flow	1. The Agency clicks the delete/remove button icon next to a specific airline entry. 2. The Agency should confirm to delete the specific airline. 3. The System removes the selected airline relation entity from the active operational tables.
Postcondition	The airline entry is safely removed from the system visibility.

Table 3.21 Agency Manage commissions Use Case Description

#	21
Name	Manage commissions
Actor	Agency
Precondition	The Agency is logged into the platform workspace dashboard.
Basic Flow	1. The Agency opens the unified pricing configuration panel. 2. The System renders current dynamic markup rules. 3. The System branches out to include the core processing logic (Add Commission, Edit Commission, Delete Commission).
Postcondition	Rule matrices are set to dynamically process live flight.

Table 3.22 Agency Add Commission Use Case Description

#	22
Name	Add Commission
Actor	Agency
Precondition	<<include>> by Manage commissions invoked from the commission configuration section.
Basic Flow	1. The Agency clicks "Add New Commission". 2. The Agency defines criteria by two steps: a. chooses the Target Airline and sets Trip Type (One-Way / Round-Trip). b. selects Type (Fixed / Percentage) and inputs the price value. 3. The Agency clicks "Save Commission". 4. The System Check if the new Commission for the airline is already exists: a. If it exists they will notify the agency that the airline already has a commission and if the agency wants to continue the system will replace it by the new commissions. b. If it does not exist The System registers the rule to apply markup overrides during live search.
Postcondition	New payout settings are available and logged into the database.

Table 3.23 Agency Edit Commission Use Case Description

#	23
Name	Edit Commission
Actor	Agency
Precondition	<<include>> by Manage commissions invoked from the commission rule panel.
Basic Flow	1. The Agency selects an existing active margin rule row. 2. The Agency clicks the Edit icon. 3. The Agency modifies criteria constraints or updates values. 4. The System overwrites structural entries inside the configuration log database.
Postcondition	The pricing rules are updated and refreshed.

Table 3.24 Agency Delete Commission Use Case Description

#	24
Name	Delete Commission
Actor	Agency
Precondition	<<include>> by Manage commissions invoked from the rule row layout interface.
Basic Flow	1. The Agency triggers deletion for an outdated markup configuration rule. 2. The Agency clicks the delete/remove button icon next to a specific commission. 3. The Agency should confirm to delete the specific commission. 4. The System drops the rule row, resetting the default baseline pricing model for that flight segment.
Postcondition	The pricing rules are updated and refreshed.

Table 3.25 Agency Manage Travel Offers Use Case Description

#	25
Name	Manage Travel Offers
Actor	Agency
Precondition	The Agency is logged into the platform workspace dashboard.
Basic Flow	1. The Agency enters the promotional offers management panel. 2. The System provides structural options to perform actions via inclusion links (Add Offer, Edit Offer, Delete Offer).
Postcondition	The promotion pipeline configuration is updated for public display.

Table 3.26 Agency Add Offer Use Case Description

#	26
Name	Add Offer
Actor	Agency
Precondition	<<include>> by Manage Travel Offers The Agency is logged into the dashboard session and has active access to the promotions workspace.
Basic Flow	1. The Agency clicks the "Add New Offer" button inside the travel offers panel. 2. The System renders a clean, empty creation form with required fields (e.g., Offer Title, Flight Paths/Destinations, Promotional Description, Discounted Price, and Expiration Date). 3. The Agency fills in the details and uploads promotional imagery if applicable. 4. The Agency clicks the "Save Offer" button. 5. The System validates the input fields to ensure all mandatory data points are correct and complete. 6. The System inserts a new entry row into the promotional offers database table. 7. The System automatically pushes the new offer item live onto the public Customer-facing portal.

Postcondition	A new promotional flight package is successfully created, saved in the database, and displayed instantly for public customers to browse.
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Table 3.27 Agency Edit Offer Use Case Description

#	27
Name	Edit Offer
Actor	Agency
Precondition	<<include>> by Manage Travel Offers The Agency wants to update an existing promotional travel offer package.
Basic Flow	1. The Agency navigates to the active offers table and clicks the "Edit" button icon next to a specific offer. 2. The System intercepts the request, queries the database, and loads an input form pre-populated with the current details of that specific offer (e.g., Title, Flight paths, Discount price, and Expiration date). 3. The Agency modifies the required fields or updates the promotional media assets. 4. The Agency clicks the "Save Changes" button. 5. The System validates the newly entered data to ensure compliance with input constraints. 6. The System updates the database record instantly and clears any cached public data. 7. The System refreshes the dashboard view and displays a success notification.
Postcondition	The promotional offer details are successfully modified, and the updated deal goes live immediately on the Customer-facing portal.

Table 3.28 Agency Delete Offer Use Case Description

#	28
Name	Delete Offer
Actor	Agency
Precondition	<<include>> by Manage Travel Offers An active travel offer has expired, or the Agency decides to cancel a promotional deal.

Basic Flow	1. The Agency browses their offers list and clicks the "Delete" button icon on the target offer row. 2. The System triggers a confirmation dialogue box (Modal) to prevent accidental deletions. 3. The Agency confirms the deletion action. 4. The System intercepts the confirmation and executes a database query to remove (or softly archive) the specific offer row. 5. The System purges the offer data cache so it immediately disappears from public view. 6. The dashboard view updates dynamically to reflect the removal, displaying a success alert notice.
Postcondition	The promotional offer is successfully deleted from the database tables and is no longer visible to customers on the platform.

Table 3.29 Agency Receive Customer WhatsApp Leads Use Case Description

#	29
Name	Receive Customer WhatsApp Leads
Actor	Agency
Precondition	A Customer clicks the contact button inside the public search platform.
Basic Flow	1. The System initiates external routing hooks. 2. The Agency receives an inbound notification/message containing pre-compiled user intent directly on their WhatsApp endpoint. 3. The Agency reviews negotiation parameters and pricing expectations.
Postcondition	Communication parameters are captured external to the main app layout box.

Table 3.30 Agency Submit Admin Inquiry (Contact Form) Use Case Description

#	30
Name	Submit Admin Inquiry (Contact Form)
Actor	Agency

Precondition	The Agency is logged into the dashboard session.
Basic Flow	1. The Agency navigates to the "Contact Admin" section from the dashboard side menu. 2. The System renders a unified support form containing an inquiry type selector and a text message area. 3. The Agency opens the dropdown select menu and chooses the specific nature of their message (e.g., Complaint, Suggestion, or Feedback). 4. The Agency types the details of their message into the input field and clicks the "Submit" button. 5. The System processes the form inputs, saves the data to the database, and routes it instantly to the Admin.
Postcondition	The form data is successfully transmitted to the Admin panel, and a confirmation message is displayed to the Agency.

Table 3.31 Agency Logout Use Case Description

#	31
Name	Logout
Actor	Agency
Precondition	The Agency is currently logged in and has an active dashboard session.
Basic Flow	1. The Agency clicks on the "Logout" button from the dashboard side menu. 2. The System intercepts the request and instantly terminates the active session. 3. The System destroys the connection between the backend server and the frontend client to secure the account. 4. The System automatically redirects the Agency back to the primary Login page.
Postcondition	The active session is completely closed, and the dashboard workspace becomes inaccessible without logging in again.

3.2.2.3 Admin use case diagram

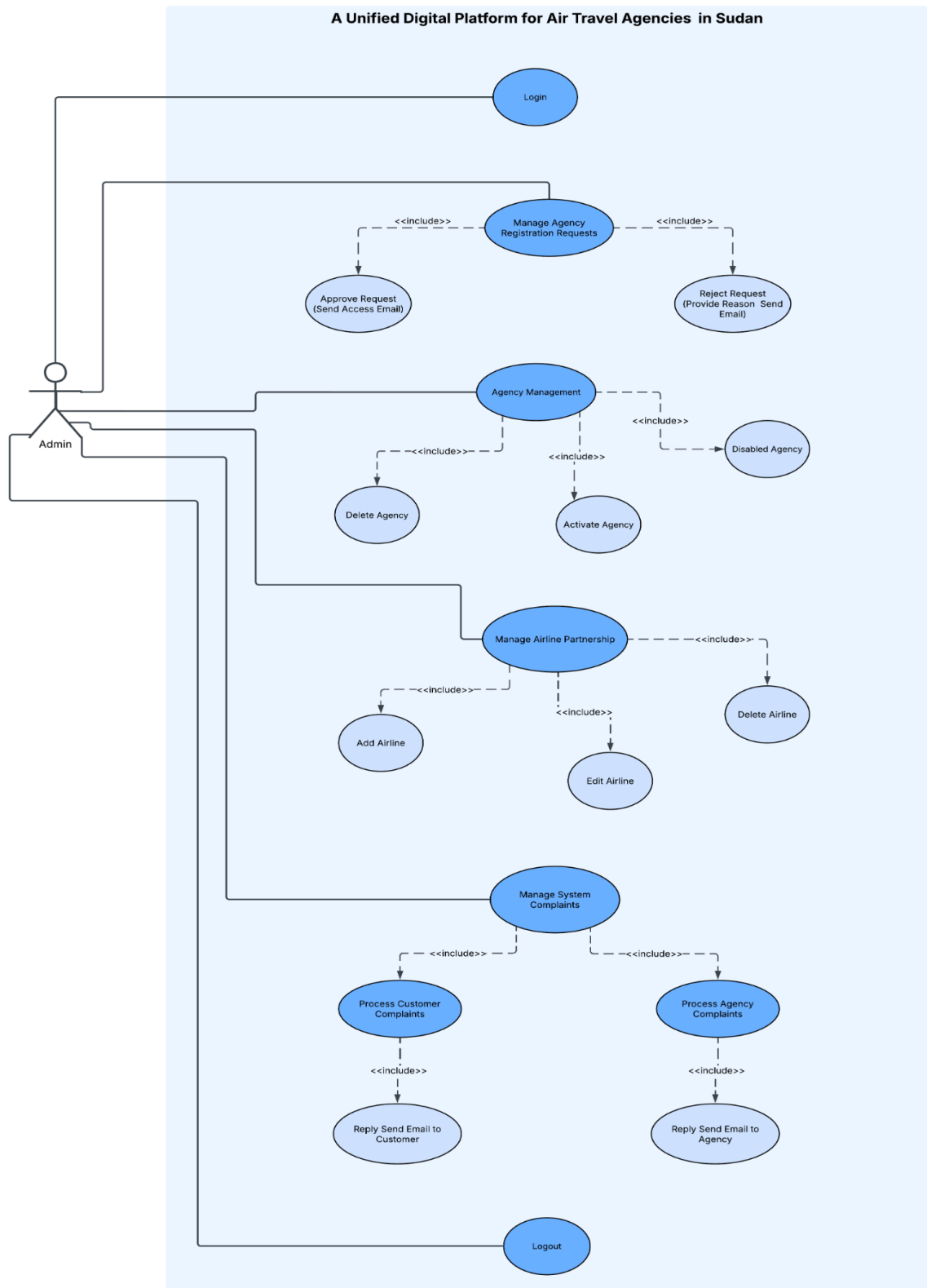


Figure 3.3 Use Case Diagram for Admin Dashboard

- **Admin Use Case Descriptions:**

Table 3.32 Admin Login Use Case Description

#	32
Name	Login
Actor	Admin
Precondition	The Admin possesses valid system administrator credentials.
Basic Flow	1. The Admin enters their master Email and Password on the platform secure login portal. 2. The System authenticates the credentials against the administrator user tables. 3. The System redirects the Admin to the central dashboard overview page.
Postcondition	The Admin session is initialized, granting administrative privileges across the platform.

Table 3.33 Admin Manage Agency Registration Requests Use Case Description

#	33
Name	Manage Agency Registration Requests
Actor	Admin
Precondition	The Admin is securely logged into the administration panel.
Basic Flow	1. The Admin navigates to the registration triage area to view incoming sub-agency applications. 2. The System displays a structured list containing sub-agency document inputs and profiles. 3. The Admin evaluates an application and invokes the integrated decision processing handlers (Approve Request or Reject Request).
Postcondition	The System updates the database and triggers notifications based on the Admin's decision.

Table 3.34 Admin Approve Request (Send Access Email) Use Case Description

#	34
Name	Approve Request (Send Access Email)
Actor	Admin
Precondition	<<include>>by Manage Agency Registration Requests The Admin is reviewing a pending registration request from a travel agency and decides to accept them into the platform.
Basic Flow	1. The Admin clicks the "Approve" button on the target agency's pending application file. 2. The System intercepts the click and changes the agency's status flag in the database from Pending to Active. 3. The System automatically triggers the backend Laravel Mail service via the configured SMTP server (using Laravel's Mail facade with TLS/SSL transport). 4. The System dynamically generates a professional onboarding email containing: a. A welcome message to the platform. b. A link to access their new dashboard. 5. The System sends the email directly to the registered agency's email address.
Postcondition	The travel agency account is securely whitelisted in the database, their dashboard becomes ready for use, and they receive an automated email.

Table 3.35 Admin Reject Request (Provide Reason, Send Email) Use Case Description

#	35
Name	Reject Request (Provide Reason, Send Email)
Actor	Admin
Precondition	<<include>> by Manage Agency Registration Requests The Admin is reviewing a pending application and finds that the agency does not meet the platform's criteria or has missing documents.
Basic Flow	1. The Admin clicks the "Reject" button on the target agency's pending file. 2. The System immediately opens a pop-up dialog box (Modal) requesting the Admin to type a mandatory

	rejection reason. 3. The Admin inputs the specific reason (e.g., "Missing valid operating license") and clicks "Confirm Rejection". 4. The System updates the database, setting the agency's status to Rejected. 5. The System triggers the mail service to construct an automated email that contains the custom rejection reason typed by the Admin. 6. The System dispatches the email to the agency's registered address.
Postcondition	The agency's request is turned down, the database record is updated, and the agency receives an email explaining exactly why they were not accepted, allowing them to fix the issue.

Table 3.36 Admin Agency Management Use Case Description

#	36
Name	Agency Management
Actor	Admin
Precondition	The Admin wants to review or update the status of agencies already working on the platform.
Basic Flow	1. The Admin navigates to the "Agencies Management" section from the dashboard side menu. 2. The System queries the database and fetches a complete list of all onboarded agencies in Sudan. 3. The System displays these agencies in a structured table showing their current status (Active, Disabled, etc.). 4. The Admin selects a specific agency to execute further administrative actions (Delete Agency , Activate Agency , or Disabled Agency).
Postcondition	The Admin views the current state of all agencies and is ready to modify their system permissions.

Table 3.37 Admin Delete Agency Use Case Description

#	37
Name	Delete Agency
Actor	Admin

Precondition	<<include>> by Agency Management The Admin decides to permanently remove an agency due to a violation of terms or business closure.
Basic Flow	1. The Admin clicks the "Delete" button icon next to the targeted agency. 2. The System prompts a confirmation warning message. 3. Upon confirmation, the System executes a database query to delete the agency's profile and relational data. 4. The System refreshes the directory view instantly to show the updated list.
Postcondition	The agency's record is completely removed from the operational platform tracking.

Table 3.38 Admin Activate Agency Use Case Description

#	38
Name	Activate Agency
Actor	Admin
Precondition	<<include>> by Agency Management An agency account was previously disabled, and the Admin wants to restore their full access.
Basic Flow	1. The Admin clicks the "Activate" toggle or button icon on a disabled agency profile. 2. The System modifies the status flag in the database back to Active. 3. The System restores all login rights and dashboard functionalities for this specific agency instantly.
Postcondition	The travel agency is fully re-activated and can resume configuring flight commissions and publishing offers.

Table 3.39 Admin Disabled Agency Use Case Description

#	39
Name	Disabled Agency
Actor	Admin
Precondition	<<include>> by Agency Management An agency has

	violated rules, and the Admin wants to temporarily block them without deleting their data.
Basic Flow	1. The Admin clicks the "Disable" button icon on an active agency row. 2. The System updates the database status flag to Disabled. 3. The System immediately revokes the agency's active authentication token, stopping them out of their current dashboard session if they are logged in.
Postcondition	The agency is blocked from logging into the platform until the Admin decides to activate them again.

Table 3.40 Admin Manage Airline Partnership Use Case Description

#	40
Name	Manage Airline Partnership
Actor	Admin
Precondition	The Admin needs to add, update, or clear airline company entries available for global system flight searches.
Basic Flow	1. The Admin navigates to the "Manage Airline Partnership" section from the dashboard side menu. 2. The System loads and displays the current list of global and local airline carriers operating within the system. 3. The Admin chooses to make modifications through the system's core functional branches (Add Airline, Edit Airline, or Delete Airline).
Postcondition	The system's baseline airline catalog data is structured and ready for structural modification.

Table 3.41 Admin Add Airline Use Case Description

#	41
Name	Add Airline
Actor	Admin
Precondition	<<include>> by Manage Airline Partnership A new airline carrier needs to be integrated into the system lookup tables.

Basic Flow	1. The Admin clicks "Add New Airline" and fills in the creation form fields (.e.g., Airline Name, Code, the image logo path). 2. The Admin clicks "Submit". 3. The System validates the entries, and appends the new airline row into the global dataset.
Postcondition	The new airline becomes instantly available as an option when agencies configure their pricing models.

Table 3.42 Admin Edit Airline Use Case Description

#	42
Name	Edit Airline
Actor	Admin
Precondition	<<include>> by Manage Airline Partnership The Admin needs to correct or update an existing airline's information.
Basic Flow	1. The Admin selects a specific airline row and clicks the "Edit" icon. 2. The System populates the form fields with the current database values for that airline. 3. The Admin modifies the fields (such as updating the company name, the code, and the logo image path) and clicks "Save changes". 4. The System overwrites the database row and refreshes the catalog view.
Postcondition	The updated airline data is synchronized globally across the platform.

Table 3.43 Admin Delete Airline Use Case Description

#	43
Name	Delete Airline
Actor	Admin
Precondition	<<include>> by Manage Airline Partnership An airline is no longer operating or associated with the platform.
Basic Flow	1. The Admin clicks the "Delete" action button icon next to

	an airline entry. 2. The System displays a confirmation pop-up window to avoid accidental clicks. 3. Upon confirmation, the System drops the entity from the database tracking rows.
Postcondition	The airline carrier is removed from global configuration views.

Table 3.44 Admin Manage System Complaints Use Case Description

#	44
Name	Manage System Complaints
Actor	Admin
Precondition	The Admin is logged into the system and navigates to the admin panel.
Basic Flow	1. The Admin clicks on the "Complaints and Suggestions" section from the dashboard side menu. 2. The System queries the database and fetches all incoming messages from both customers and agencies. 3. The System displays the messages in a comprehensive data table where each row contains: Name, Category (Type), and Status (Replied / Pending). 4. The Admin can use filter toggles to sort and filter the messages specifically by "Agencies" or "Customers". 5. The Admin selects a specific row to view and manage through the inclusion branches (Process Customer Complaints or Process Agency Complaints).
Postcondition	The Admin views and filters the unified support list, ready to drill down into details.

Table 3.45 Admin Process Customer Complaints Use Case Description

#	45
Name	Process Customer Complaints
Actor	Admin
Precondition	<<include>> by Manage System Complaints The Admin

	clicks on a specific row belonging to a regular customer/user.
Basic Flow	1. The Admin clicks on the customer's message row from the main list. 2. The System opens an expanded details view (Modal or dedicated page) showing the full description, confirming that the sender is a Customer, and loading their full message text. 3. The Admin reads the complaint and triggers the action to write a response (Reply Send Email to Customer).
Postcondition	The customer's message details are fully reviewed, and the system readies the reply workspace.

Table 3.46 Admin Reply Send Email to Customer Use Case Description

#	46
Name	Reply Send Email to Customer
Actor	Admin
Precondition	<<include>> by Process Customer Complaints The Admin has reviewed the customer's expanded details and types a response.
Basic Flow	1. The Admin types the official answer into the response input box and clicks the "Save / Send" button. 2. The System automatically triggers the backend mail service to dispatch the formatted reply directly to the Customer's email inbox. 3. The System instantly updates the database, changing the ticket status from Pending to Replied. 4. The main data table refreshes to display the updated status for that row.
Postcondition	The customer receives the resolution via email, and the row status is visually updated to "Replied" in the dashboard.

Table 3.47 Admin Process Agency Complaints Use Case Description

#	47
Name	Process Agency Complaints
Actor	Admin

Precondition	<<include>> by Manage System Complaints The Admin clicks on a specific row belonging to a registered travel agency.
Basic Flow	1. The Admin clicks on the agency's message row from the main list. 2. The System opens the expanded details view, explicitly stating that the sender type is an Agency, and displays the full context of their inquiry or suggestion. 3. The Admin reviews the text and initiates the response workflow (Reply Send Email to Agency).
Postcondition	The agency's corporate message parameters are fully opened and ready for resolution.

Table 3.48 Admin Logout Use Case Description

#	48
Name	Logout
Actor	Admin
Precondition	The Admin is currently authenticated and has an active administrative session in the system panel.
Basic Flow	1. The Admin clicks the "Logout" button at the bottom of the navigation workspace panel. 2. The System destroys the active administrative session context on the backend. 3. The System clears all high-privilege authorization cookies or local tokens stored in the browser window. 4. The System redirects the browser view back onto the default system Login landing interface.
Postcondition	The Admin session is completely ended, preventing any unauthorized browser back-button access to administrative dashboards.

3.2.3 Non-Functional Requirements:

- **Performance:**

- Response Time: Flight search results from the Duffel API must be retrieved and displayed within 3 to 5 seconds. This threshold is consistent with industry-standard UX research, which indicates that users begin to abandon web tasks after 3 seconds of loading time [25].
- Concurrency: The platform must be capable of handling at least 100 concurrent users performing searches simultaneously without performance degradation [23, 26]. This target is estimated based on the expected initial user base of Sudanese travel agency customers during peak hours [3], and is provisioned and optimized according to the selected hosting infrastructure [23].
- Resource Optimization: Media assets and scripts should be optimized (minified) to ensure fast loading times, even on slower internet connections common in the region.

- **Security:**

- Authentication and Authorization: Secure access is mandatory. Agencies can only access the dashboard after their identity is verified and "Approved" by the Admin.
- API Security: API credentials (API Keys/Tokens) must be handled strictly on the server side (Backend) to prevent exposure to the client side.
- Input Validation: All user inputs must be sanitized to protect the system against common web vulnerabilities like SQL Injection and Cross-Site Scripting (XSS).

- **Availability:**

- Hybrid Deployment Strategy: The platform adopts a hybrid deployment approach to ensure operational efficiency. The React.js

frontend is hosted on Netlify's global edge network, ensuring high availability and rapid access for users. The Laravel RESTful API and MySQL database are currently operating in a **local development environment (localhost)**, facilitating rigorous functional validation and testing during this graduation project phase. This architecture provides a stable foundation for a future transition to a production-ready PHP-compatible cloud hosting environment to ensure full public availability.

- Continuous Deployment (CI/CD): For the frontend, the system leverages GitHub for version control and Netlify for automated deployment, supporting a zero-downtime update process for the client-side interface. This ensures that UI improvements or frontend fixes are deployed seamlessly.
- System Uptime and Scalability: The current architecture is designed to support 24/7 accessibility once the backend is migrated to a cloud-based server. During the development phase, the local environment maintains consistent performance for testing purposes, with a clear roadmap for scaling the infrastructure to handle peak flight search traffic upon final deployment.

- **Reliability:**

- Graceful Error Handling: Since the system integrates with the external Duffel API, it is designed to handle potential API timeouts or connectivity issues gracefully. Instead of a system crash, users are presented with clear, helpful notifications (e.g., "Service temporarily unavailable, please try again") to maintain a professional user experience.
- Data Consistency: The system ensures that flight information, pricing, and agency details remain consistent throughout the user's search session, preventing any discrepancies between the flight selected and the flight details shared via WhatsApp.

- Calculation Precision (Accuracy): The commission engine is engineered for financial accuracy. It guarantees that the final price comprising the base fare plus the agency's specific commission (fixed or percentage) is calculated correctly without any decimal or rounding errors.
- **Usability:**
 - Responsive Design (Mobile-First): Since the majority of travelers in the region use smartphones for searching and comparing flights, the interface is built with a Mobile-First approach. It adapts seamlessly to all screen sizes, ensuring a consistent and functional experience across Smartphones, Tablets, and Desktops.
 - Native Arabic Support (RTL): The platform provides full, native support for Arabic (Right-to-Left) layout. This includes specialized typography and alignment designed for maximum readability, comfort, and cultural relevance for the local target audience.
 - Accessibility: The UI uses high-contrast colors and clear font sizes to ensure that the system is usable for people with different visual abilities.
- **User Experience (UX):**
 - Intuitive Navigation (3-Click Rule): The search-to-agency-contact flow is optimized for speed and simplicity. A customer should be able to reach an agency via WhatsApp in no more than 3 clicks from the homepage (Search → Select Flight → Contact Agency). This requirement is verified through usability walkthrough testing documented in Chapter 5.
 - Real-Time Feedback: The system provides immediate visual feedback for user actions such as loading spinners during flight searches and clear success/error notifications to reduce user anxiety and clarify the system's state.

- Minimalist Interface: To prevent "information overload," the flight results display only the most critical data (Price, Airline, Duration, and Agency) in a clean, card-based layout, allowing users to make quick and informed decisions.
- **Maintainability:**
 - Modular Architecture (Component-Based): The system is built using React's modular architecture, where the UI is broken down into independent, reusable components. This structure allows developers to update or fix specific parts of the platform (e.g., the Search Form or Agency Dashboard) without affecting the entire system.
 - Clear Documentation: To facilitate long-term maintenance, the source code is written with meaningful naming conventions and comments. Additionally, the database schema is documented to ensure that future developers can understand the data relationships and logic quickly.
 - Clean Code Practices: The project follows modern JavaScript (ES6+) and Tailwind CSS standards, making the codebase readable and easy to debug.
- **Compatibility:**
 - Cross-Browser Support: The system must function correctly across all modern web browsers, including Chrome, Safari, Edge, and Firefox.
 - Integration: The automated WhatsApp link generation must be compatible with the official WhatsApp application on both Android and iOS platforms.

3.3 Design Analysis:

3.3.1 Architecture design:

3.3.1.1 AirBridge General View

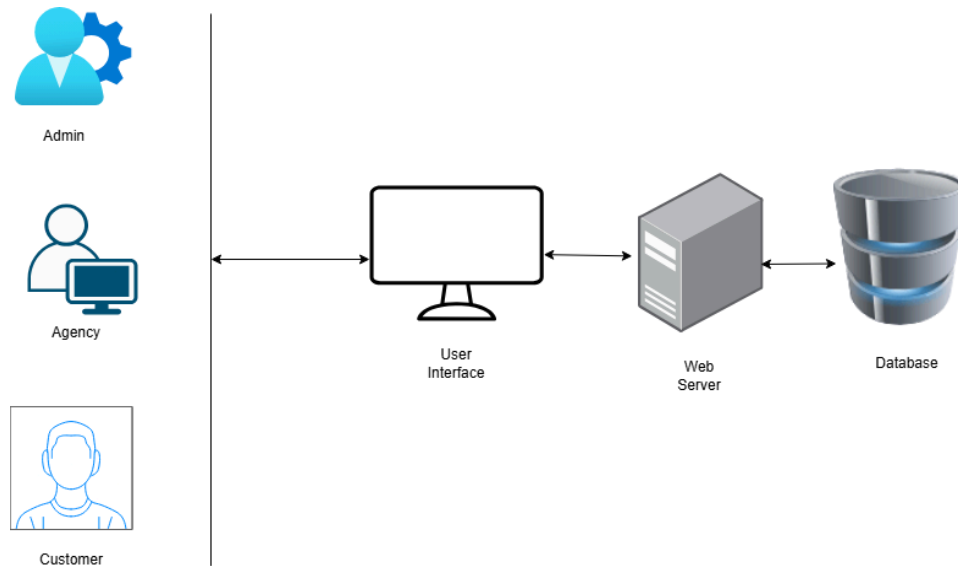


Figure 3.4 AirBridge General View

3.3.1.2 High-Level Architecture

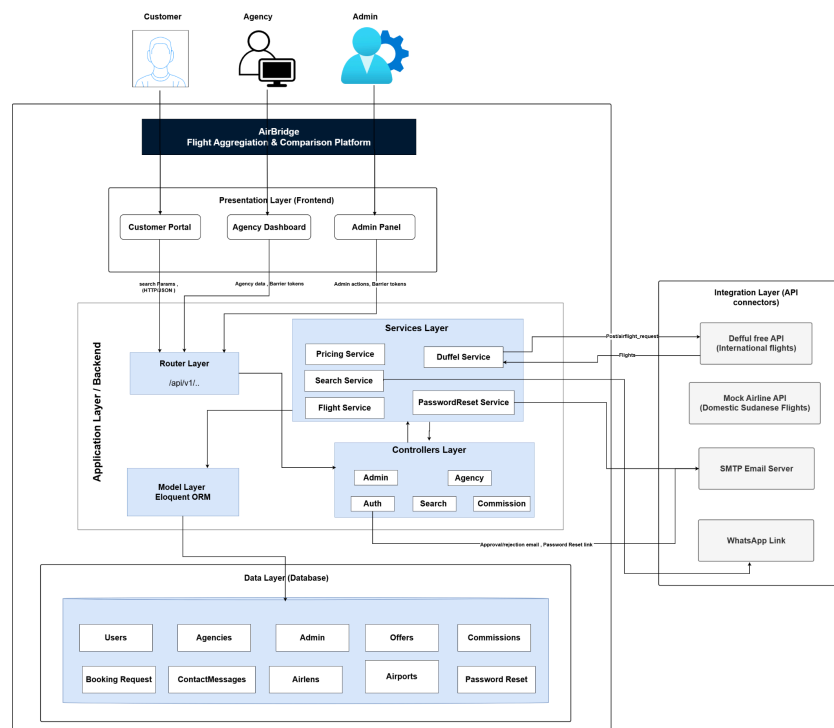


Figure 3.5 High-Level Architecture

3.3.1.3 Laravel MVC Model Architecture

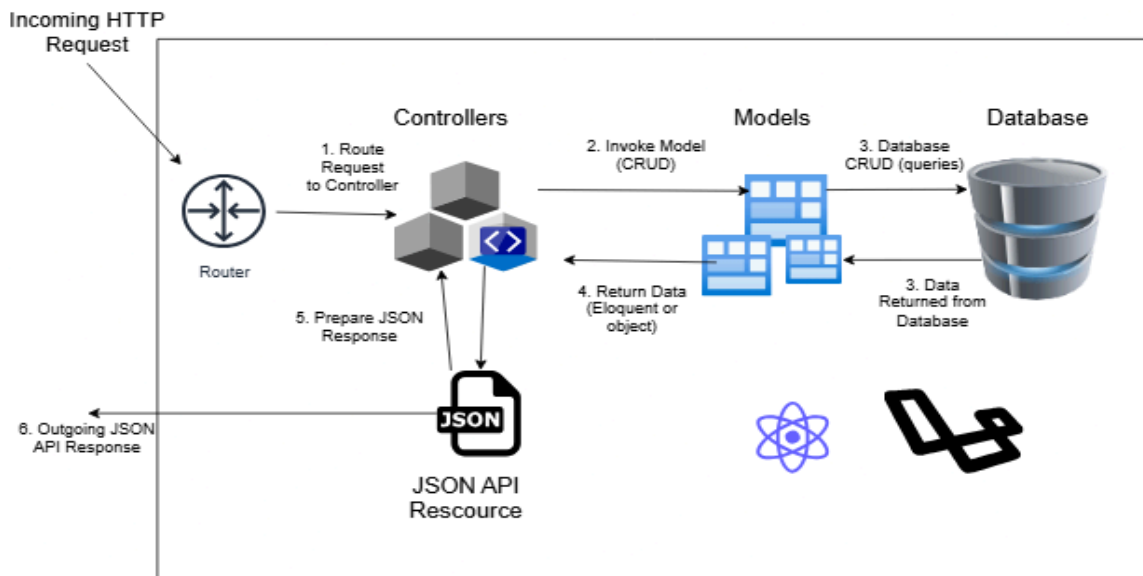


Figure 3.6 Laravel MVC Model Architecture

Backend MVC Architecture Flow:

The system's server-side logic is structured around the MVC (Model-View-Controller) architectural pattern, adapted specifically for an API-driven environment. Essentially, it represents the data journey from the initial request to the final response:

1. **The Request Entry (Router):** Every request originates from the React frontend client. Once the server receives this, the **Router** acts as the traffic controller, directing the request to the appropriate **Controller** assigned to handle that specific functionality.
2. **The Brain of the Operation (Controller):** The Controller is the core manager of the process. It interprets what the user needs, executes the necessary business logic, and determines exactly which data is required to fulfill the request.
3. **Data Handling (Model and Database):** If the request requires data such as flight details the Controller interacts with the **Model**. The Model acts as the interface to the database using **Eloquent ORM**, ensuring that all data operations are handled efficiently, securely, and in an organized manner.

4. **Preparing the Response (API Resources):** This is a key part of our architecture. Since the system implements a Single Page Application (SPA) architecture, we don't send complete HTML pages to the client. Instead, we use **API Resources** to transform the raw data retrieved from the database into structured, clean **JSON** objects.
5. **Delivering the Data:** Finally, this JSON response is sent back to the React application. The frontend then takes over, dynamically rendering the UI and presenting the data to the user instantly, without ever needing to reload the page, which provides a seamless user experience without full page reloads.

3.3.1.4 Client-Side Rendering (CSR) and API Interaction Flow Architecture

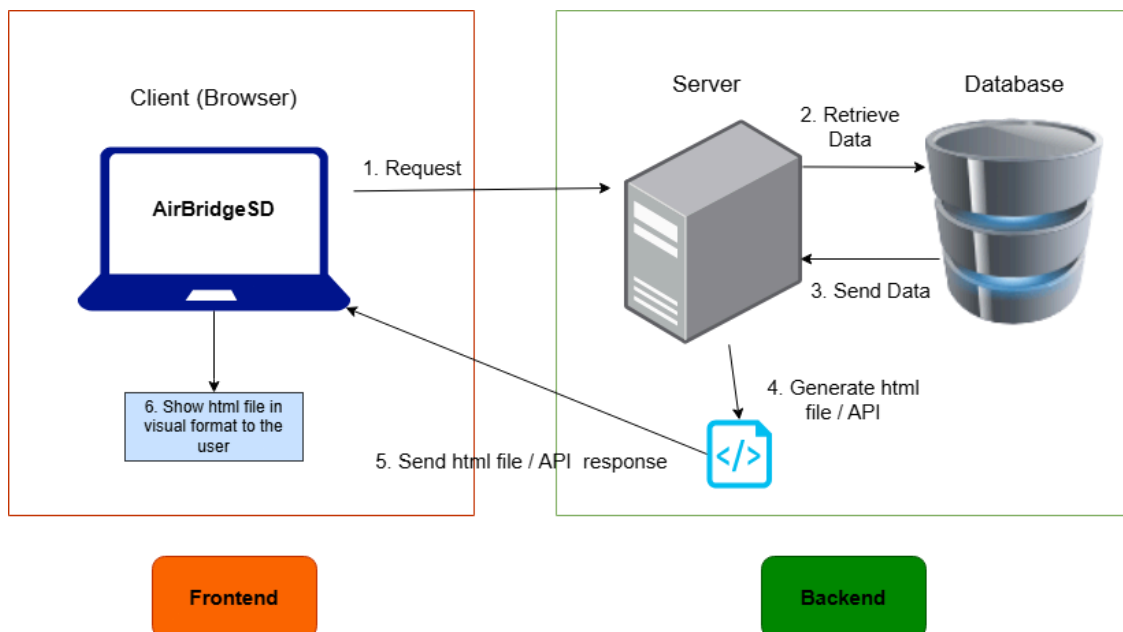


Figure 3.7 Client-Side Rendering (CSR) and API Interaction Flow Architecture

This diagram illustrates the Request/Response Cycle within the AirBridgeSD system, representing the high-level interaction between the client and the server:

1. **Request:** The user initiates an interaction through the browser, sending a request to the server (e.g., accessing the booking dashboard).
2. **Retrieve Data:** The server processes the request and queries the database to fetch the necessary information.
3. **Send Data:** The database retrieves the raw data and sends it back to the server.
4. **Generate Response:** The server processes the data and prepares the appropriate response (either HTML or an API payload).
5. **Send Response:** The final response is transmitted back to the user's browser over the network.
6. **Show Visual Format:** Finally, the browser renders the data into a user-friendly, visual interface for the user to interact with.

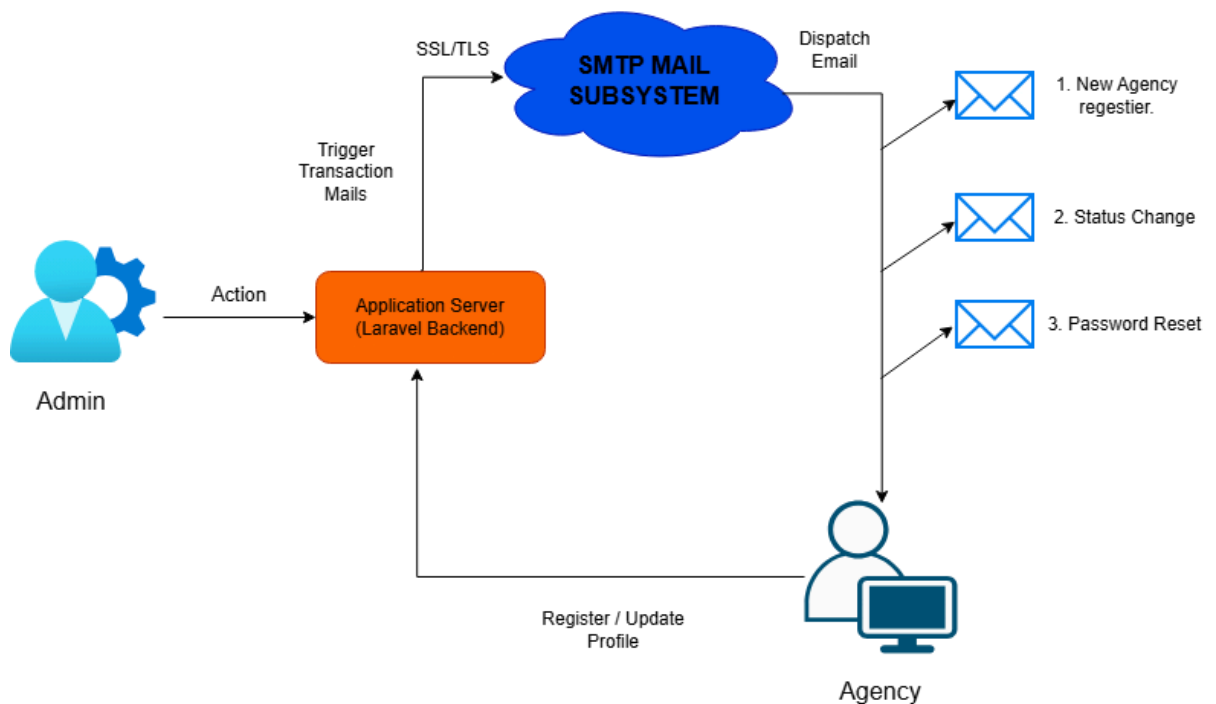


Figure 3.8 AirBridgeSD SMTP Mail Subsystem Integration and Workflow

To automate platform communications and ensure secure account onboarding, an **SMTP (Simple Mail Transfer Protocol)** server was integrated into the Laravel backend. The mail subsystem handles automated transactional emails through secure transport layers (TLS/SSL).

Key Functionalities Implemented:

- **Administrator Notifications:** Automatically dispatches email alerts to system administrators whenever a new travel agency registers and requires document review.
- **Agency Status Updates:** Triggers instant notification emails to travel agencies once their account status shifts from "Pending" to "Approved" by the admin.
- **Security and Verification:** Facilitates secure password resets and account confirmation workflows, protecting platform access.

3.3.2 Detailed Design:

This section consists of:

- **Activity Diagram:** The Activity Diagram serves as a visual roadmap of the user's journey within the AirBridgeSD platform. It tracks the logical sequence of actions from the moment a Customer enters their search criteria until a WhatsApp booking request is successfully generated. This diagram is essential for identifying the "Decision Points" such as verifying if flights match the search or checking an agency's approval status ensuring a smooth and error-free flow between the frontend interface and the backend processing logic.
- **Sequence Diagram:** The Sequence Diagram models the dynamic interaction between system objects over time. It specifically details how messages are exchanged between the Frontend (React), the Backend (Laravel), and external services like the Duffel API. This diagram is crucial for visualizing the "Search" and "Commission Calculation" processes, showing exactly how

a user's request triggers a series of calls to the database and external flight providers to return a final result.

- **Class Diagram:** The Class Diagram represents the static structure of the system. It defines the system's entities (such as Customer, Agency, Admin, Flight, and Commission) along with their attributes and methods. Furthermore, it outlines the relationships and cardinalities between these classes (e.g., an Agency can set multiple Commission Rules), providing a foundation for the database schema and the object-oriented implementation in the Backend.

3.3.2.1 Activity Diagram:

Customer Activity Diagram

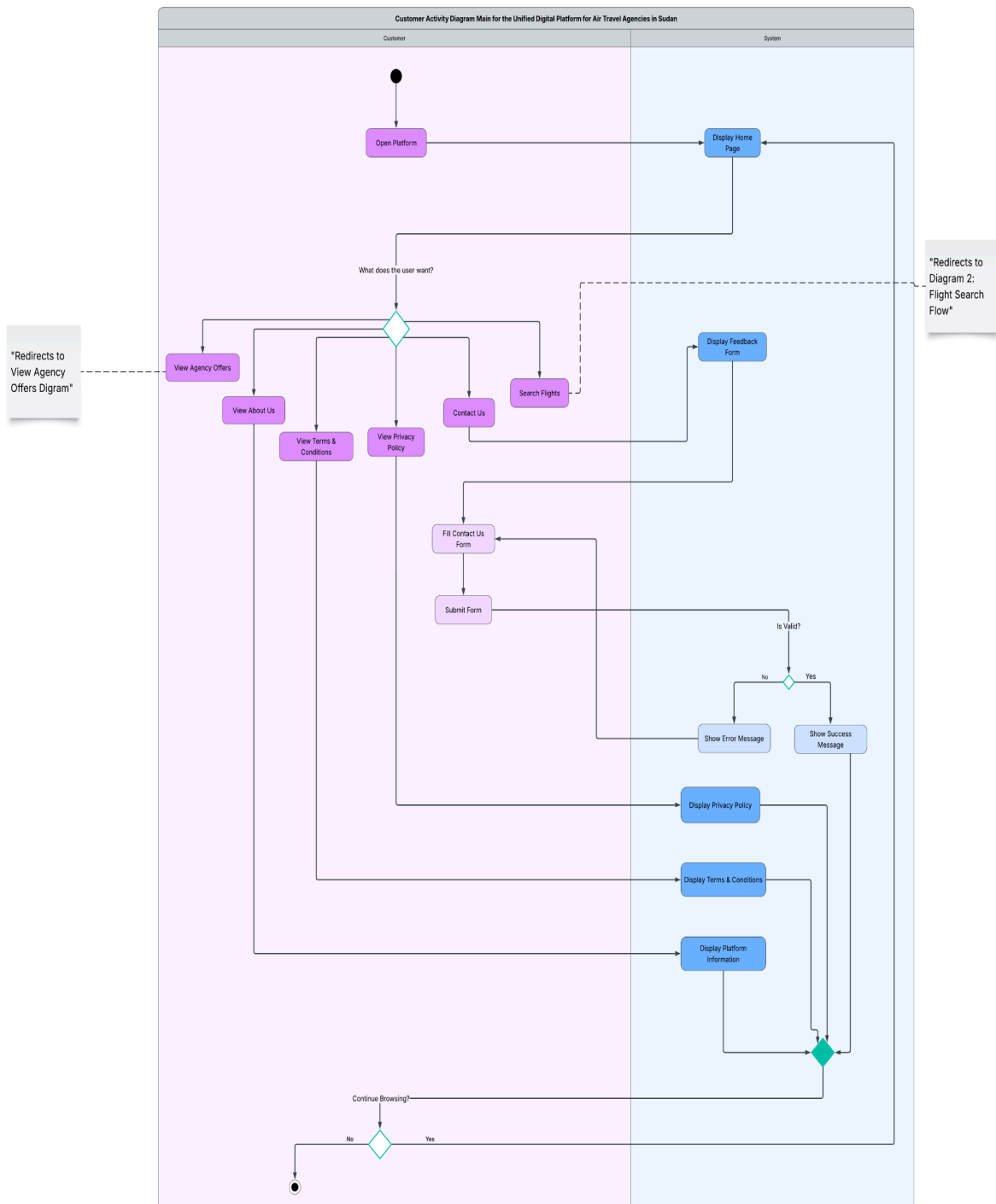


Figure 3.9 Customer Activity Diagram (Main Page)

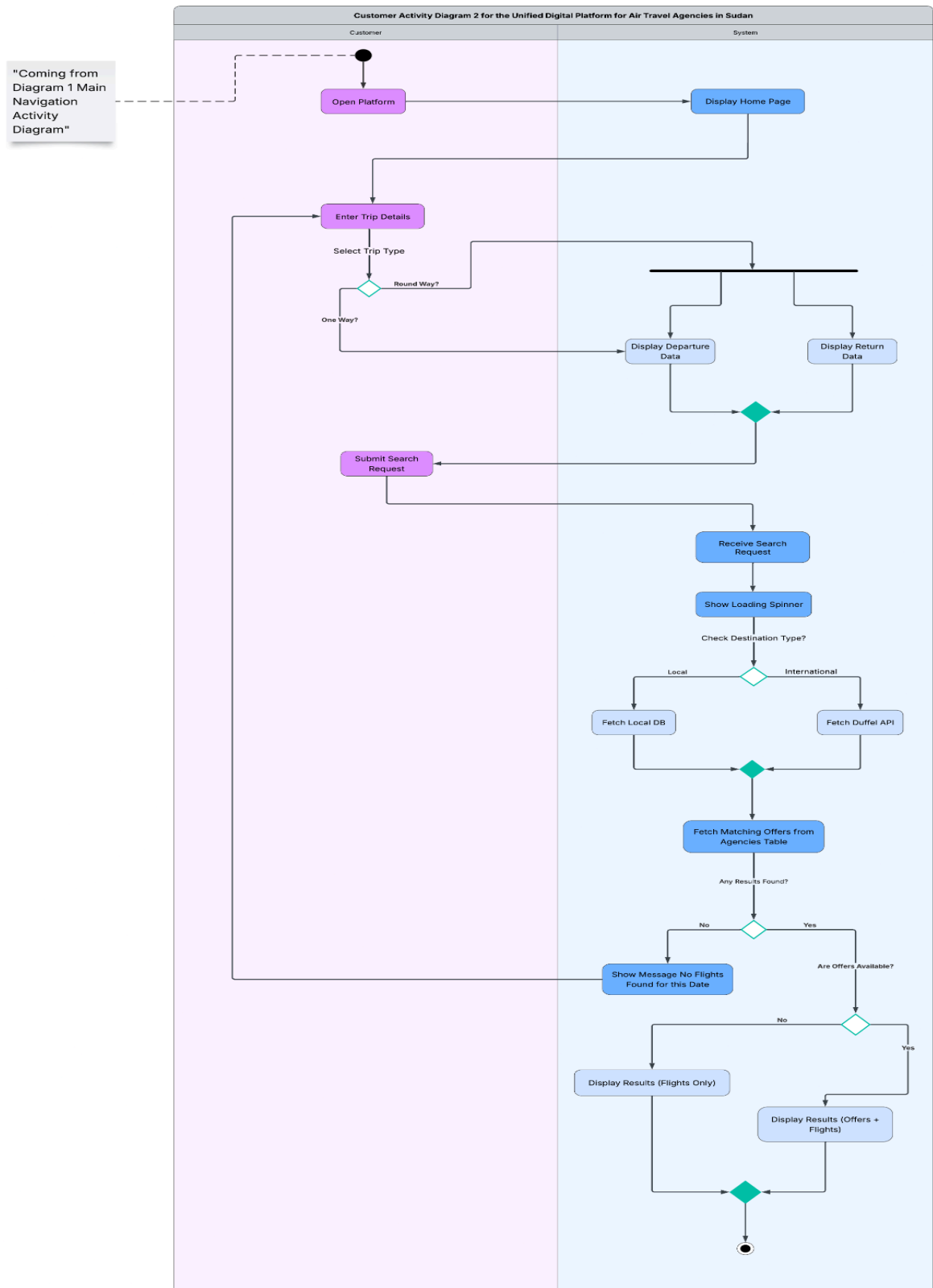


Figure 3.10 Customer Activity Diagram (Search Flight)

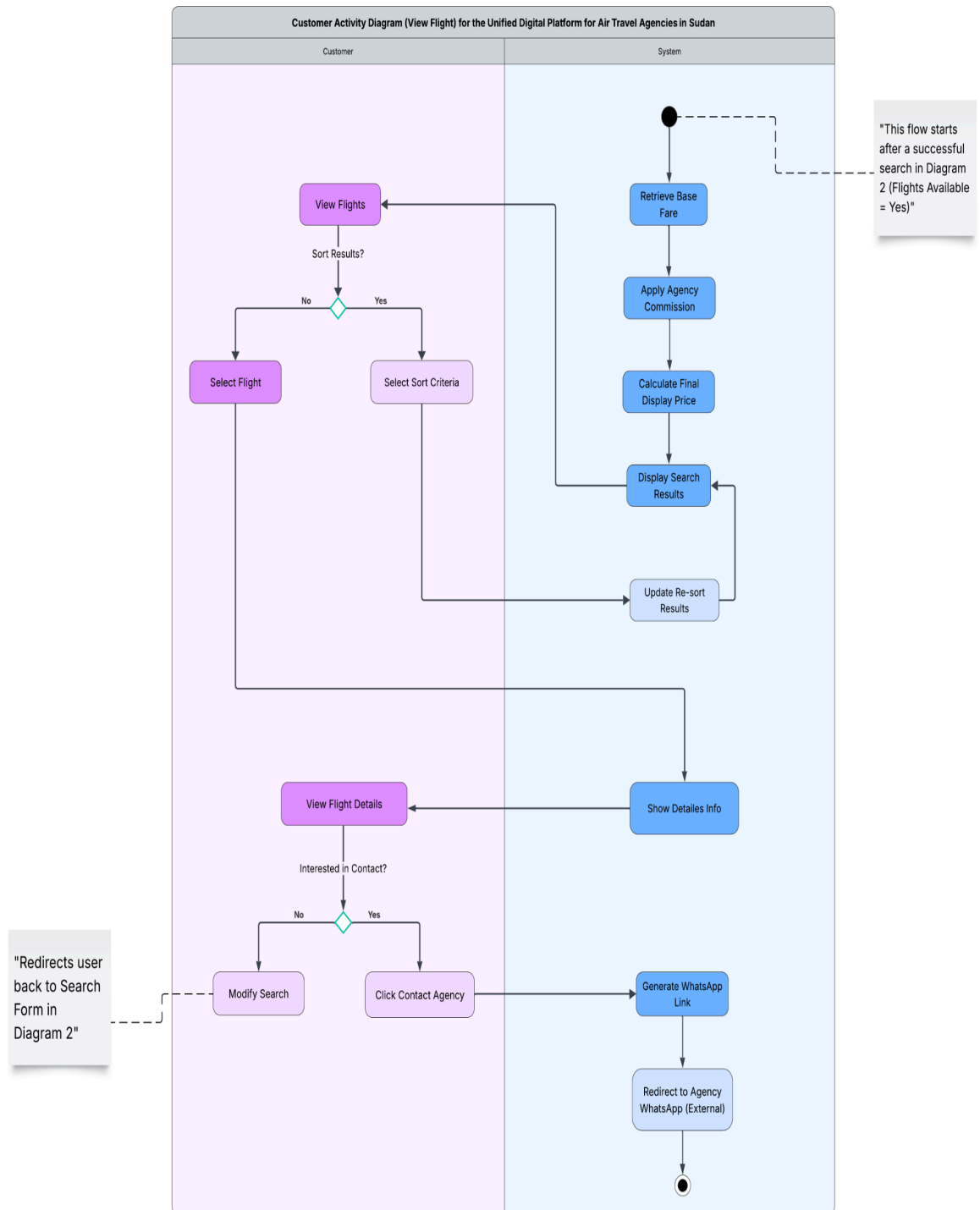


Figure 3.11 Customer Activity Diagram (View Flights)

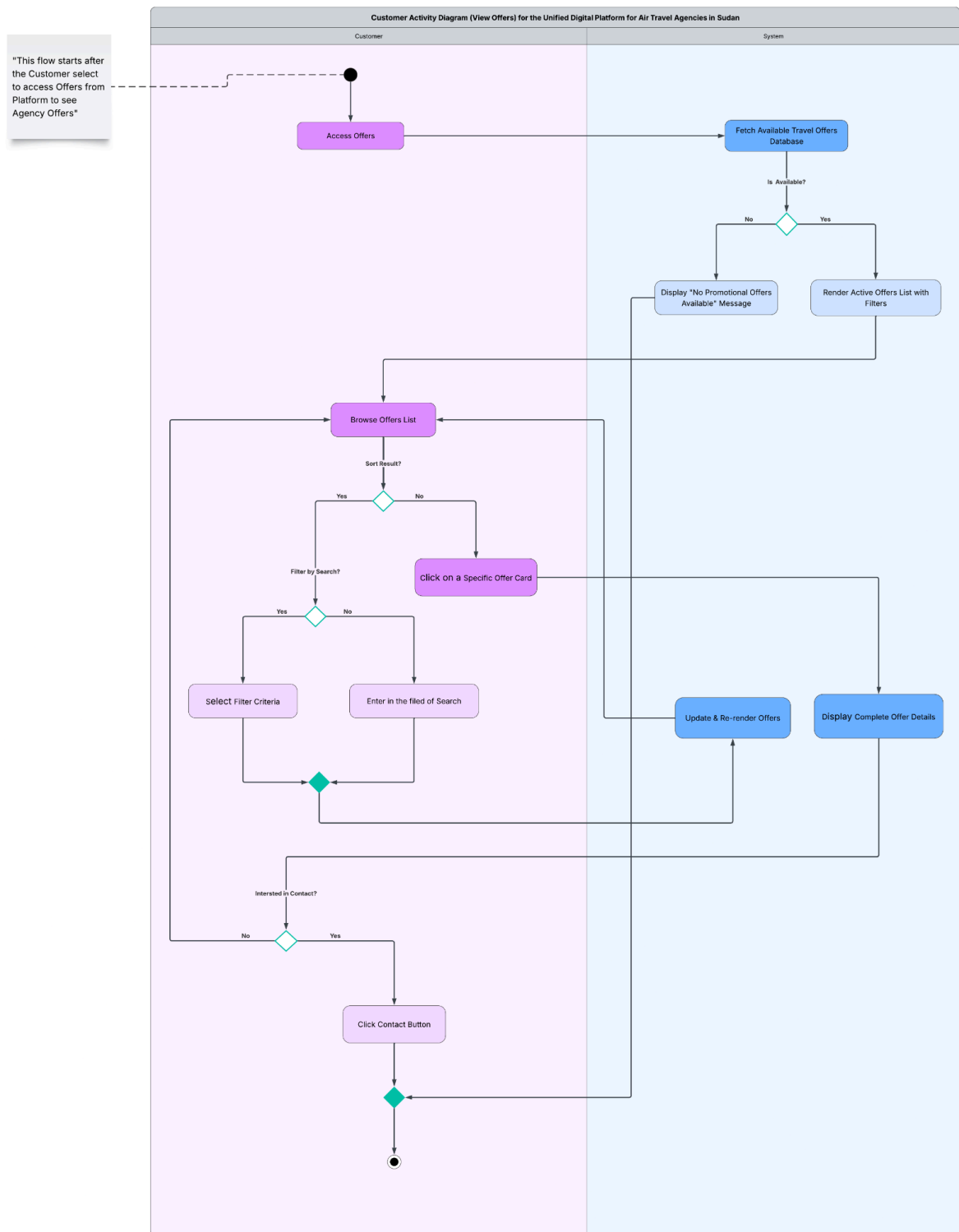


Figure 3.12 Customer Activity Diagram (View Offers Page)

Agency Activity Diagram

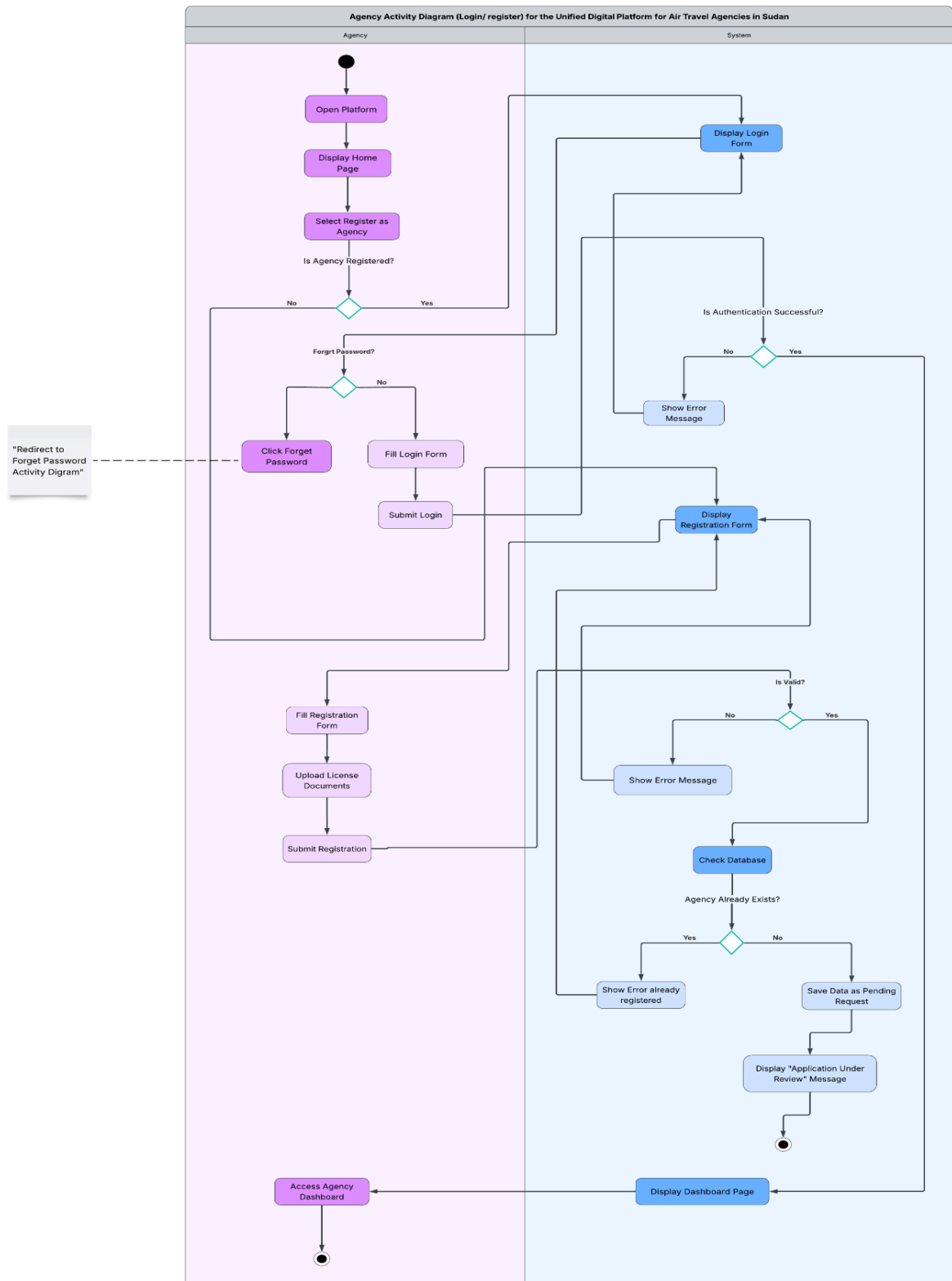


Figure 3.13 Agency Activity Diagram (Login/Register) in System

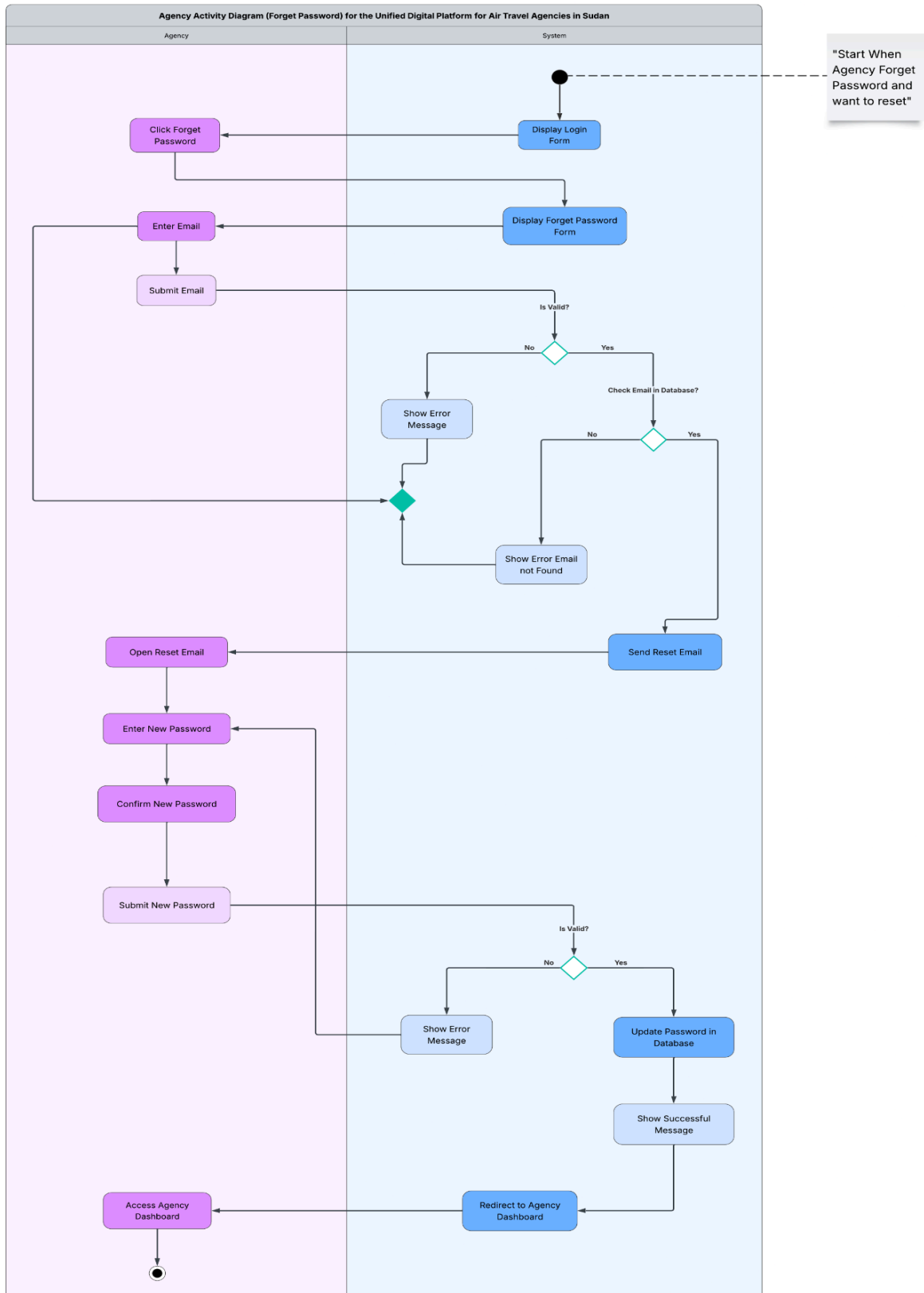


Figure 3.14 Agency Activity Diagram (forgot Password) in System

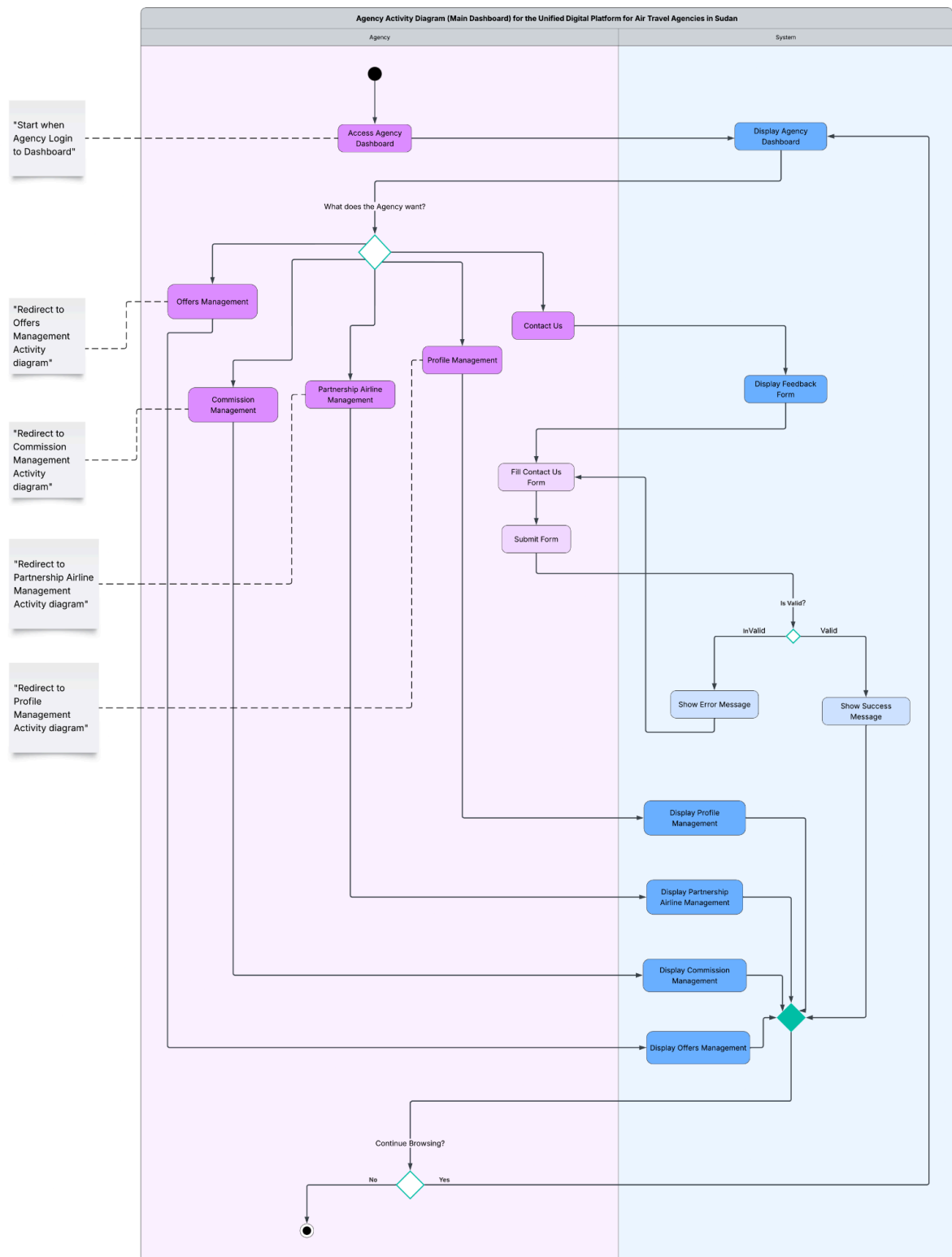


Figure 3.15 Agency Main Dashboard Activity Diagram in System

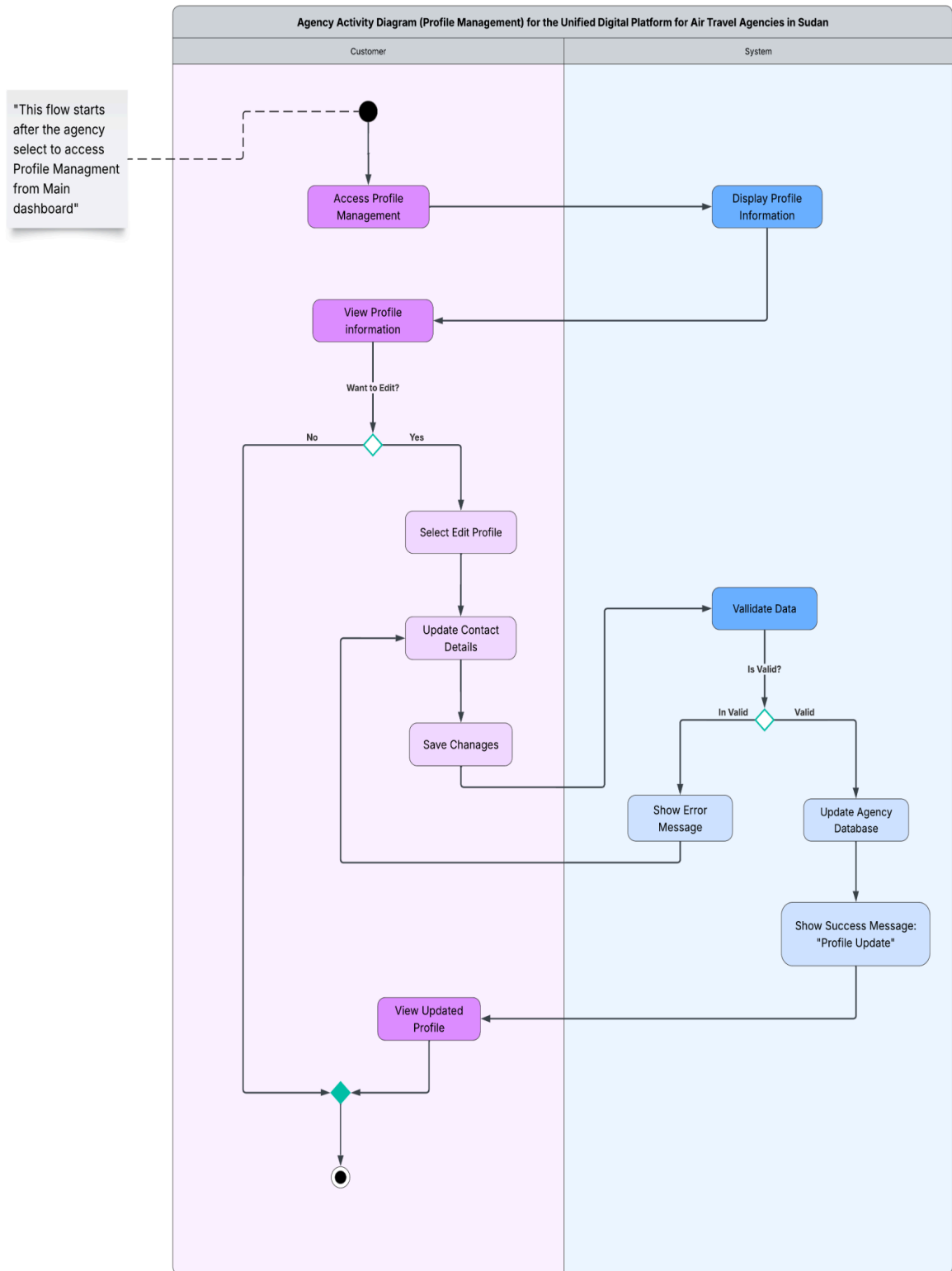


Figure 3.16 Agency Activity Diagram (Profile Management) in System

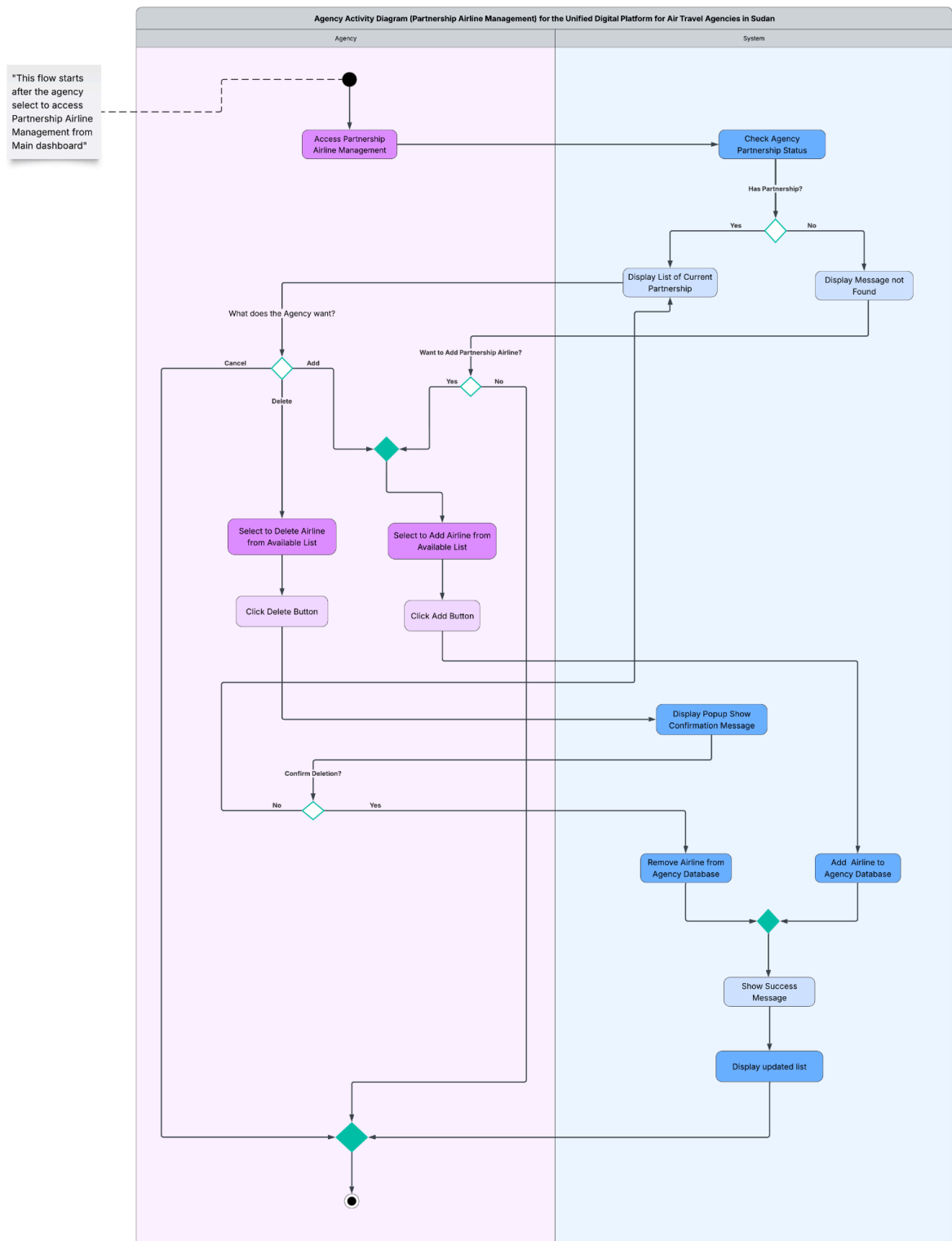


Figure 3.17 Agency Activity Diagram (Partnership Airline Management) in System

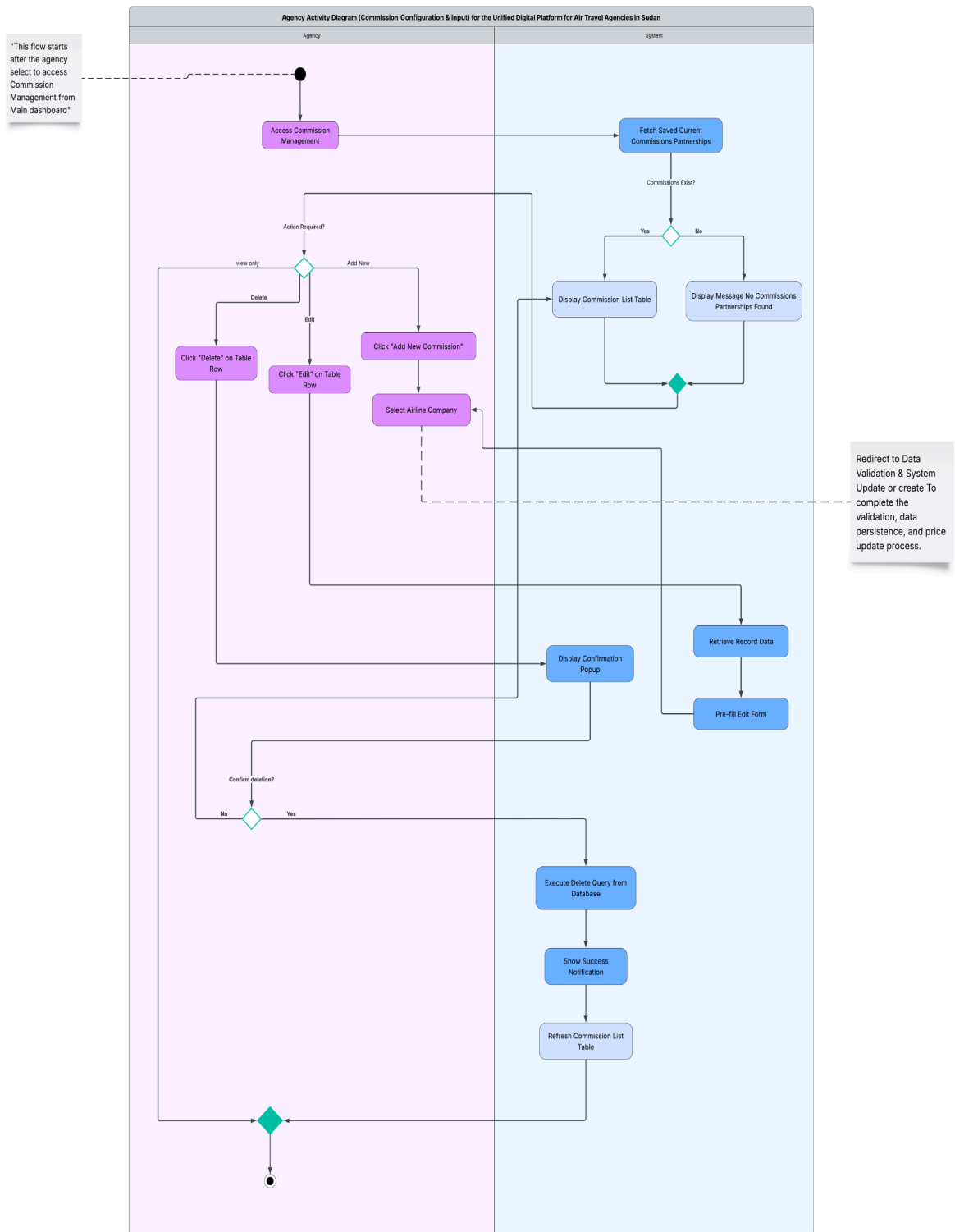


Figure 3.18 Agency Activity Diagram (Commission Configuration and Input) in System

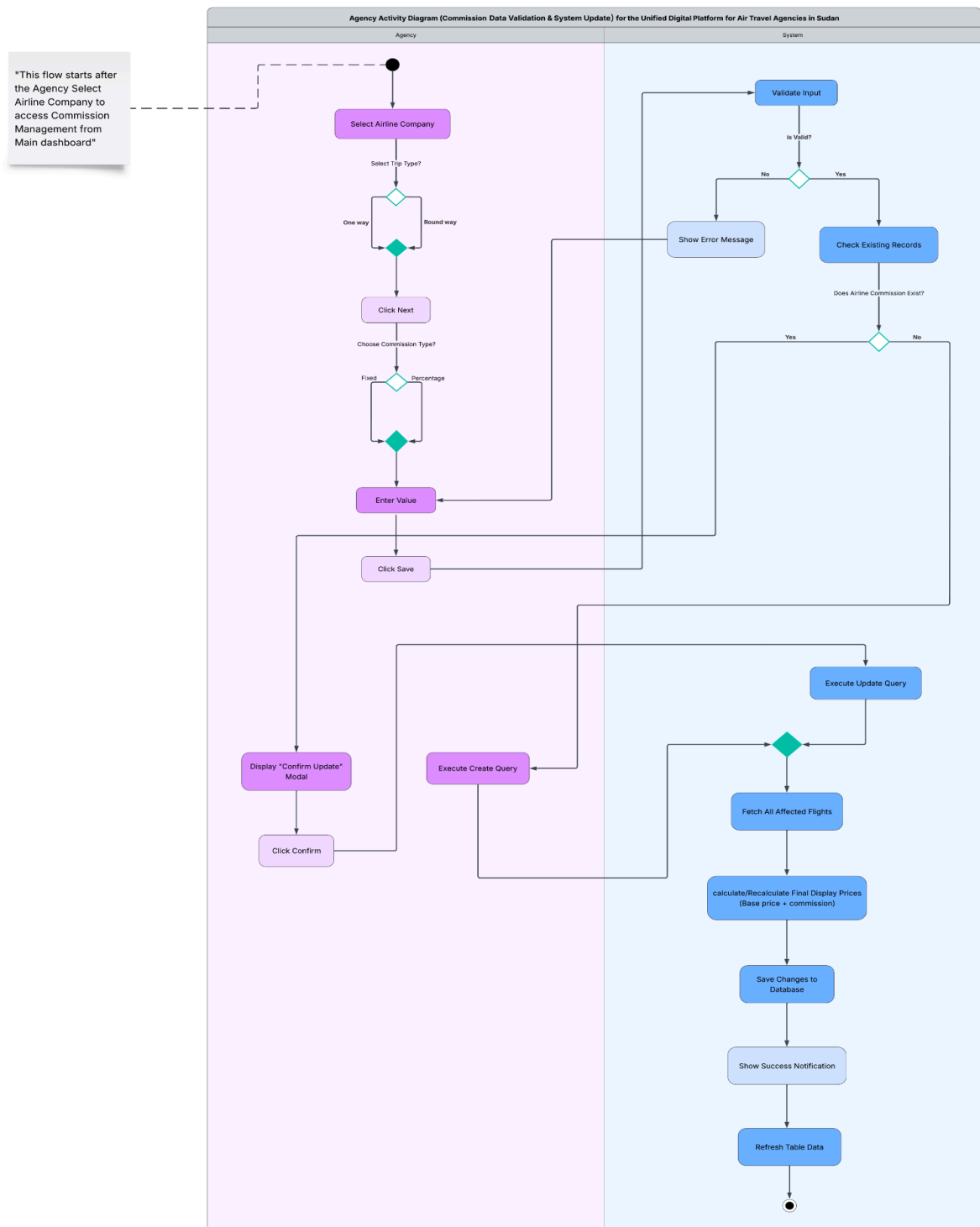


Figure 3.19 Agency Activity Diagram (Commission Data Validation and Update) in System

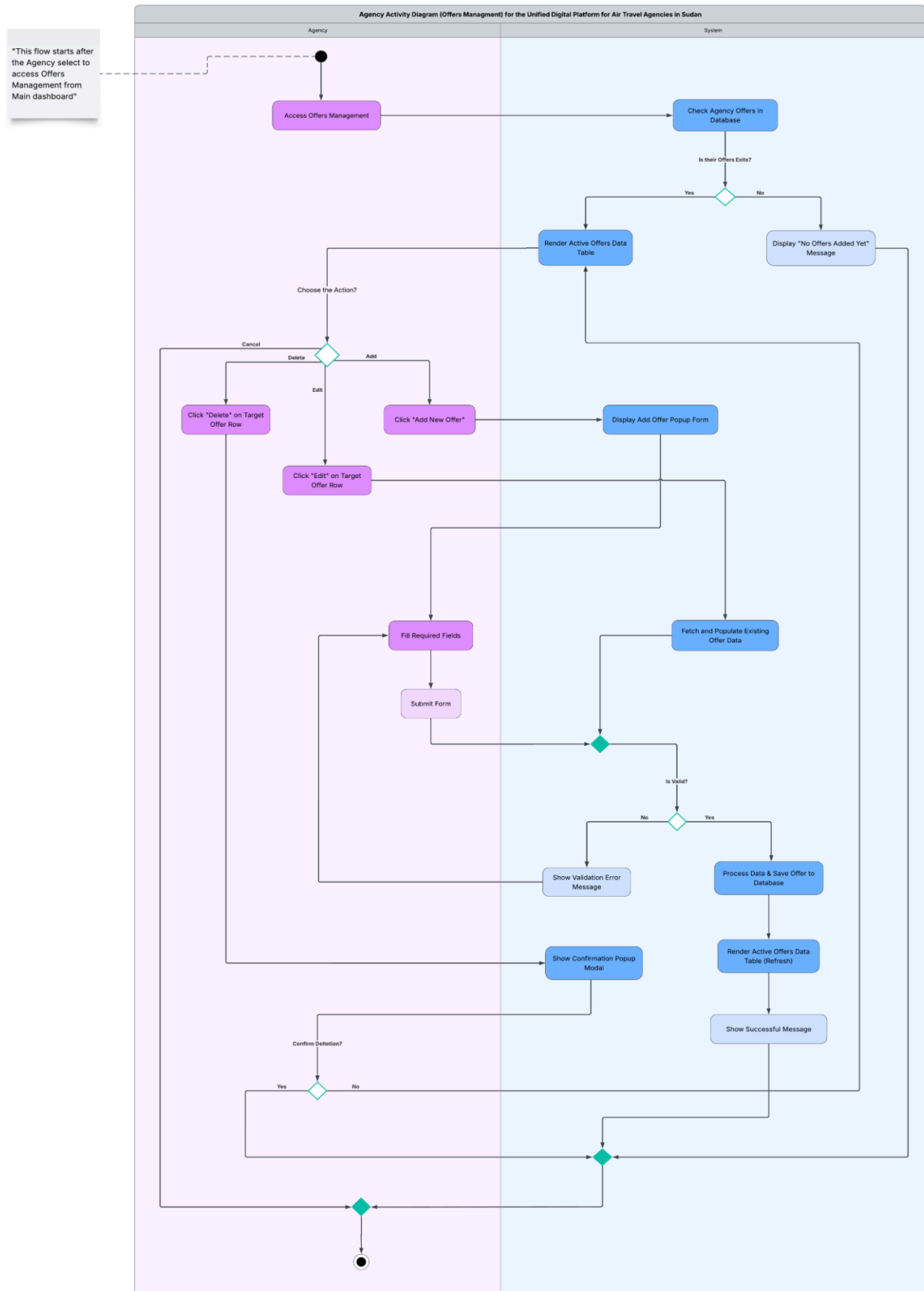


Figure 3.20 Agency Activity Diagram (Offers Management) in System

Admin Activity Diagram

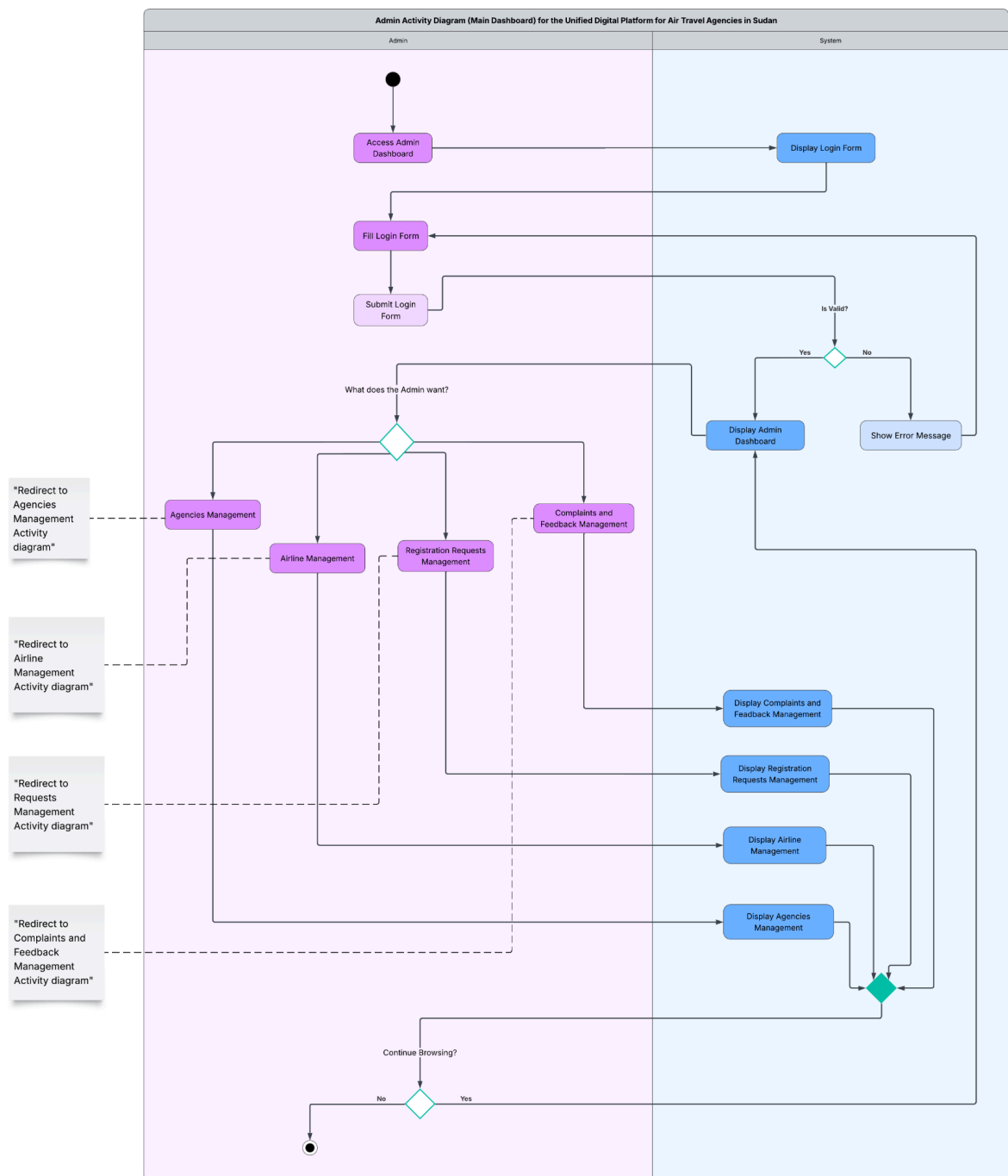


Figure 3.21 Admin Main Dashboard Activity Diagram

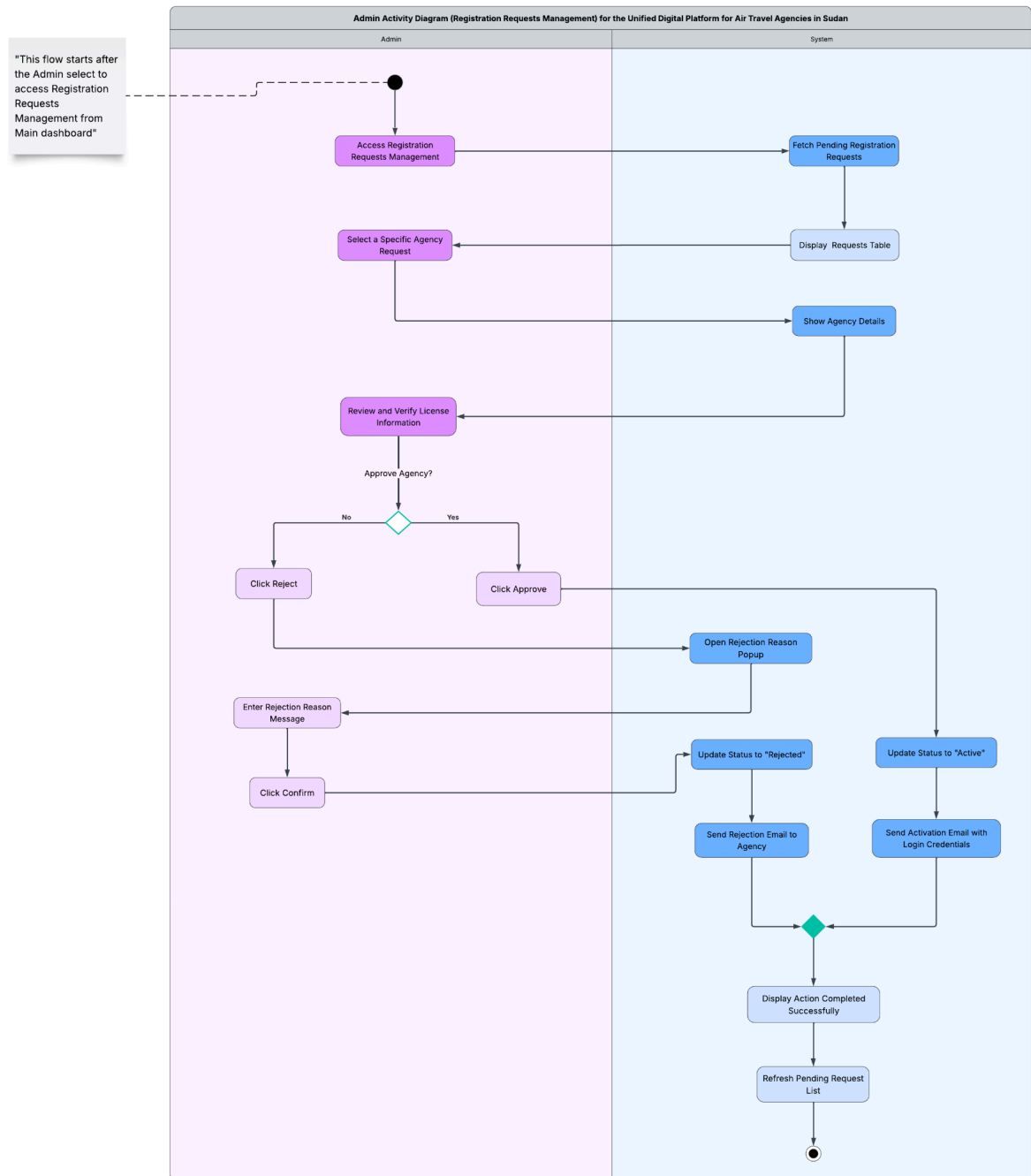


Figure 3.22 Admin Activity Diagram (Registration Requests Management)

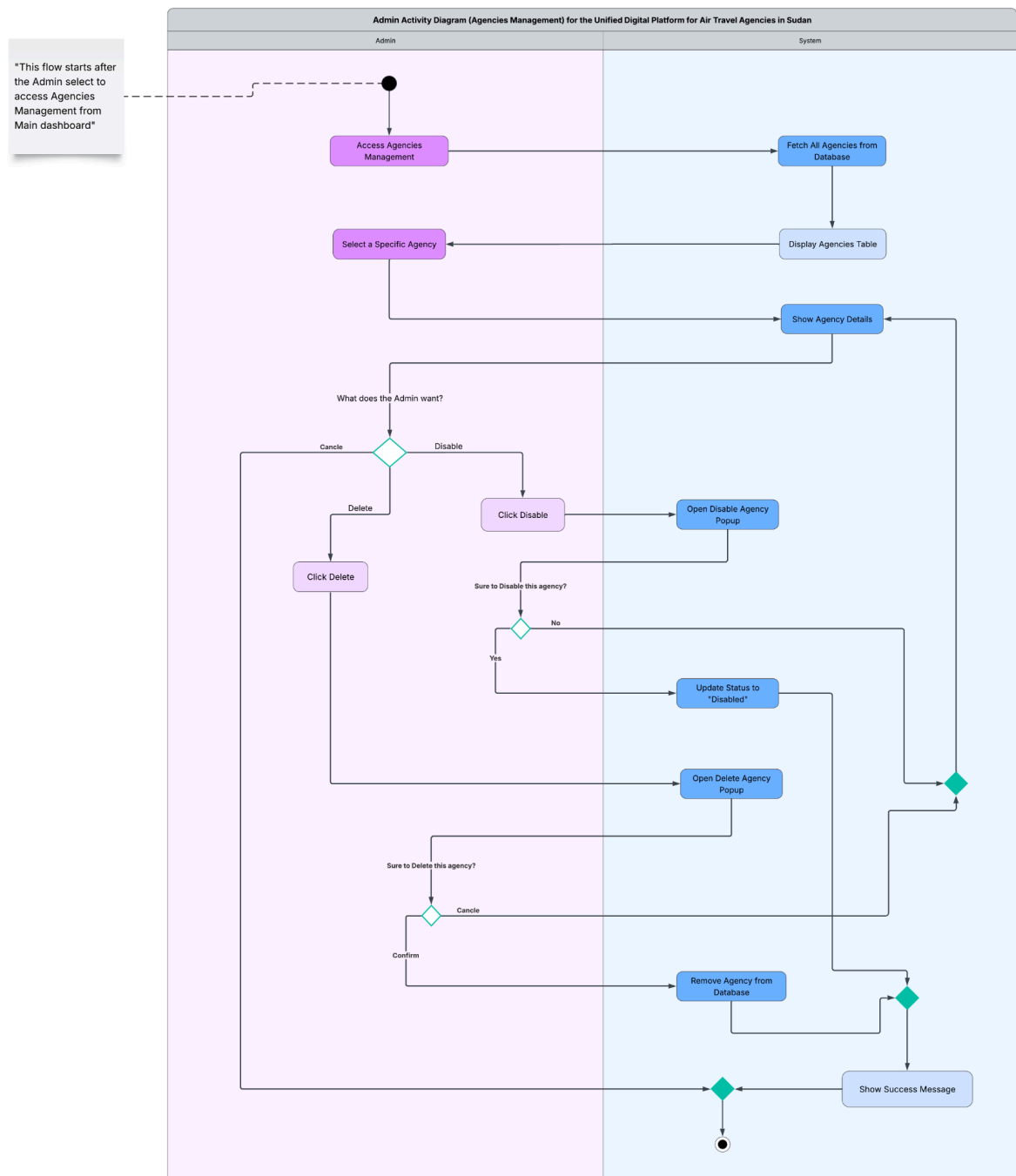


Figure 3.23 Admin Activity Diagram (Agencies Management)

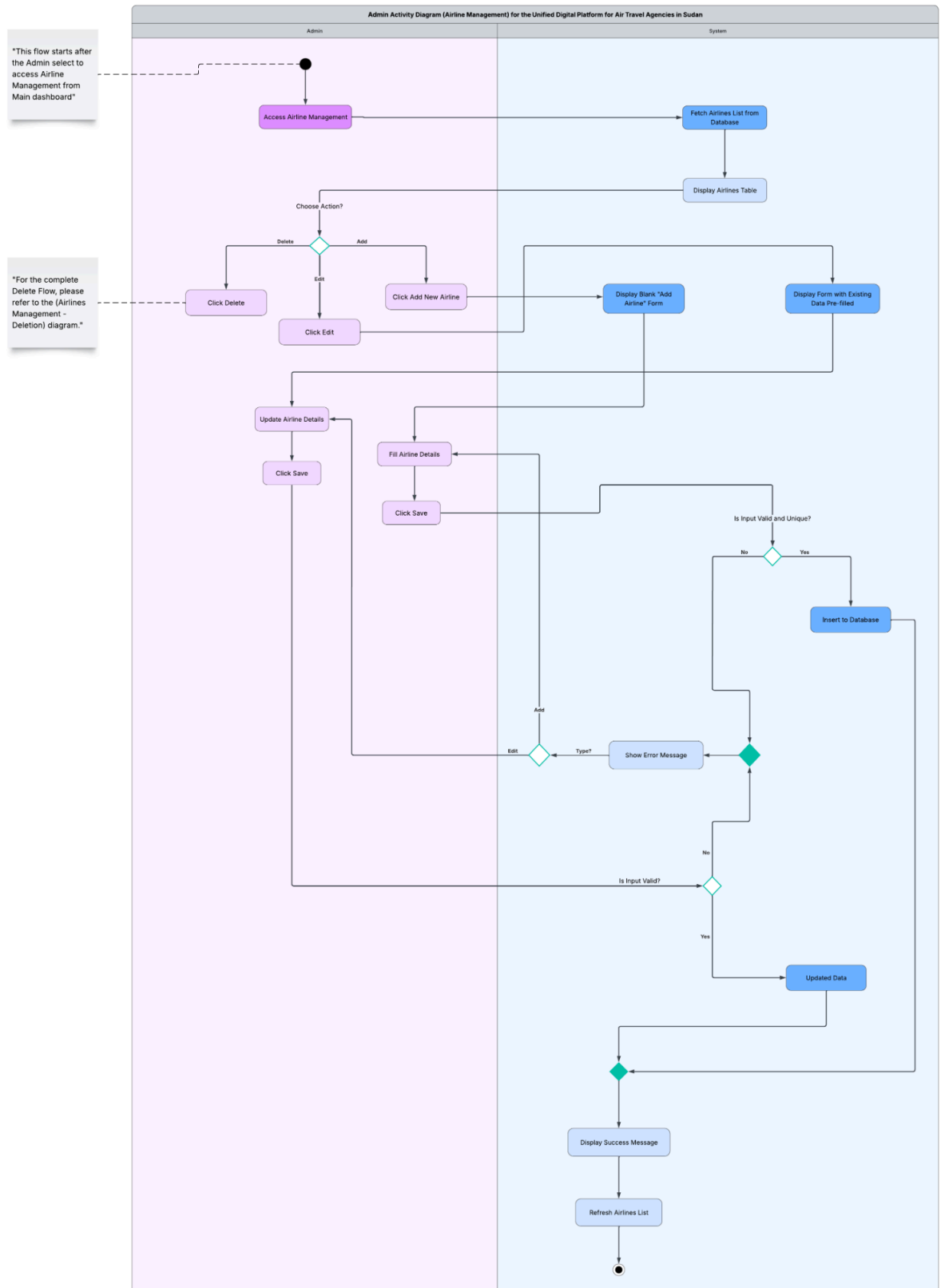


Figure 3.24 Admin Activity Diagram (Airlines Management)

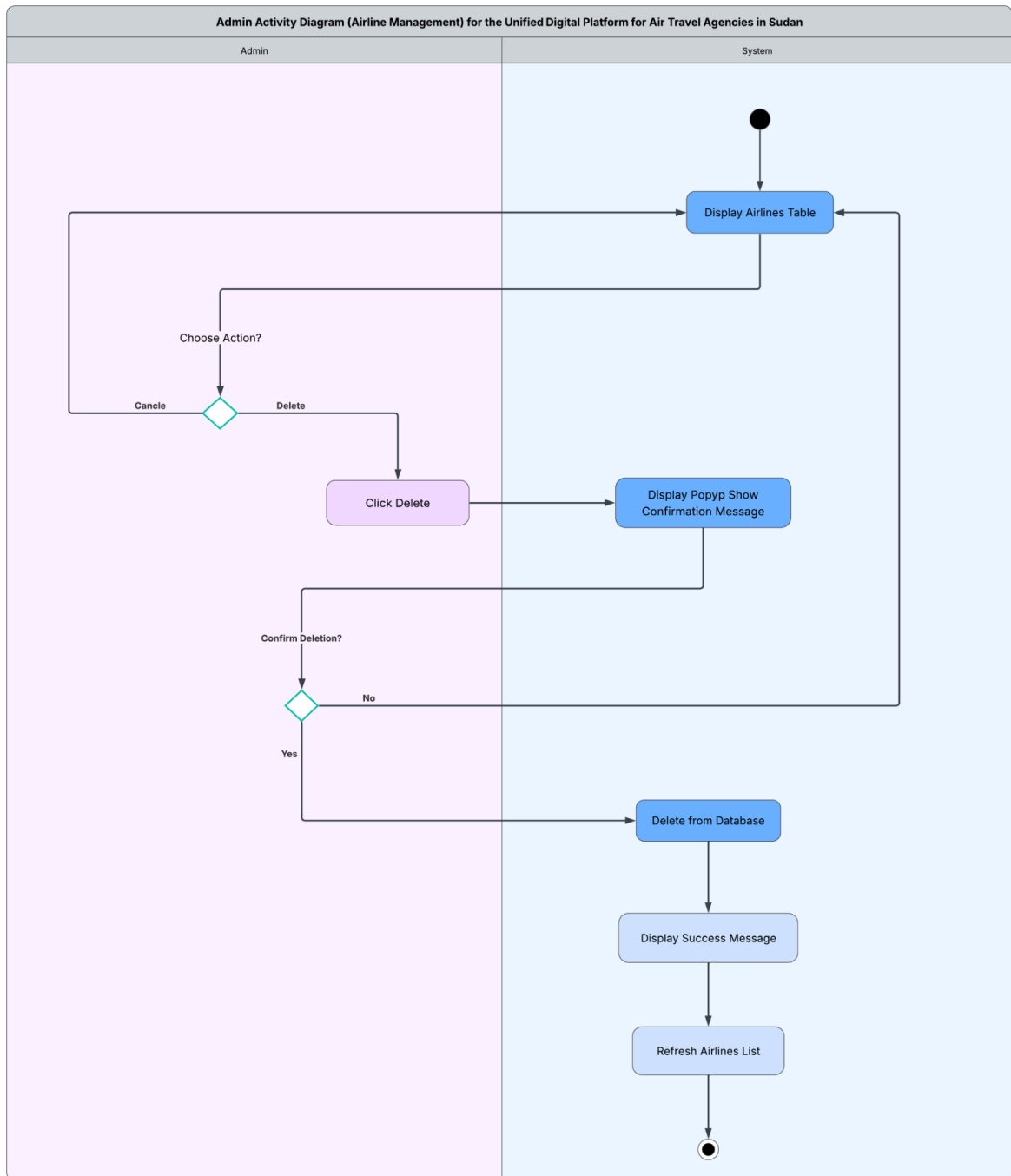


Figure 3.25 Admin Activity Diagram (Airlines Management - Deletion)

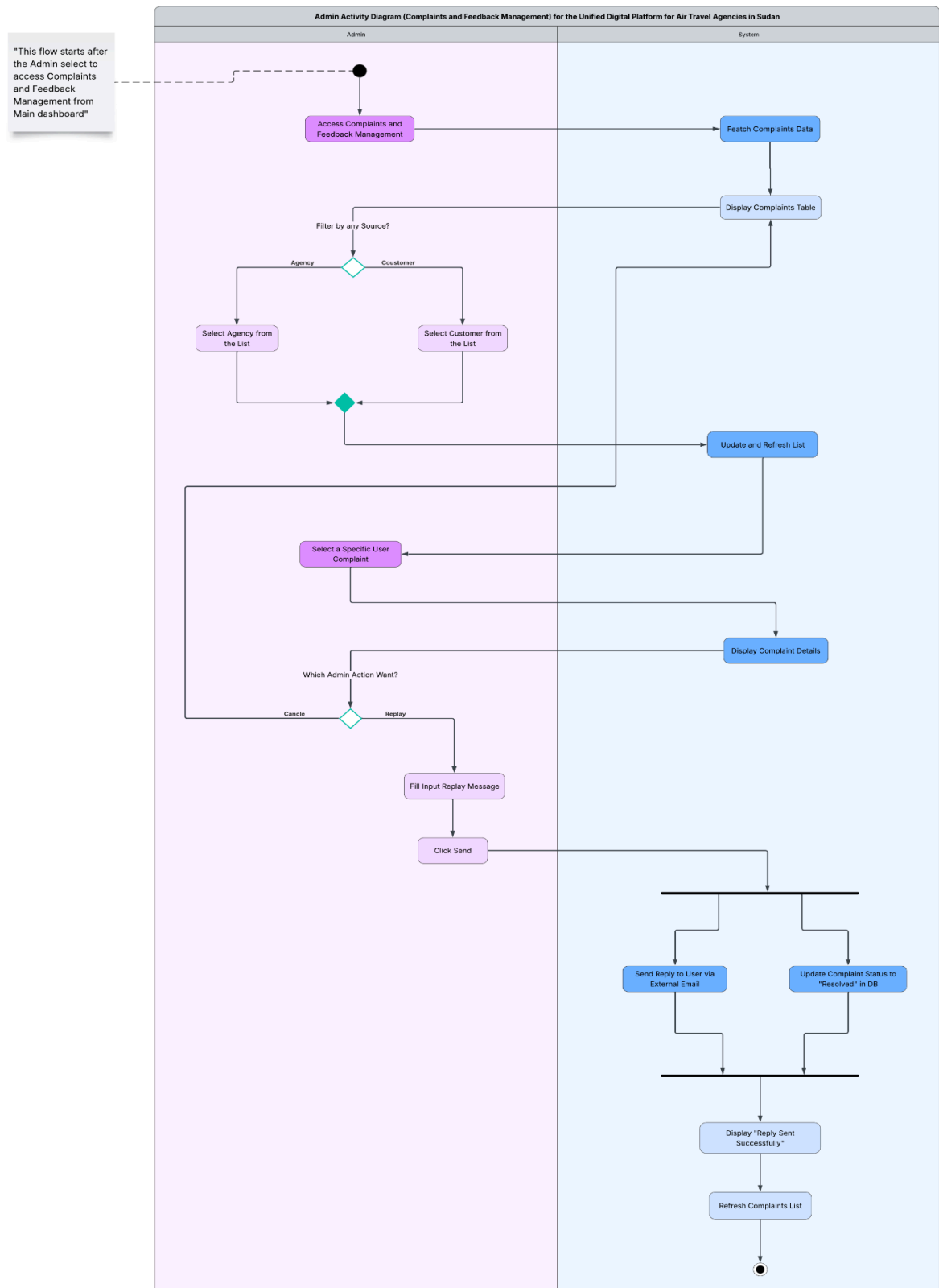


Figure 3.26 Admin Activity Diagram (Complaints and Feedback Management)

3.3.2.2 Sequence Diagram:

Authentication Sequence Diagrams:

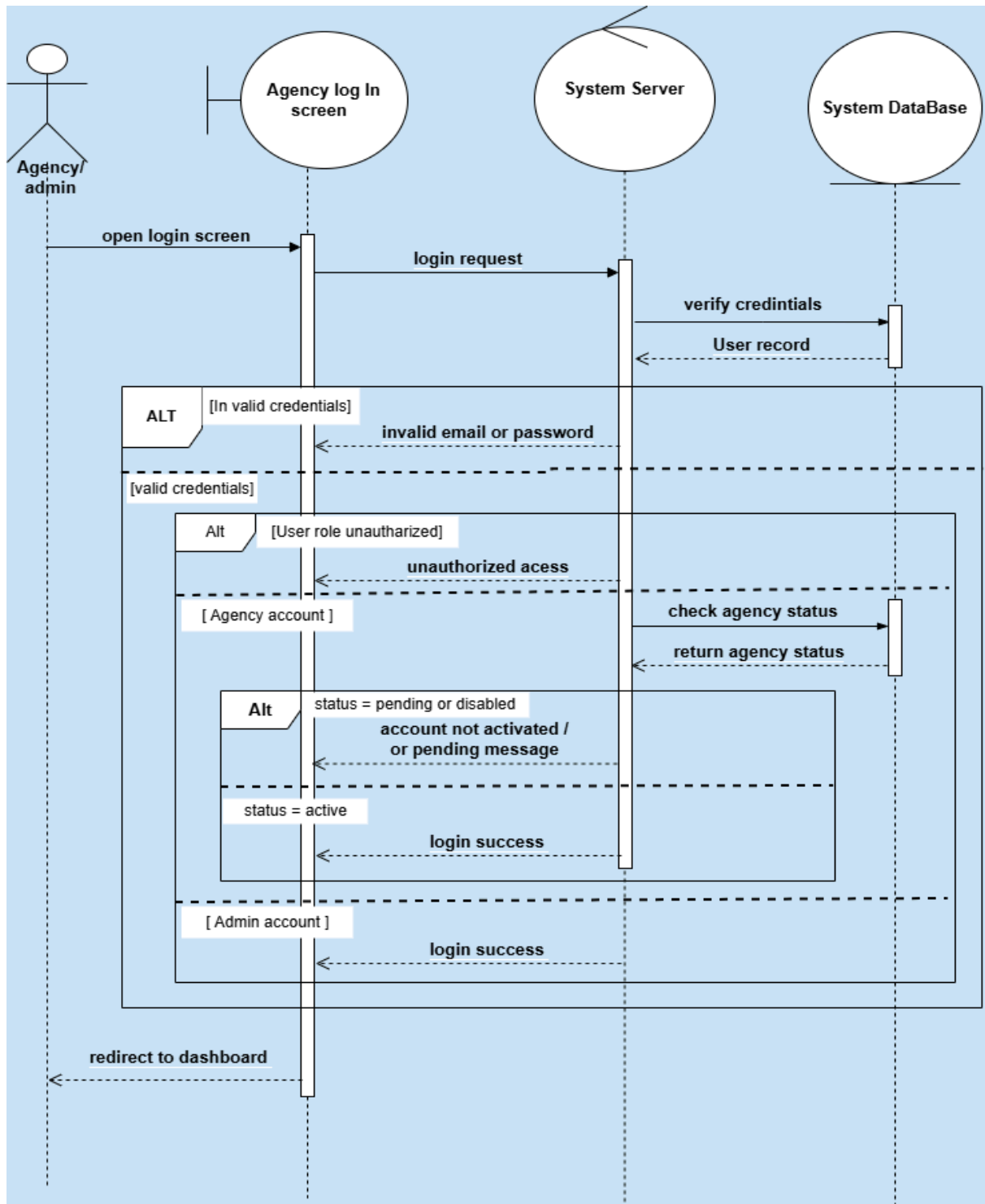


Figure 3.27 Admin and Agency LogIn Sequence Diagram

Admin and Agency LogOut Sequence Diagram:

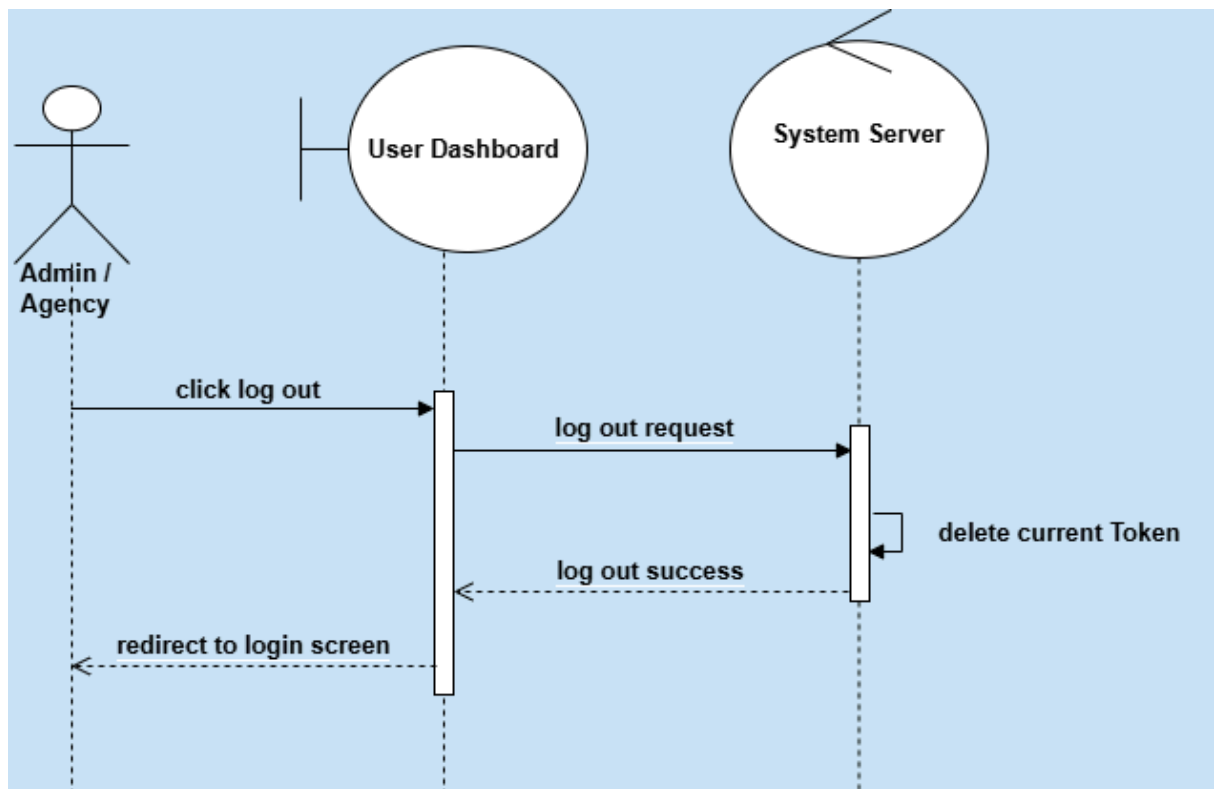


Figure 3.28 Admin and Agency LogOut Sequence Diagram

Admin and Agency Reset Password Sequence Diagram:

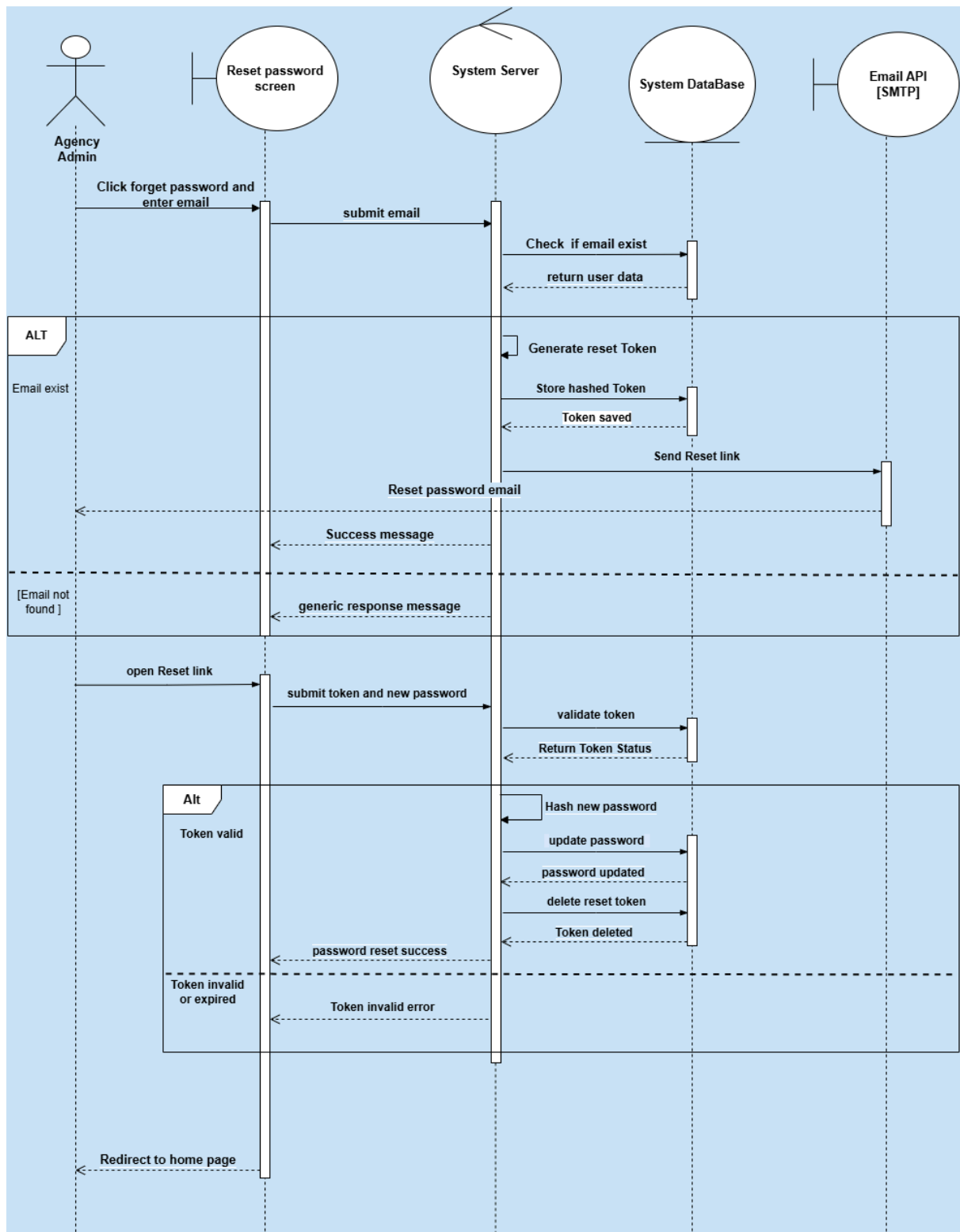


Figure 3.29 Admin and Agency Reset Password Sequence Diagram

Customer Search Flights Sequence Diagram:

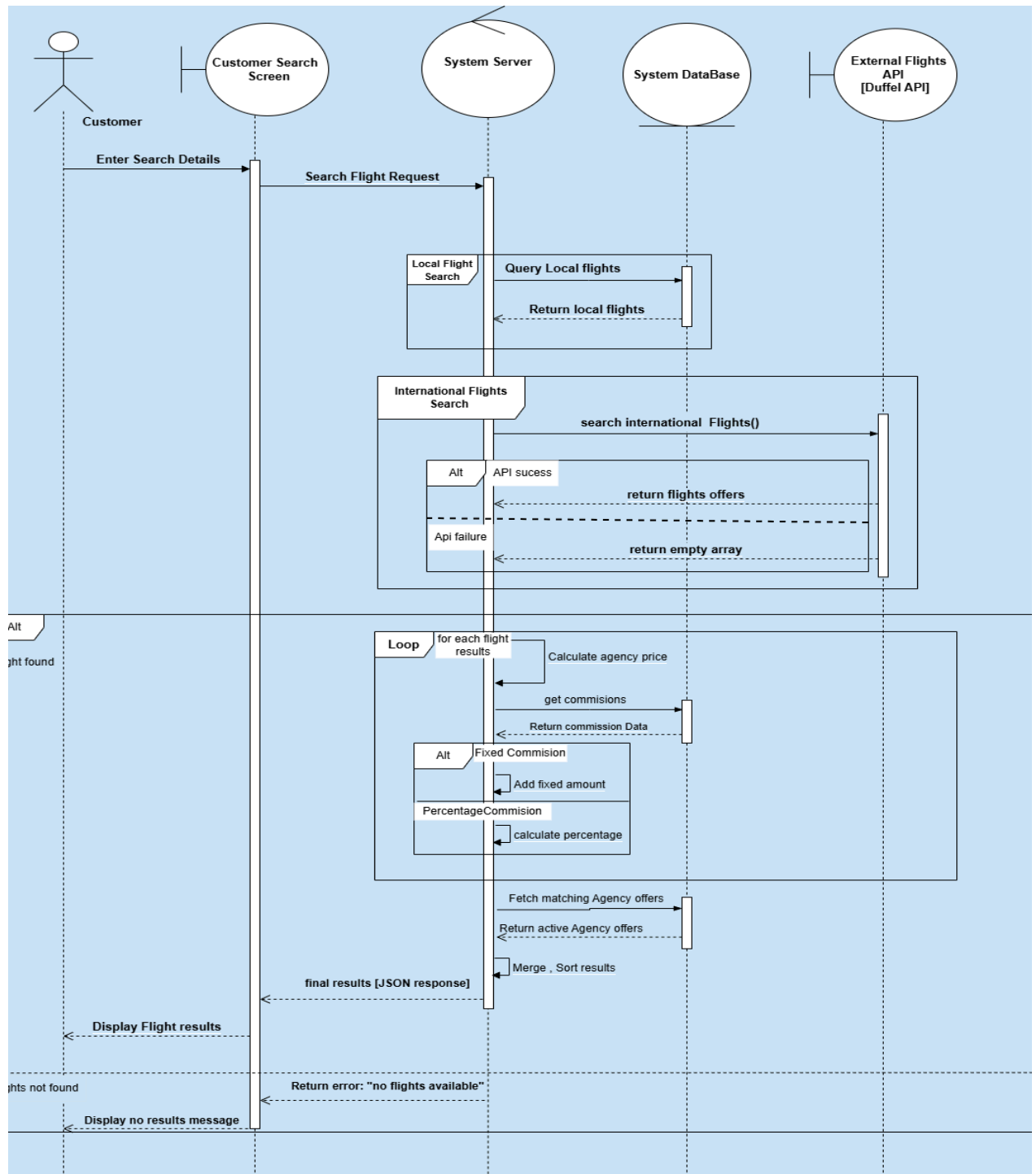


Figure 3.30 Customer Search for Flights Sequence Diagram

Customer Search Special Offer Sequence Diagram:

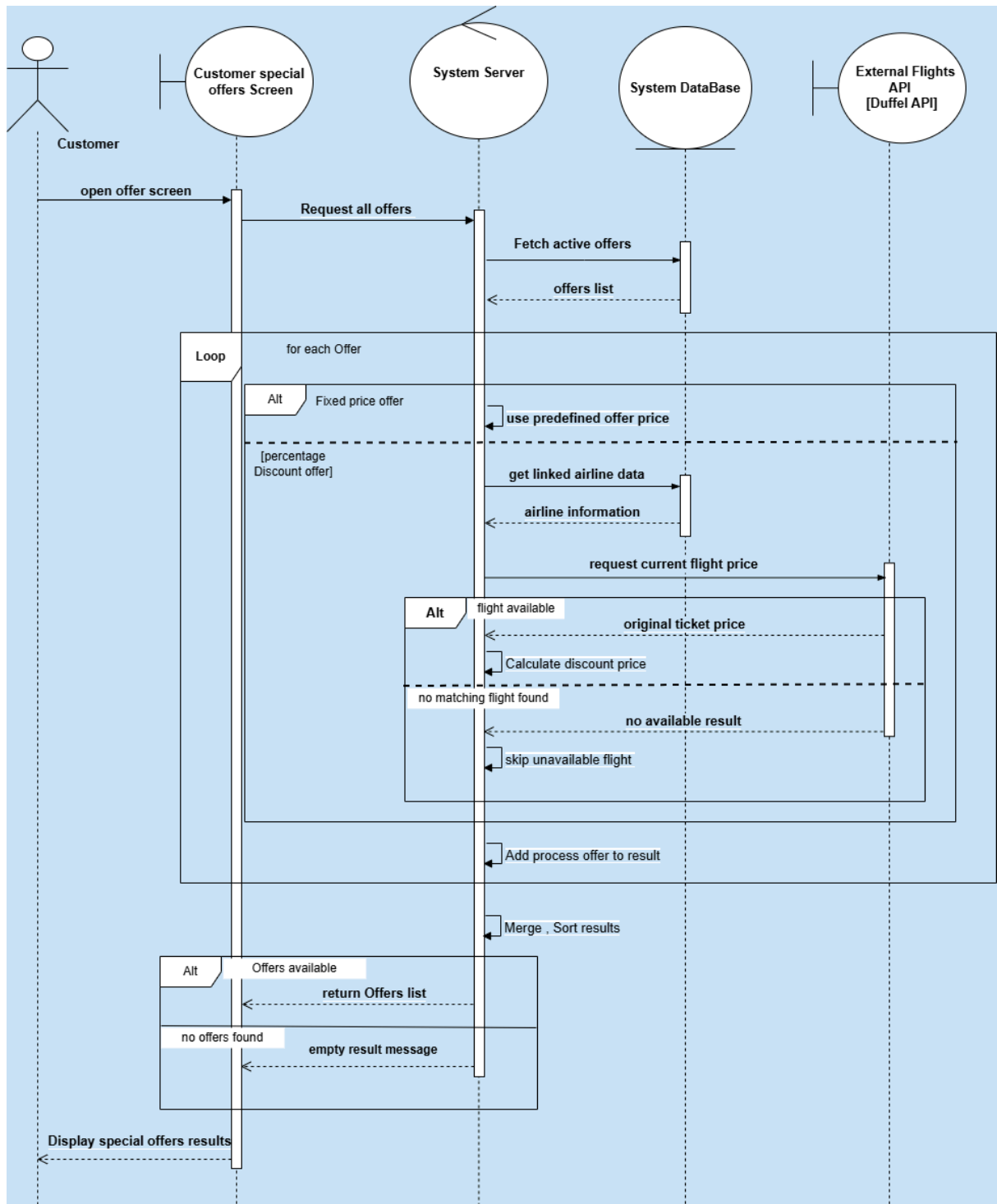


Figure 3.31 Customer Search for Special Offers Sequence Diagram

Customer booking request for Flight Sequence Diagram :

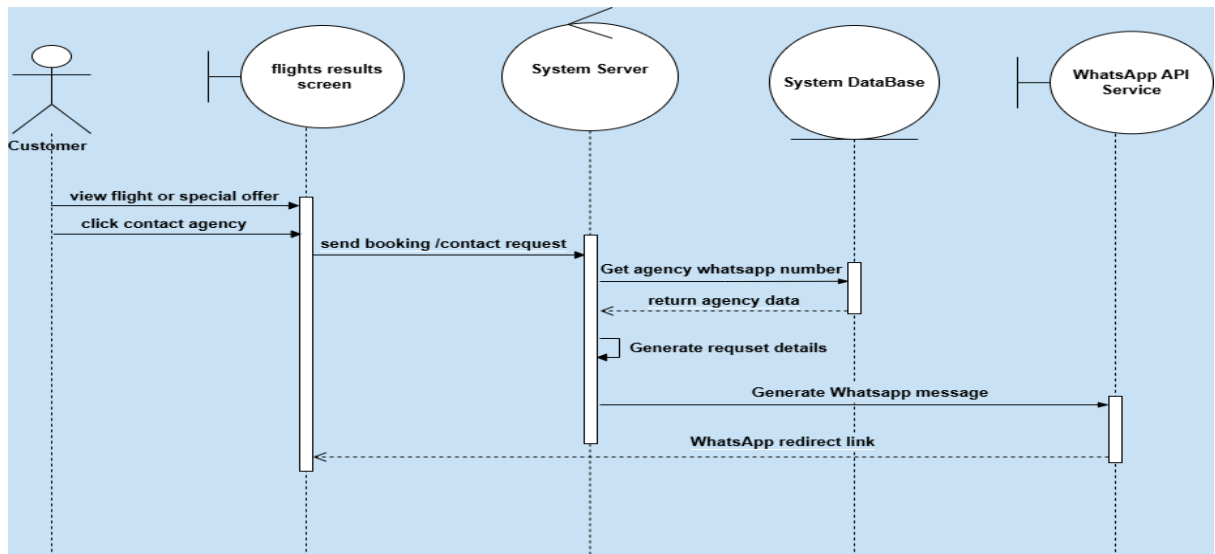


Figure 3.32 Customer booking request for Flight Sequence Diagram

Agency Registration Sequence Diagram:

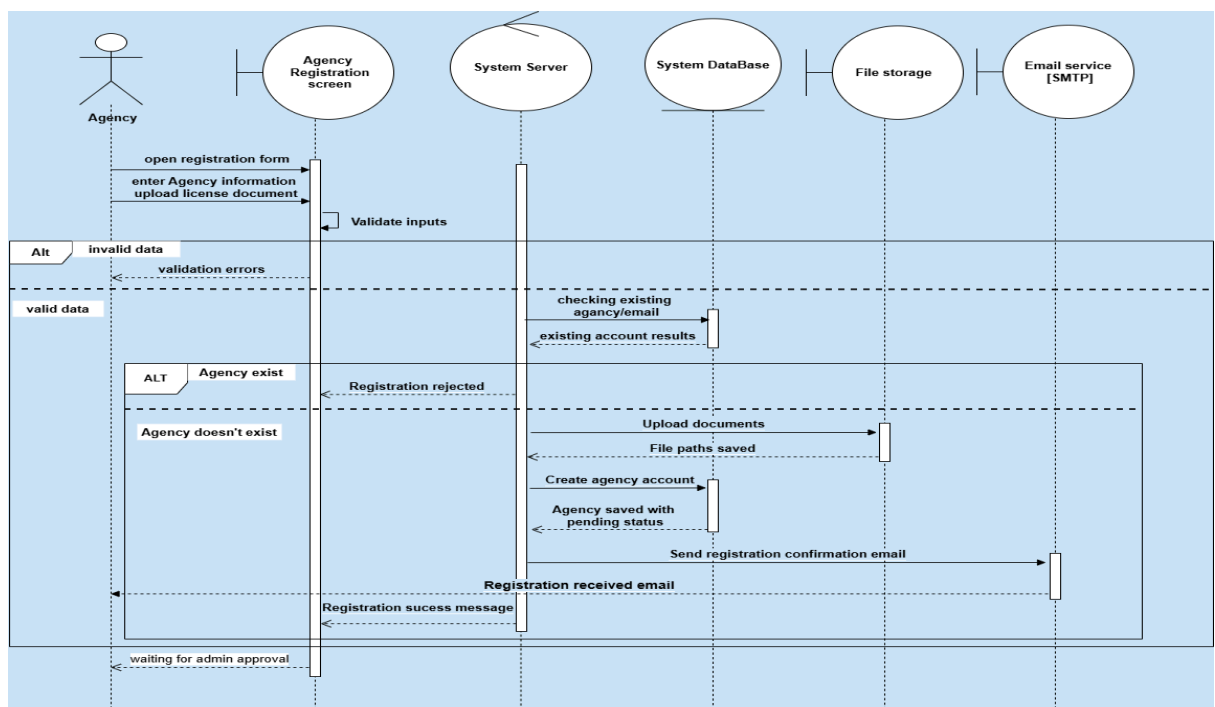


Figure 3.33 Agency Registration Sequence Diagram

Agency Manage partnered Airlines Sequence Diagram:

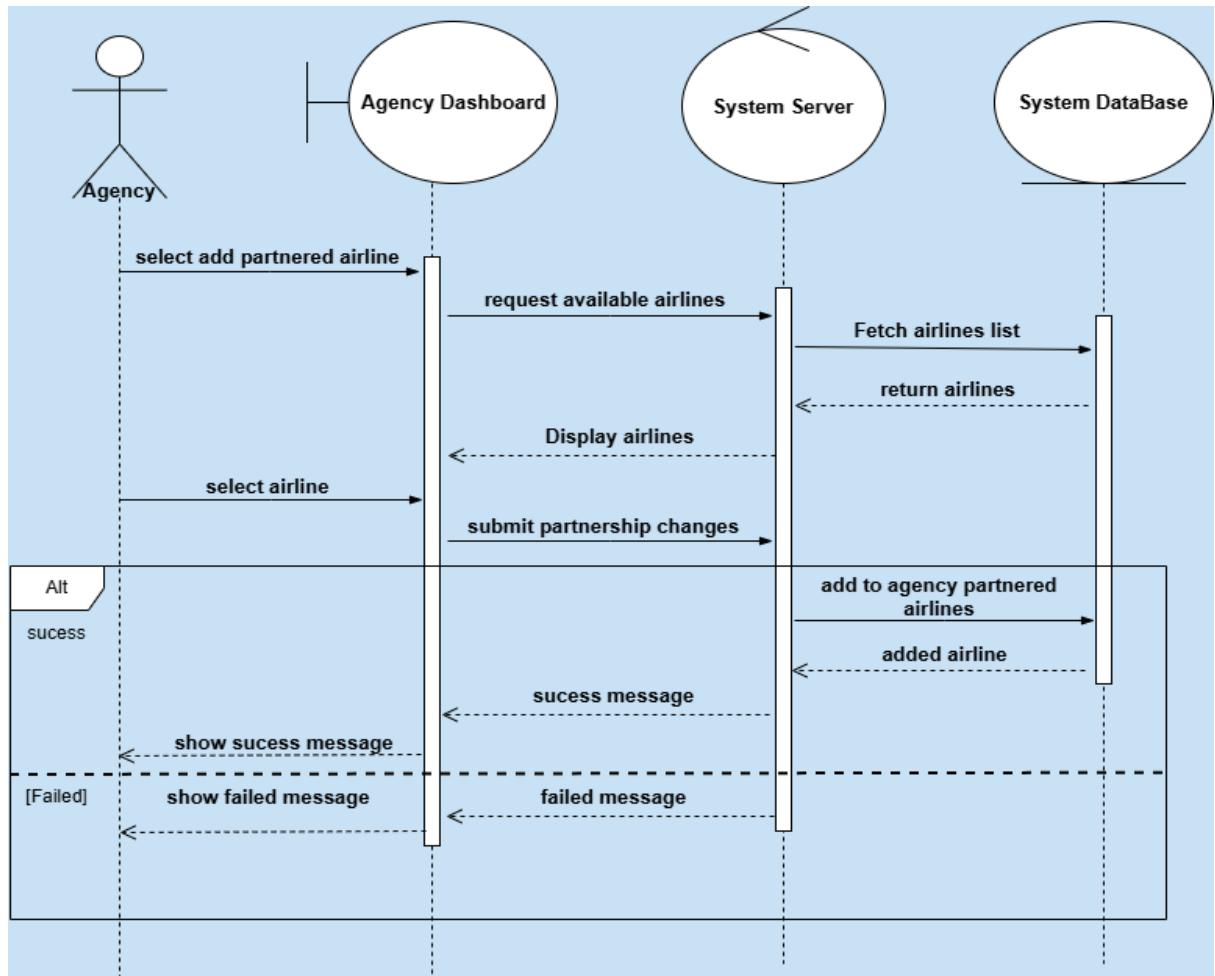


Figure 3.34 Agency manage partnered airlines Sequence Diagram

Agency Remove partnered Airline Sequence Diagram:

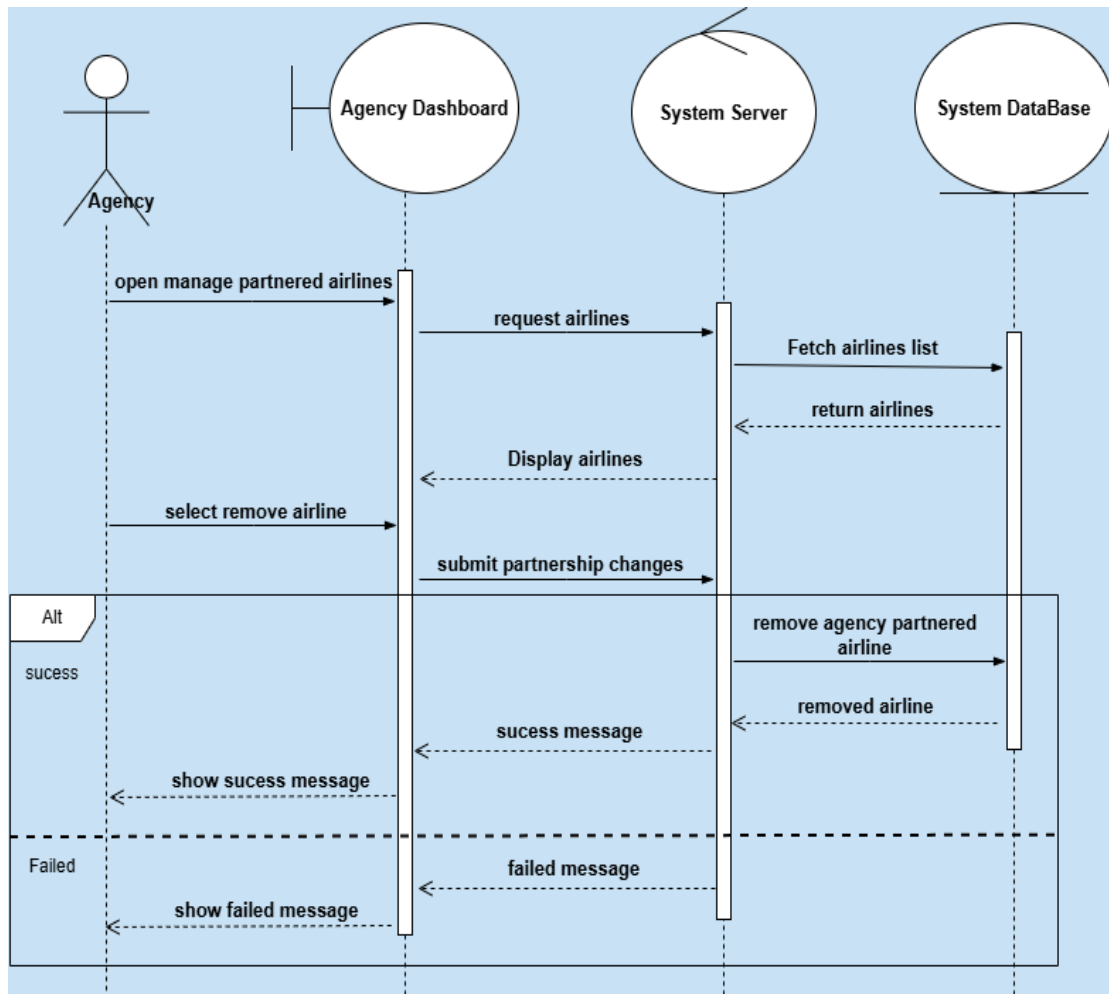


Figure 3.35 Agency remove partnered airline Sequence Diagram

Agency Manage commission Sequence Diagram:

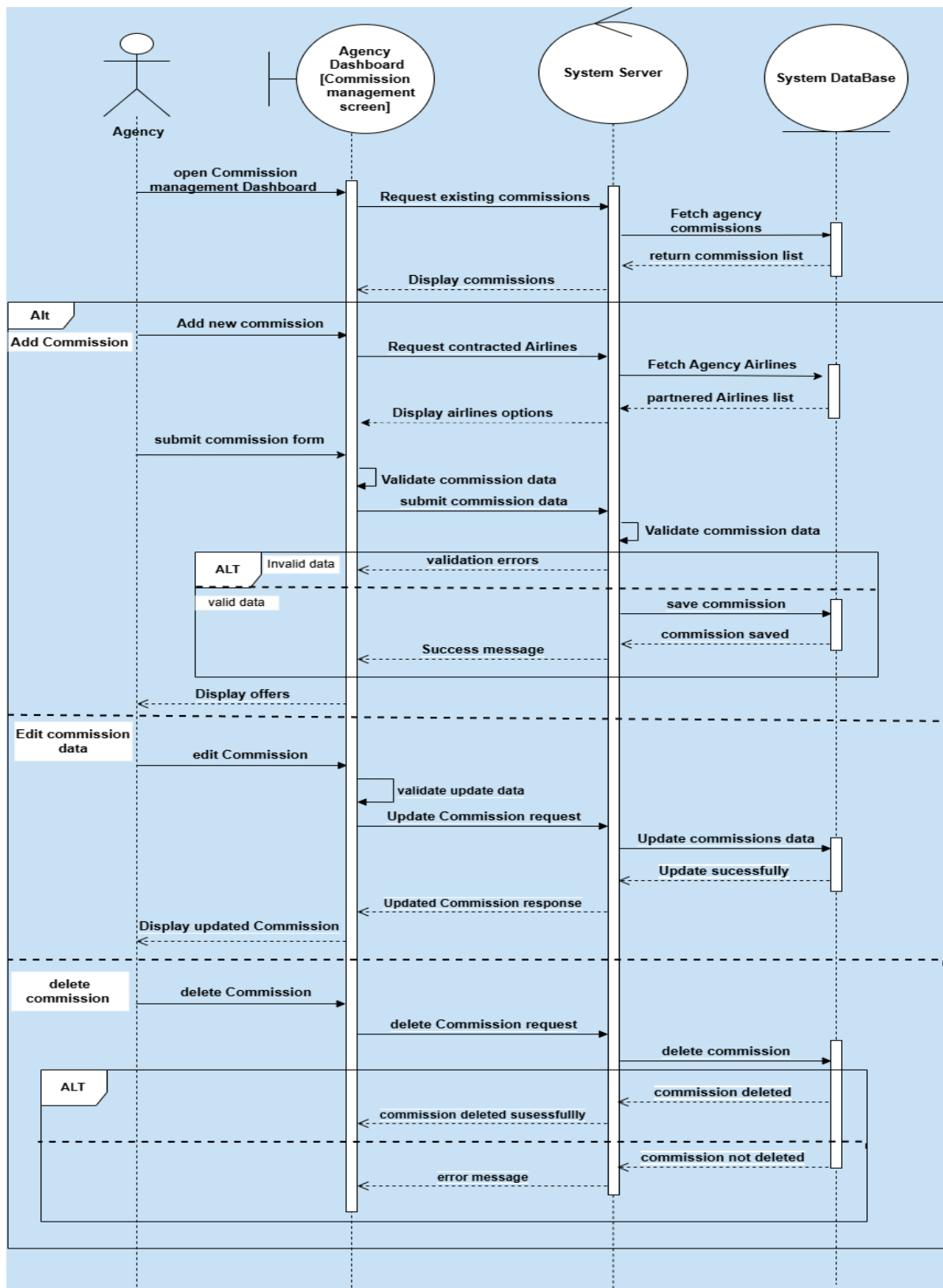


Figure 3.36 Agency manage Commissions Sequence Diagram

Agency Manage Offers Sequence Diagram:

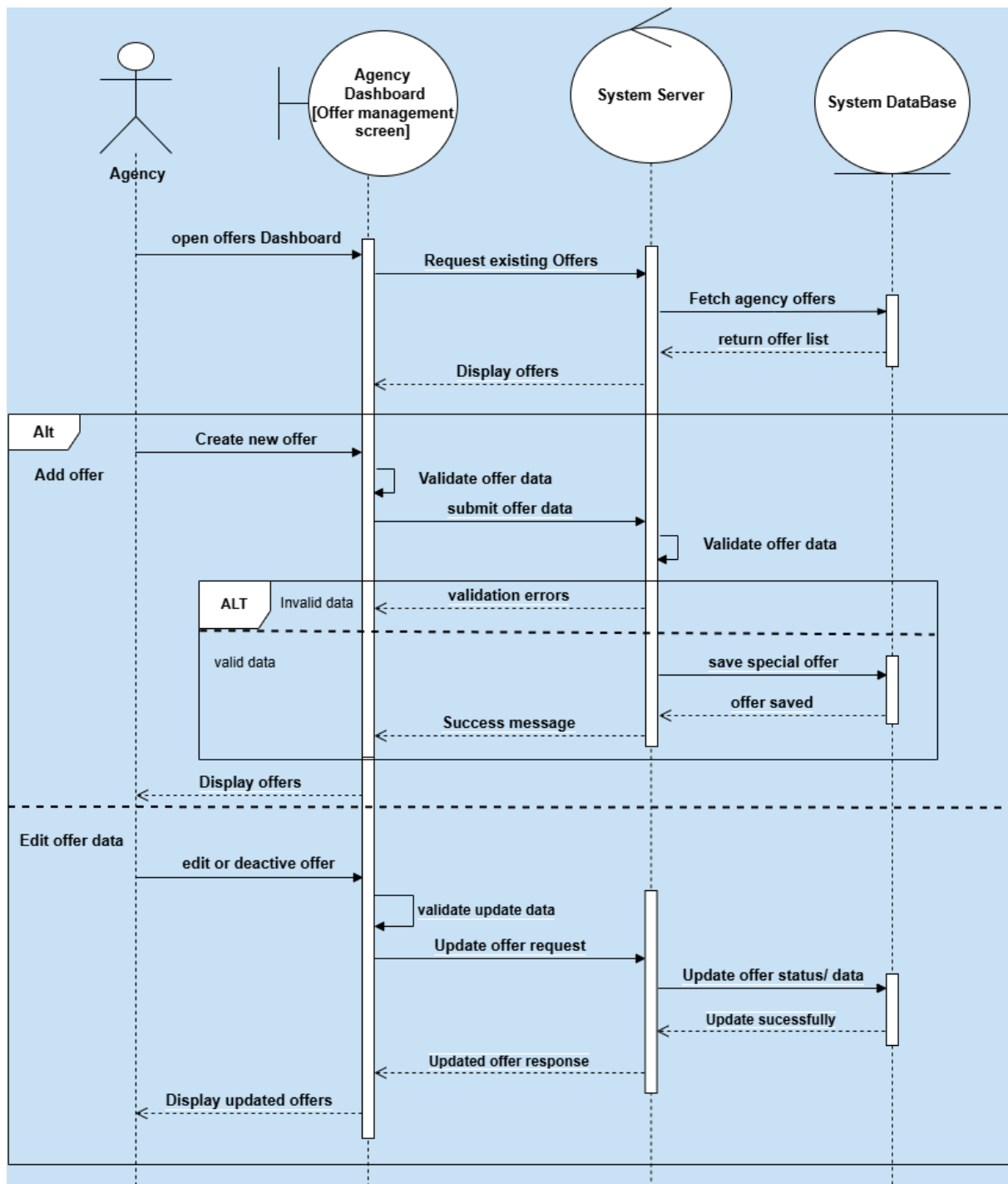


Figure 3.37 Agency manage Offers Sequence Diagram

Admin Manage Registration Request Sequence Diagram:

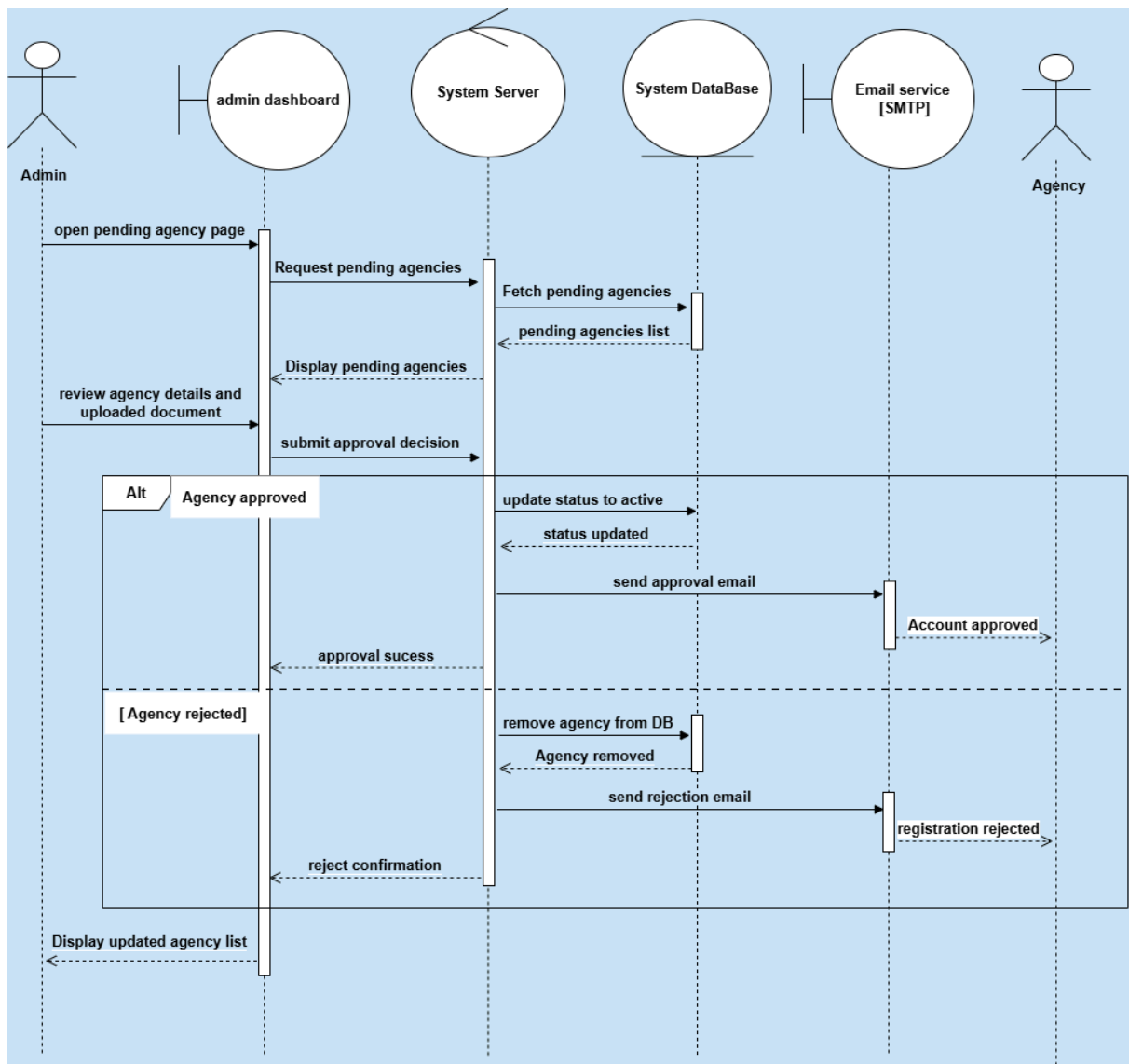


Figure 3.38 Admin manage Registration requests Sequence Diagram

Admin Manage agencies: view agency Sequence Diagram:

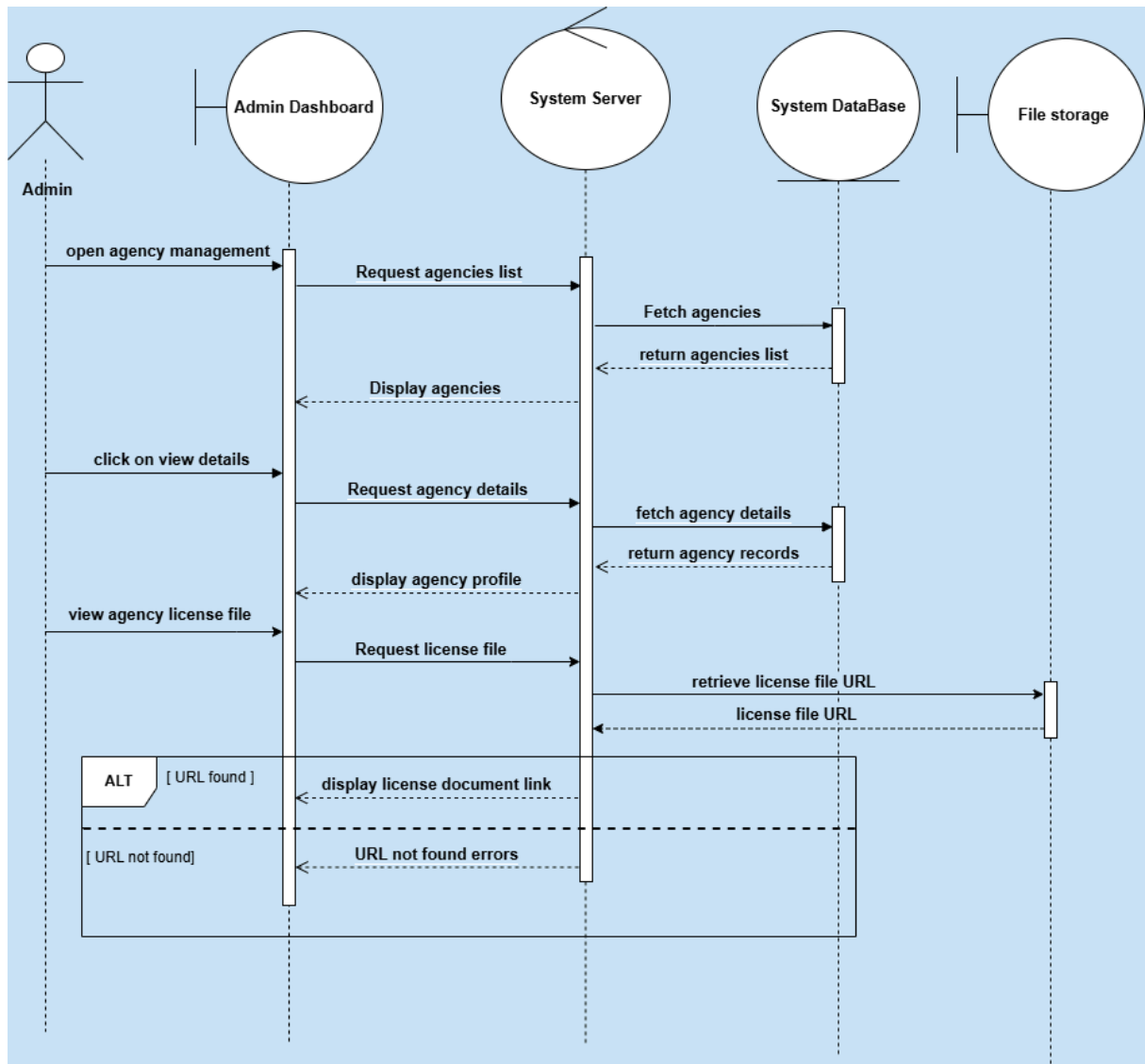


Figure 3.39 Admin manage Agencies (view agency)Sequence Diagram

Admin Manage agencies: change status , remove agency Sequence Diagram:

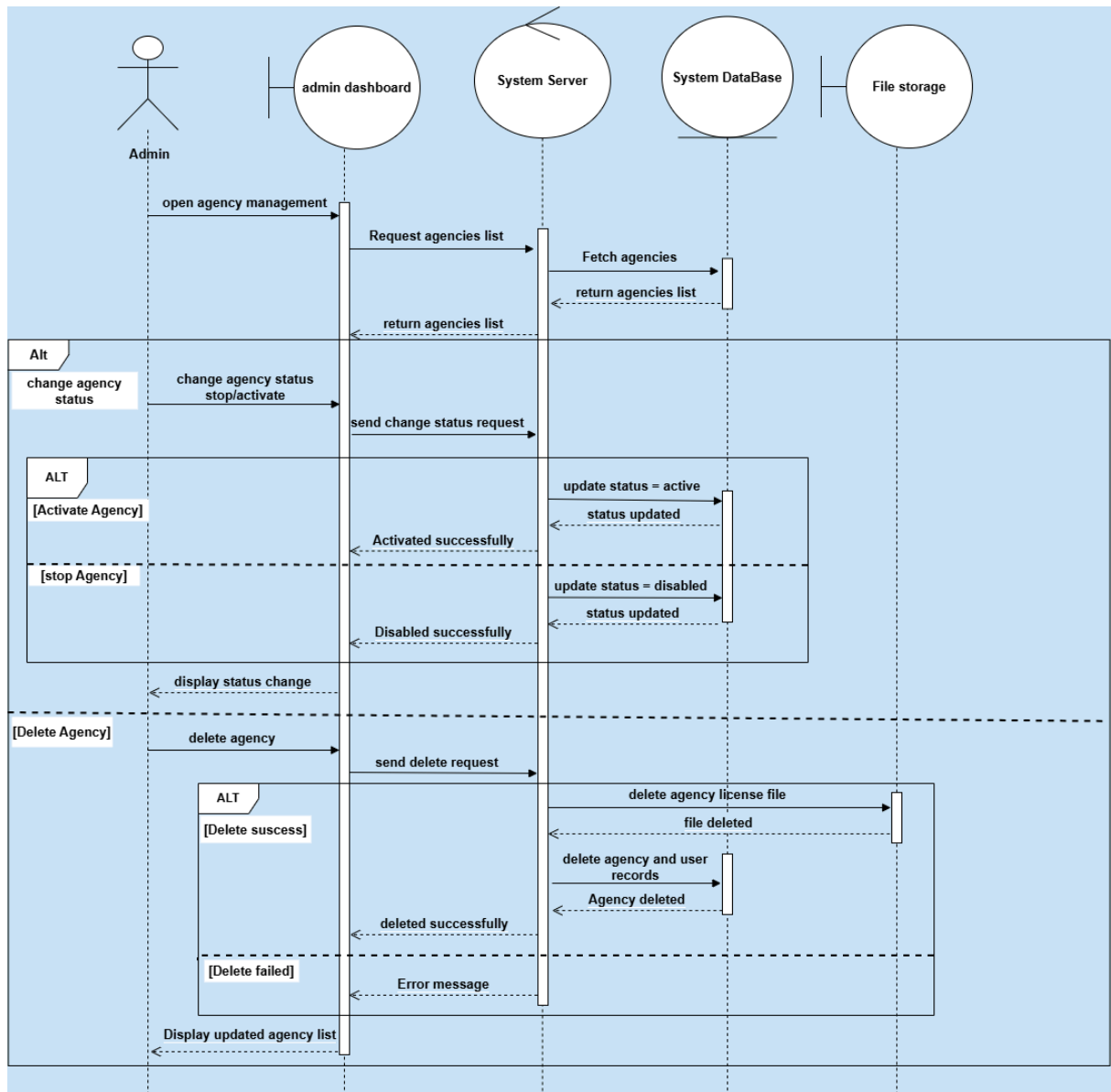


Figure 3.40 Admin manage agencies (change status , delete agency) Sequence Diagram

Admin manage Contact Messages and complains Sequence Diagram:

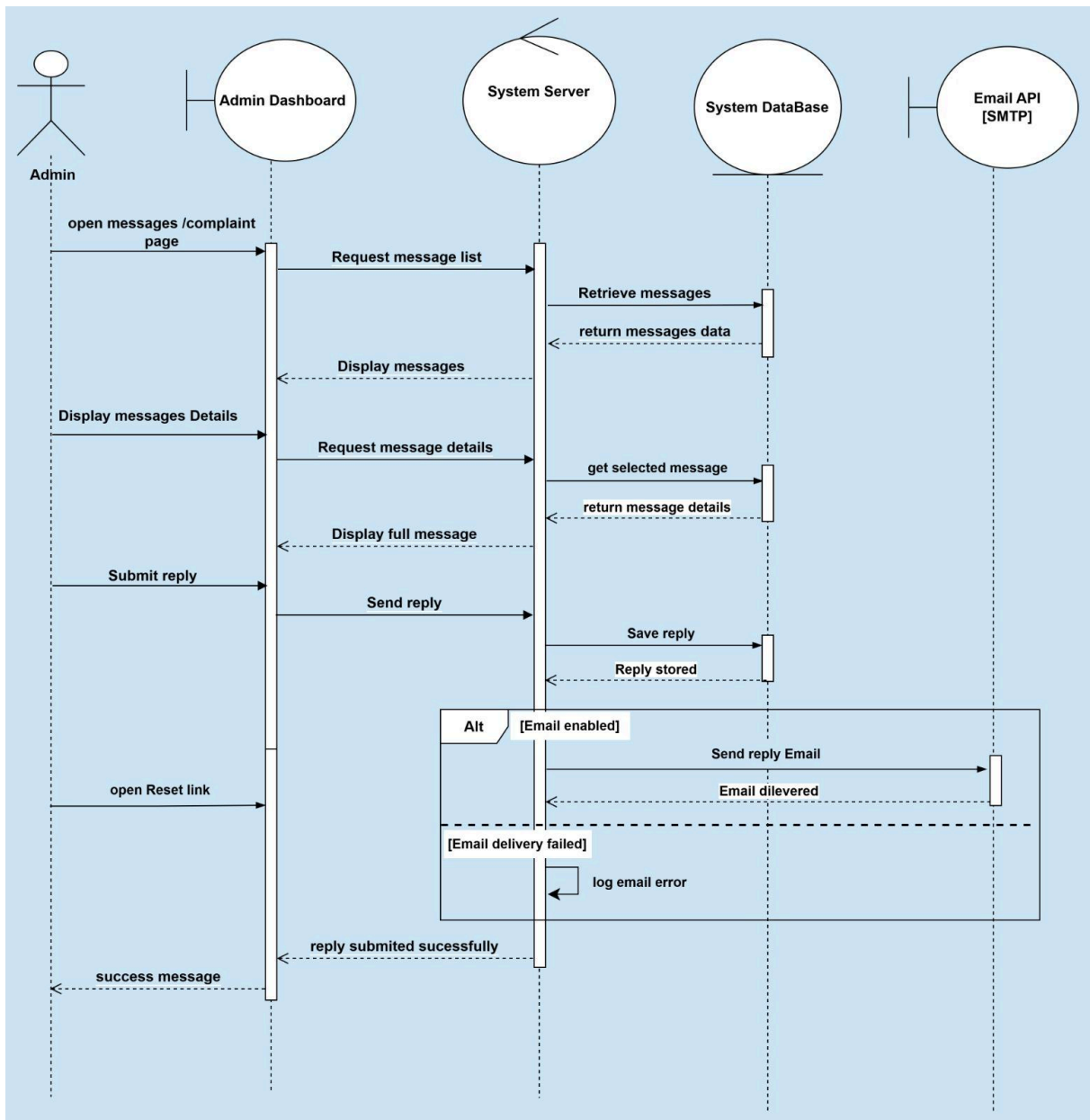


Figure 3.41 Admin manage contact messages and complains Sequence Diagram

3.3.2.3 Class Diagram:

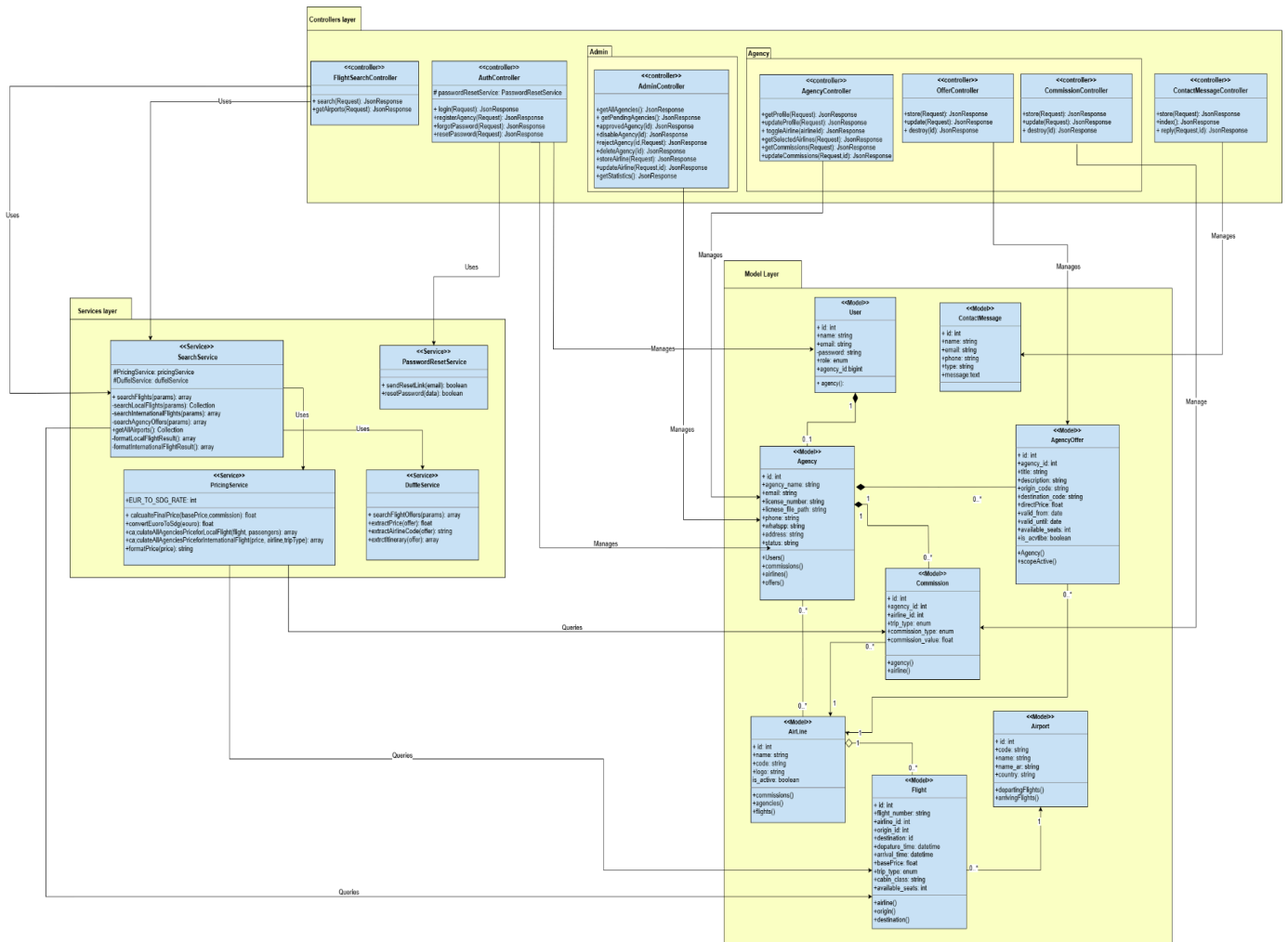


Figure 3.42 Class Diagram

Figure 3.41 presents the static structural model of the AirBridgeSD system. The diagram defines the primary entities (classes), their attributes, methods, and the relationships between them.

The **Agency** class is the central operational entity, associated with one or more **Commission** records (one-to-many) and one or more **AgencyOffer** records (one-to-many). Each Commission references a specific **Airline**, and each AgencyOffer similarly references an Airline. The **Admin** class maintains a supervisory relationship over Agency instances, managing their lifecycle status transitions.

The **Customer** actor interacts with the system as a transient user — not stored as a registered entity — and initiates **BookingRequest** events that are routed externally via WhatsApp. The **Flight** entity is associated with both an **Airline** and two **Airport** instances (origin and destination), with commission pricing calculated at query time by combining the Flight's base price with the relevant Agency Commission rule.

3.4 Database Analysis:

3.4.1 ER Diagram:

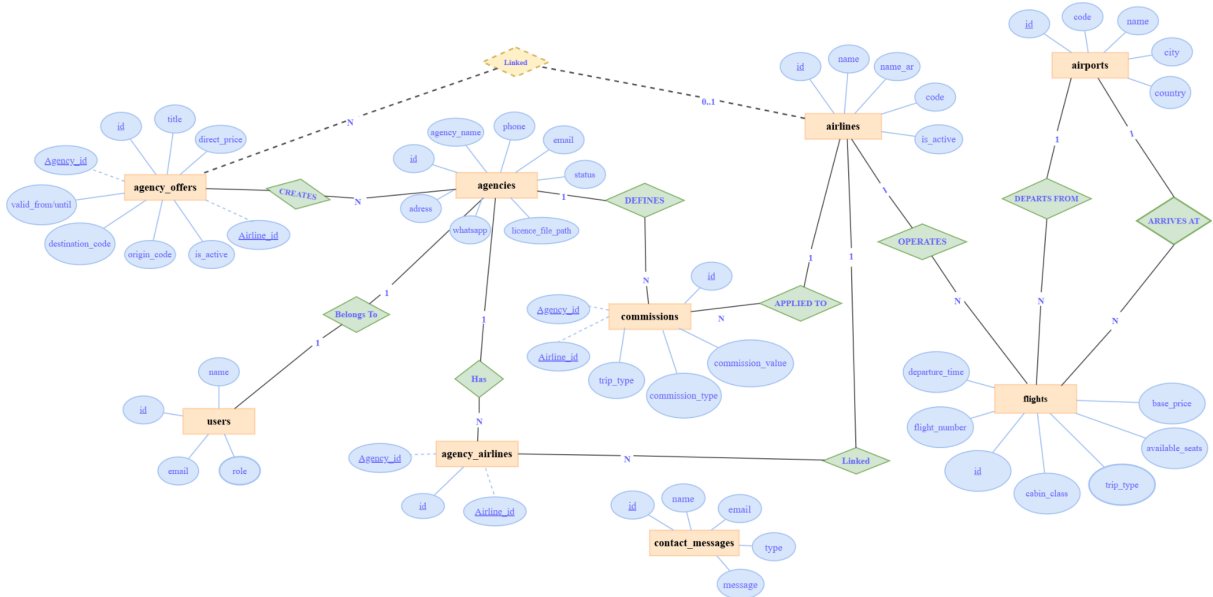


Figure 3.43 ER Diagram

Figure 3.42 illustrates the Entity-Relationship (ER) diagram for the AirBridgeSD database, depicting the logical data model prior to physical implementation in MySQL.

The primary entities are: **Agencies**, **Airlines**, **Airports**, **Flights**, **Commissions**, **agency_offers**, and **contact_messages**. The key relationships are as follows:

- An **Agency** may partner with many **Airlines** (many-to-many, resolved through the Agency_Airlines associative table).

- An **Agency** may define many **Commission** rules, each scoped to one **Airline** and one trip type (one-to-many).
- An **Agency** may publish many **agency_offers**, each linked to one **Airline** (one-to-many).
- A **Flight** is operated by one **Airline** and connects one origin **Airport** to one destination **Airport**.
- **contact_messages** are submitted by external actors (Customers or Agencies) and are not entity-linked by foreign key — they store sender information as plain text fields.

3.4.2 Database Schema Description:

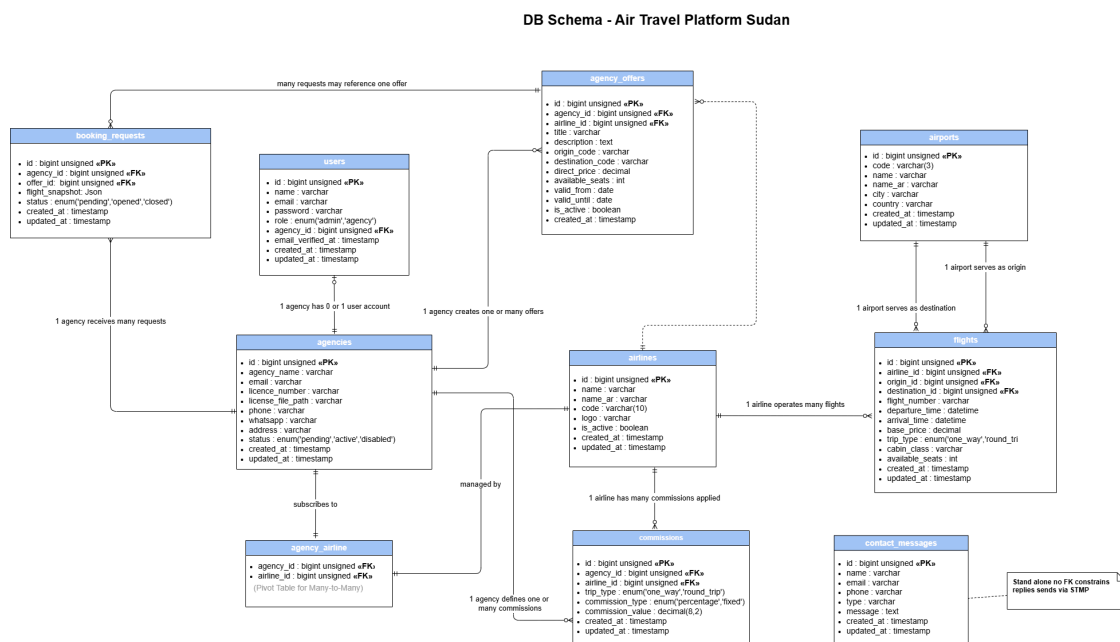


Figure 3.44 Database Schema

The AirBridgeSD relational database is implemented in MySQL and comprises eight core tables designed to support the platform's three-actor operational model. The schema follows Third Normal Form (3NF) principles to minimise data redundancy and ensure referential integrity.

The Agencies table: serves as the central entity for registered travel agencies, storing credentials, contact information, and account status. The status field is implemented as an enumerated type (pending, active, disabled) to represent the agency lifecycle managed by the platform administrator.

The Agency_Airlines:Pivot table implements a many-to-many associative relationship between agencies and airlines, recording which airlines each agency has declared as active partners. This table references both the Agencies and Airlines tables via foreign keys.

The Agency-offers table: stores promotional travel packages published by agencies, including pricing, routing codes, validity dates, and seat capacity. It references both Agencies and Airlines.

The Airlines table: maintains the global airline catalogue administered by the platform. It stores both English and Arabic name fields to support the platform's bilingual requirement.

The Airports table: stores airport data including city, country, and IATA code fields, referenced by the Flights table for origin and destination identification.

The Commissions table: stores per-agency, per-airline commission rules. Each record defines the commission type (percentage or fixed) and value for a specific airline and trip type, scoped to a single agency.

The contact_messaggies table: stores inbound support messages from both customers and agencies, with a type field to distinguish the sender category.

The Flights table: stores locally managed flight records. Full column definitions are provided in).

3.4.3 Database Tables:

Table 3.49 Database Users

Column Name	Type	Null	Default	Key	Description
-------------	------	------	---------	-----	-------------

id	bigint	No		PK	Auto-increment primary key
name	varchar	No			Full name of the user
email	varchar	No		UNIQUE	Login email must be unique across all users
password	varchar	No			bcrypt-hashed password plaintext never stored
role	enum(admin, agency)	No			Role-based access: admin or agency
agency_id	bigint	Yes	NULL	FK → agencies.id	Links user to their agency; NULL for admin accounts
created_at	timestamp	Yes	NULL		Record creation timestamp (managed by Laravel)
updated_at	timestamp	Yes	NULL		Record last-update timestamp (managed by Laravel)

Table 3.50 Database Agencies

Column Name	Type	Null	Default	Key
id	bigint	No		PK
agency_name	varchar	No		
email	varchar	Yes	NULL	
licence_number	varchar	Yes	NULL	
license_file_path	varchar	No		
phone	varchar	No		
whatsapp	varchar	No		
address	varchar	Yes	NULL	

status	enum(pending, active, disabled, rejected)	No	pending	
created_at	timestamp	Yes	NULL	
updated_at	timestamp	Yes	NULL	

Table 3.51 Database Agency_Airlines

Column Name	Type	Null	Default	Key	Description
id	bigint	No		PK	Auto-increment primary key
agency_id	bigint	No		FK → agencies.id	References the agency linked to this airline
airline_id	bigint	No		FK → airlines.id	References the airline linked to this agency
created_at	timestamp	Yes	NULL		Record creation timestamp
updated_at	timestamp	Yes	NULL		Record last-update timestamp

Table 3.52 Database Agency-offers

Column Name	Type	Null	Default	Key	Description
id	bigint	No		PK	Auto-increment primary key
agency_id	bigint	No		FK → agencies.id	References the agency that published this offer
title	varchar	No			Display title of the promotional offer
description	text	No			Detailed description of

					the offer
origin_code	varchar	No			IATA code of the departure airport (string, not FK — supports mixed Duffel/local routes)
destination_code	varchar	No			IATA code of the arrival airport
pricing_mode	enum(direct, discount)	No			Pricing strategy: direct = fixed SDG price; discount = percentage off cheapest flight
direct_price	decimal	Yes	NULL		Fixed price in SDG — populated when pricing_mode = direct; NULL otherwise
available_seats	int	Yes	0		Number of seats available for this offer
valid_from	date	Yes	NULL		Date from which this offer is available to customers
valid_until	date	Yes	NULL		Date after which this offer expires
is_active	tinyint	No	1		Visibility flag: 1 = visible to customers; 0 = hidden
created_at	timestamp	Yes	NULL		Record creation timestamp
updated_at	timestamp	Yes	NULL		Record last-update timestamp

Table 3.53 Database Airlines

Column Name	Type	Null	Default	Key	Description
id	bigint	No		PK	Auto-increment primary key

name	varchar	No			Airline full English name
name_ar	varchar	Yes	NULL		Airline Arabic name — supports bilingual display
code	varchar	No		UNIQUE	IATA 2-letter airline code — stored as uppercase; must be unique
logo	varchar	Yes	NULL		URL or storage path to the airline logo image
is_active	tinyint	No	1		Availability flag: 1 = available for agency partnerships
created_at	timestamp	Yes	NULL		Record creation timestamp
updated_at	timestamp	Yes	NULL		Record last-update timestamp

Table 3.54 Database Airports

Column Name	Type	Null	Default	Key	Description
id	bigint	No		PK	Auto-increment primary key
name	varchar	No			Airport full English name
name_ar	varchar	No			Airport Arabic name
code	varchar	No		UNIQUE	letter IATA airport-3 code — must be unique
city	varchar	Yes	NULL		City where the airport is located
country	varchar	Yes	NULL		Country where the airport is located
created_at	timestamp	Yes	NULL		Record creation timestamp
updated_at	timestamp	Yes	NULL		Record last-update timestamp

Table 3.55 Database Commissions

Column Name	Type	Null	Default	Key	Description
id	bigint	No		PK	Auto-increment primary key
agency_id	bigint	No		FK → agencies.id	References the agency this commission rule belongs to
airline_id	bigint	No		FK → airlines.id	References the airline this commission rule applies to
trip_type	enum(one_way , round_trip)	No			Scope of the commission: one-way or round-trip flights
commission_type	enum(percentage, fixed)	No			Calculation method: percentage of base price or fixed SDG amount
commission_value	decimal	No			Numeric value of the commission (percentage or SDG amount)
created_at	timestamp	Yes	NULL		Record creation timestamp
updated_at	timestamp	Yes	NULL		Record last-update timestamp

Table 3.56 Database contact_messaggies

Column Name	Type	Null	Default	Key	Description
id	bigint	No		PK	Auto-increment primary key

sender_id	bigint	Yes	NULL	FK → users.id	User who sent the message (Admin or Agency). Nullable for customer messages
receiver_id	bigint	Yes	NULL	FK → users.id	User intended to receive the message
name	varchar	No			Name of the person who submitted the message
email	varchar	No			Email address for admin reply delivery
phone	varchar	Yes	NULL		Optional phone number of the sender
type	varchar	No			Message category distinguishes customer from agency messages
message	text	No			Full content of the support message
sent_at	timestamp	No	CURRENT_TIMESTAMP		Actual message sending timestamp
created_at	timestamp	Yes	NULL		Record creation timestamp
updated_at	timestamp	Yes	NULL		Record last-update timestamp

Table 3.57 Database Flights

Column Name	Type	Null	Default	Key	Description
id	bigint	No		PK	Auto-increment primary key
airline_id	bigint	No		FK → airlines.id	References the airline operating this flight

flight_number	varchar	No			Flight identifier code (e.g. SD101) — not a FK
origin_id	bigint	No		FK → airports.id	References the departure airport
destination_id	bigint	No		FK → airports.id	References the arrival airport
trip_type	enum(one_way, round_trip)	No	one_way		Flight type: one-way or round-trip
departure_time	datetime	No			Scheduled departure date and time
arrival_time	datetime	No			Scheduled arrival date and time
base_price	decimal	No			Base fare in SDG before any agency commission is applied
available_seats	int	No			Number of available seats on this flight
cabin_class	enum(ECONOMY, PREMIUM_ECONOMY, BUSINESS, FIRST)	No	ECONOMY		Cabin class of this flight record
created_at	timestamp	Yes	NULL		Record creation timestamp
updated_at	timestamp	Yes	NULL		Record last-update timestamp

Table 3.58 booking_requests

Column Name	Type	Null	Default	Key	Description
id	bigint	No		PK	Auto-increment primary key
agency_id	bigint	No		FK → agencies.id	Identifies the agency that receives the booking request
offer_id	bigint	Yes		FK → agency_offers.id	References the selected local promotional offer if applicable; nullable for international Duffel flights
flight_snapshot	json	No			Stores a snapshot of flight information at request generation time (origin, destination, airline, departure date, displayed price, trip type) to preserve historical request details even if flight data changes later
status	enum	No			Tracks the current request state: generated, viewed by agency, or completed/closed
created_at	timestamp	Yes			Record creation timestamp

updated_at	timestamp	Yes			Record last-update timestamp
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CHAPTER 4

IMPLEMENTATION

4.1 Introduction:

The implementation phase translates the system design specifications produced in Chapter 3 into a fully functional software application. This chapter documents the technical execution of the AirBridgeSD platform across four areas: (1) the development environment and technology stack, including the frontend, backend, and integration layers; (2) the tools and services used for development, testing, quality assurance, design, and project management; (3) the system's implemented screens, demonstrating the realised user interface for all three actor roles; and (4) the version control and deployment pipeline used to publish the platform. Where relevant, implementation decisions are linked back to the corresponding functional and non-functional requirements established in Chapter 3.

4.2 Tools and Technologies:

To achieve the project goals, a modern technology stack was selected to ensure high performance, security, and ease of maintenance.

4.2.1 Development and Programming Languages:

- **HTML5 and CSS3 (Tailwind CSS):** For the visual presentation and layout, **Tailwind CSS** was adopted as a utility-first styling framework. The framework was instrumental in achieving a **Mobile-First responsive design**, utilizing responsive utility classes to ensure layout integrity across various screen dimensions, while maintaining modern aesthetic standards.
- **JavaScript (React.js):** Application state including flight search parameters, retrieved results, active filters, and user selections is managed using **React's built-in useState and useContext hooks**, enabling shared state across the search form, results list, filter panel, and flight detail components without requiring a third-party state management library. Routing between platform pages (Home, Offers, About, Agency Dashboard, Admin Panel) is handled via **React Router**, enabling client-side navigation without full page reloads.

- **PHP (Laravel):** Laravel 12.48.1 was employed as the core backend framework to manage the system's logic and security. It serves as a **RESTful API provider** that bridges the gap between the client-side interface and the **Duffel flight services**. Authentication is implemented using **Laravel Sanctum**, which issues API tokens upon successful login. These tokens are transmitted in the Authorization header of all subsequent API requests from the React frontend, ensuring that agency dashboard routes and admin panel routes are accessible only to authenticated and authorised users.

4.2.2 Database Management System

- **SQL (MySQL):** MySQL was implemented as the primary relational database management system (RDBMS). It provides a structured environment for storing critical data, ensuring **ACID compliance**, which is vital for maintaining data integrity during commission calculations and agency management.

4.2.3 Development and Debugging Tools

- **Visual Studio Code** was utilized as the main Integrated Development Environment (IDE) for this project. It was selected for its versatility and robust support for **React.js** and **PHP** through specialized extensions. The environment facilitated efficient development by providing built-in tools for code debugging, formatting, and integrated terminal management, which streamlined the overall implementation process.
- **Postman:** This tool was essential for testing and documenting API endpoints. It was used to verify the requests and responses between the Laravel backend and the **Duffel API**, ensuring that flight data is correctly retrieved and formatted before being sent to the frontend.
- **SonarQube:** was used to ensure the quality and cleanliness of the code. It automatically scans the project to detect errors, security gaps, and clean code (unnecessary or messy code). By using this tool, the system's stability was improved, ensuring that the final software is secure, easy to maintain.

- **Browser Developer Tools (Chrome DevTools):** These were employed for real-time debugging of the frontend, monitoring network activity, and optimizing the performance of the React application.

4.2.4 Design and Project Management Tools

- **Figma:** This tool was used to create prototypes and UI designs. It allowed for the precise planning of the user experience (UX) and the visual identity of the **AirBridgeSD** platform.
- **Lucidchart / Draw.io:** These platforms were utilized to design the structural diagrams of the system, including the Architecture Diagram and Use Case models, ensuring a clear visualization of the system's logic.
- **ClickUp:** This project management tool was implemented to track tasks, organize the development timeline, and ensure that all project milestones were achieved according to the schedule.
- **Gantt Chart:** was implemented to manage the project's timeline. It helped in organizing tasks from design to coding and testing ensuring that each phase was completed on time and that the project followed a clear and professional schedule.

4.2.5 Version Control and Collaboration

- **Git and GitHub:** Git was used to track every change made to the code, ensuring no progress was lost. The project was hosted on **GitHub** as a central repository to manage different versions of the code, making the development process more organized and secure.
- **Netlify:** was used to host and deploy the React.js frontend of the platform. Upon each push to the main branch of the GitHub repository, Netlify's CI/CD pipeline automatically builds the React application, deploys the static output to its global CDN, and provides an HTTPS certificate. The Laravel backend API is maintained separately in a local development environment, and the React frontend communicates with it via environment-variable-configured API base URLs.

4.3 Implementation (System Screens):

4.3.1 Customer interfaces :

الرئيسيةالعروضحول المنصةالتواصل معنا

تسجيل وكالة

AirBridgeSD

سافر بكل راحة وأمان

اكتشف العالم مع AirBridgeSD

قارن الأسعار من جميع وكالات السفر

+150

وجدة عالمية

+50

شركة طيران

500+

رحلة توصية

ذهاب فقط

ذهاب وعودة

من

إلى

أخر مدينة المغادرة

أخر مدينة الوصول

تاريخ المغادرة

05/31/2026

عدد الركاب

1

مئة السفر

اقتصادية

ركاب

+

-

ابحث عن الرحلات

كيف تعمل المنصة؟

1

أدخل بيانات الرحلة

أخر وجهتك وتاريخ سفرك

2

قارن الأسعار

عروض من جميع الوكالات

3

تواصل مع الوكالة

واتساب أو هاتف أو زيارة

4

احجز خارج المنصة

الوكالة تصدر التذكرة مباشرة

تنويه هام

• المنصة تعرض ومقارنة الأسعار فقط

• الحجز يتم خارج المنصة مباشرة مع الوكالة

• الأسعار المعروضة للمقارنة وقد تتغير حسب التوفر

• الوكالات مسؤولة عن إصدار التذاكر في الخطأ الخاصة

القانونية

سياسة الخصوصية

الشروط والأحكام

المساعدة

تواصل معنا

الأسئلة الشائعة

روابط سريعة

الرئيسية

حول المنصة

AirBridgeSD

منصة لجميع ومقارنة أسعار الرحلات الجوية في السودان، توفر شفافية وخيارات متنوعة لرحلاتك.

نتائج البحث

من PZU إلى 27 - RUH رحلة مباشرة

عرض وكالة

Al-Musafer Travel

تبقى 5 مقاعد

ماتج دي: ٢٠٢٦/٧/١١

السعر الإجمالي

2,000,000.00

خارج سوداني

✈️ مطار بورتسودان الدولي الجديد — مطار الملك خالد الدولي

شركة الطيران: Saudia

أقل سعر ما يتكرر

إجرة سفر جديدة مع الخطوط السعودية

تواصل للحجز

عرض وكالة

Al-Safir Travel

تبقى 15 مقعداً

ماتج دي: ٢٠٢٦/٧/١١

السعر الإجمالي

2,100,000.00

خارج سوداني

✈️ مطار بورتسودان الدولي الجديد — مطار الملك خالد الدولي

شركة الطيران: Air Arabia

عرض خاص لرحلات الصيف

خصم على التذاكر المباشرة مع خدمة حجز سريعة ودعم 24 ساعة

تواصل للحجز

السعر الإجمالي

2,473,955.00

شامل الضرائب (SDG)

Al-Nile Travel Agency

ص 02:10

مطار الملك خالد الدولي

١ توقف

م 02:45

مطار بورتسودان الدولي الجديد

Ethiopian Airlines

ET03498

تواصل للحجز

السعر الإجمالي

2,473,955.00

شامل الضرائب (SDG)

Dar Al-Salam Travel

ص 02:10

مطار الملك خالد الدولي

١ توقف

م 02:45

مطار بورتسودان الدولي الجديد

Ethiopian Airlines

ET03498

تواصل للحجز

السعر الإجمالي

2,539,902.75

شامل الضرائب (SDG)

Al-Iazira Travel

ص 02:10

مطار الملك خالد الدولي

١ توقف

م 02:45

مطار بورتسودان الدولي الجديد

Ethiopian Airlines

ET03498

تواصل للحجز

السعر الإجمالي

2,612,471.40

شامل الضرائب (SDG)

Al-Safir Travel

ص 02:10

مطار الملك خالد الدولي

١ توقف

م 02:45

مطار بورتسودان الدولي الجديد

Ethiopian Airlines

ET03498

تواصل للحجز

السعر الإجمالي

2,685,040.05

شامل الضرائب (SDG)

Al-Musafer Travel

ص 02:10

مطار الملك خالد الدولي

١ توقف

م 02:45

مطار بورتسودان الدولي الجديد

Ethiopian Airlines

ET03498

تواصل للحجز

تصفية النتائج

ترتيب حسب

الأقل سعراً

الأقصر وقتاً

شركات الطيران

EgyptAir Express ☐Ethiopian Airlines ☒

الوكالات

Al-Musafer Travel ☐Al-Safir Travel ☐

القانونية

سياسة الخصوصية
الشروط والأحكام

المساعدة

تواصل معنا
الأسئلة الشائعة

روابط سريعة

الرئيسية
حول المنصة

AirBridgeSD

منصة للجمع ومقارنة أسعار الرحلات الجوية في السودان، توفر شفافية وخيارات متنوعة لرحلاتك.

الرئيسية العروض حول المنصة للتواصل معنا

التسجيل وكالة

AirBridgeSD

السعر الإجمالي

2,473,955.00

السعر يشمل الضرائب والرسوم ويخضع للتأكيد النهائي أثناء وقت الحجز

مقدم الخدمة

Al-Nile Travel Agency

وكالة معتمدة

إرسال طلب الحجز للوكالة

مسار رحلة الذهاب

10 من 25 د

2026/6/9

06:00 م

مطار أجيس أبابا بولي الدولي

2026/6/9

02:45 م

مطار بورسودان الدولي الجديد

2026/6/10

02:10 ص

مطار الملك خالد الدولي

2026/6/9

10:35 م

مطار أجيس أبابا بولي الدولي

معلومات الأمتعة

تخضع سياسات الأمتعة المحلية لاشروط وأحكام شركة الطيران ودرجة الحجز. يرجى التأكد من الوكيل أثناء عملية الحجز لمعرفة الوزن المسموح به محلياً.

Screen 4.3 Flight Details Page

الرئيسية العروض حول المنصة للتواصل معنا

التسجيل وكالة

AirBridgeSD

عروض الوكالات

5 عرض متاح من الوكالات المعتمدة

ابحث بالملء أو الوجهة، شركة الطيران، أو الوكالة...

عرض وكالة

Al-Musafer Travel

تبقى 3 مقاعد

السعر

2,000,000

دولة سوداني

✈️ مطار بورسودان الدولي الجديد — مطار الملك خالد الدولي

شركة الطيران: Saudia

أقل سعر ما حيزكر

تجربة سفر جديدة مع الخطوط السعودية

تواصل للحجز

عرض وكالة

Al-Nile Travel Agency

تبقى 7 مقاعد

السعر

1,999,000

دولة سوداني

✈️ مطار الخرطوم الدولي — مطار دبي الدولي

شركة الطيران: Flydubai

عرض مباشر: الخرطوم إلى دبي

رحلة مريحة ومباشرة إلى دبي مع أفضل الخدمات.

تواصل للحجز

عرض وكالة

Dar Al-Salam Travel

تبقى 20 مقاعد

السعر

990,000

دولة سوداني

✈️ مطار بورسودان الدولي الجديد — مطار القاهرة الدولي

شركة الطيران: Badr Airlines

القاهرة بأقل الأسعار

احصل على أفضل سعر الآن إلى القاهرة من بورسودان بخمسة حصري.

تواصل للحجز

عرض وكالة

Al-Safir Travel

تبقى 16 مقاعد

السعر

2,100,000

دولة سوداني

✈️ مطار بورسودان الدولي الجديد — مطار الملك خالد الدولي

شركة الطيران: Air Arabia

عرض خاص لرحلات الصيف

حزم على التذاكر المباشرة مع خدمة حجز سريعة ودعم 24 ساعة.

تواصل للحجز

عرض وكالة

Al-Nile Travel Agency

تبقى 5 مقاعد

السعر

1,150,000

دولة سوداني

✈️ مطار بورسودان الدولي الجديد — مطار الملك خالد الدولي

شركة الطيران: Saudia

عروض الصيف: بورسودان إلى الرياض

سافر الآن بأسعار مخفضة ووزن إمتاعي 10 كيلو. العرض ساري لفترة محدودة جداً!

تواصل للحجز

تصفية العروض

ترتيب حسب السعر

☒ الافتراضي
☐ الأقل سعراً
☐ الأعلى سعراً

تاريخ السفر

mm/dd/yyyy

الوكالة

☐ Al-Musafer Travel
☐ Al-Nile Travel Agency
☐ Dar Al-Salam Travel
☐ Al-Safir Travel

Screen 4.4 Offers Page

الرئيسية

العروض

حول المنصة

للتواصل معنا

تسجيل وكالة

AirBridgeSD

تفاصيل العرض

السعر

990,000

جنيه سوداني — يجمع للتأكد مع الوكالة

مقدم الخدمة

Dar Al-Salam Travel

وكالة معتمدة

إرسال طلب الحجز للوكالة

مسار الرحلة

عرض وكالة

مطار القاهرة الدولي

مطار بورسودان الدولي

القاهرة بأقل الأسعار

احصل على مقعدك الآن إلى القاهرة من بورسودان بخمسة حصري.

المقاعد المتاحة

20 مقعد

تاريخ الانتهاء

١٠ يونيو ٢٠٢٦

تاريخ البداية

١ يونيو ٢٠٢٦

Screen 4.5 Offer Details Page

الرئيسية

العروض

حول المنصة

للتواصل معنا

تسجيل وكالة

AirBridgeSD

حول منصة AirBridgeSD

مهمتنا

جسرك الآمن للعالم

منصة AirBridgeSD هي منصة رائدة لتجميع ومقارنة أسعار الرحلات الجوية في السودان. نحن نهدف إلى تبسيط عملية البحث عن الرحلات وتوفير الوقت والجهد للمسافرين.

نربط بين المسافرين ووكالات السفر المعتمدة، لنوفر لك شفافية في الأسعار وخيارات متنوعة لرحلاتك الداخلية والدولية.

كيف تعمل المنصة؟

1

ابحث

أدخل معلومات رحلتك

2

قارن

شاهد عروض جميع الوكالات

3

تواصل

اختر الوكالة واتواصل معها

4

احجز

الوكالة تصدر التذكرة

أسعار شفافة

نوفر لك أسعار شفافة ومصدقية

نتائج سريعة

مقارنة فورية بين عروض الوكالات المختلفة

وكالات معتمدة

جميع الوكالات مرخصة وذات سمعة طيبة

ملاحظة هامة

• المنصة تعرض ومقارنة الأسعار فقط

• الحجز يتم خارج المنصة مباشرة مع الوكالة

• الأسعار المعروضة للمقارنة وقد تتغير حسب التوفر

• الوكالات مسؤولة عن إصدار التذاكر في أنظمتها الخاصة

القانونية

سياسة الخصوصية

الشروط والأحكام

المساعدة

تواصل معنا

الأسئلة الشائعة

روابط سريعة

الرئيسية

حول المنصة

AirBridgeSD

منصة لتجميع ومقارنة أسعار الرحلات الجوية في السودان. نوفر شفافية وخيارات متنوعة لرحلاتك.

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Screen 4.6 About the platform Page

[الرئيسية](#) [العروض](#) [حول المنصة](#) [للتواصل معنا](#) [تسجيل وكالة](#)

AirBridgeSD

للتواصل معنا

نحن هنا لمساعدتك في أي استفسار أو دعم فني

أرسل رسالتك

الاسم الكامل

رقم الهاتف

نوع الرسالة

استفسار عام

الرسالة

أدخل رسالتك هنا...

إرسال الرسالة

معلومات التواصل

الهاتف

789 456 123 249+

البريد الإلكتروني

info@airbridgesd.com

[القانونية](#)

سياسة الخصوصية
الشروط والأحكام

[المساعدة](#)

تواصل معنا
الأسئلة الشائعة

[روابط سريعة](#)

الرئيسية
حول المنصة

AirBridgeSD

منصة للجمع ومقارنة أسعار الرحلات الجوية في السودان، توفر شفافية وخيارات متنوعة لرحلاتك.

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Screen 4.7 Contact us Page

[الرئيسية](#) [العروض](#) [حول المنصة](#) [للتواصل معنا](#) [تسجيل وكالة](#)

AirBridgeSD

الأسئلة الشائعة

بعض الأسئلة الشائعة

الأسئلة الشائعة

هل المنصة تقوم بالحجز؟

لن، المنصة تعرض ومقارنة الأسعار فقط. الحجز يتم مباشرة مع الوكالة المختارة.

هل الأسعار نهائية؟

الأسعار للمقارنة وقد تتغير حسب توفر المقاعد والوقت. الأسعار النهائية تحددتها الوكالة.

كيف أتواصل مع الوكالة؟

يمكنك التواصل عبر واتساب أو الهاتف أو زيارة مكتب الوكالة.

هل هناك رسوم استخدام للمنصة؟

لن، استخدام المنصة مجاني تماماً للمسافرين.

[القانونية](#)

سياسة الخصوصية
الشروط والأحكام

[المساعدة](#)

تواصل معنا
الأسئلة الشائعة

[روابط سريعة](#)

الرئيسية
حول المنصة

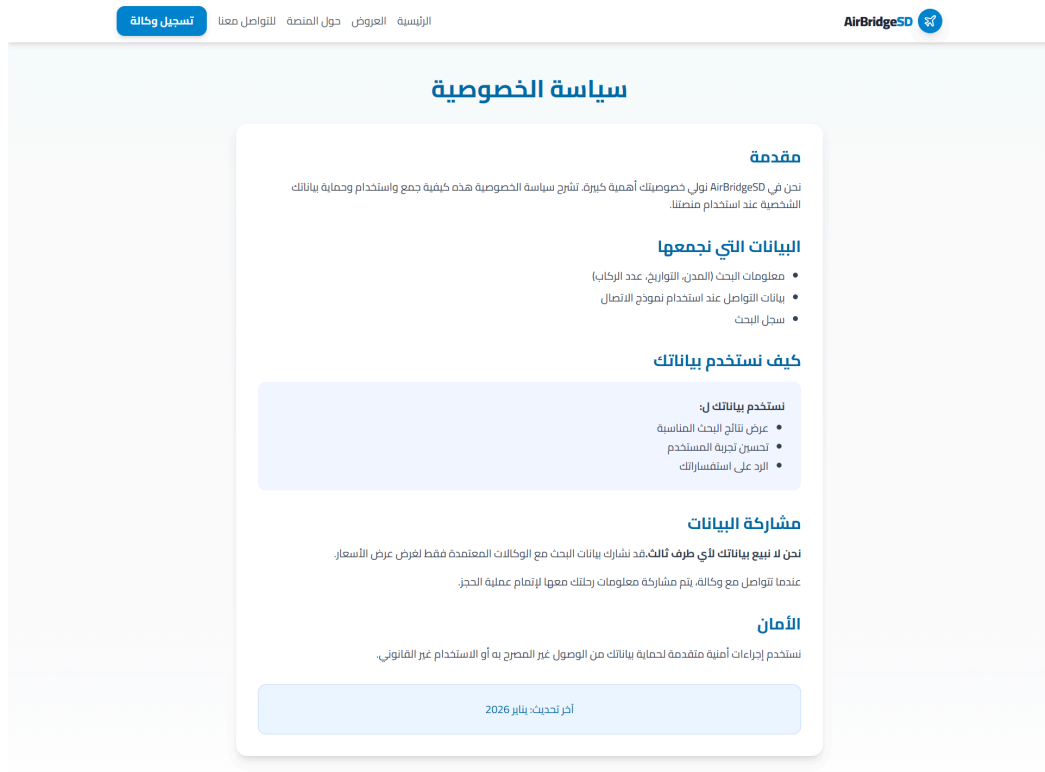
AirBridgeSD

منصة للجمع ومقارنة أسعار الرحلات الجوية في السودان، توفر شفافية وخيارات متنوعة لرحلاتك.

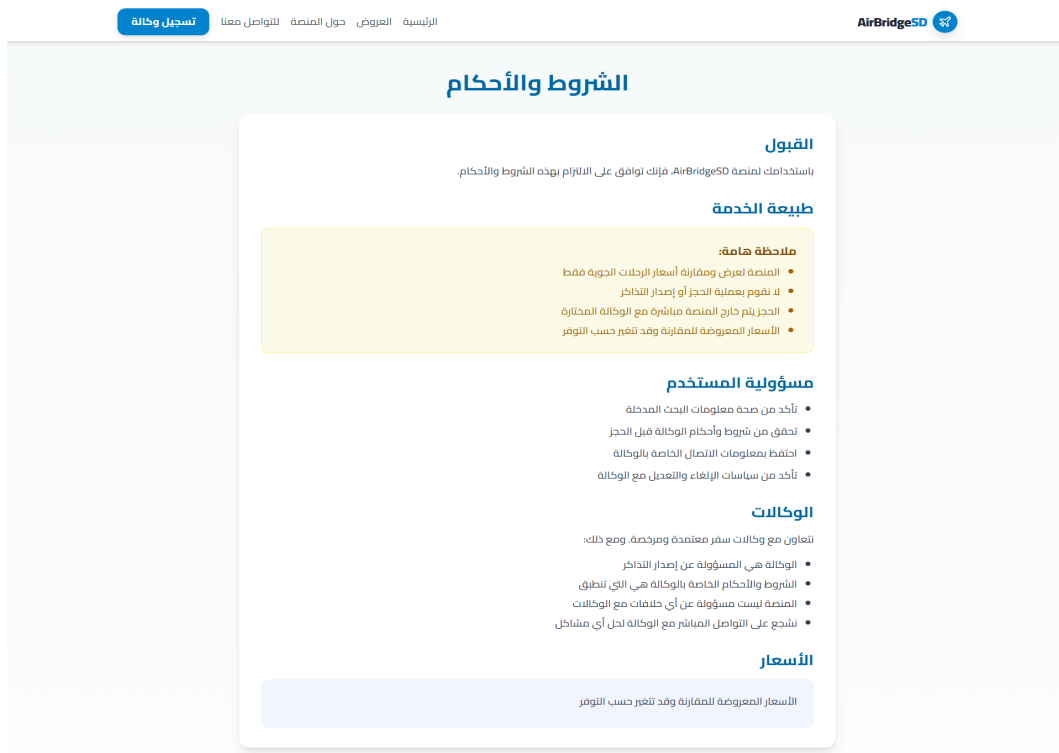
© 2026 AirBridgeSD، جميع الحقوق محفوظة.

Screen 4.8 Frequently Asked Questions Page

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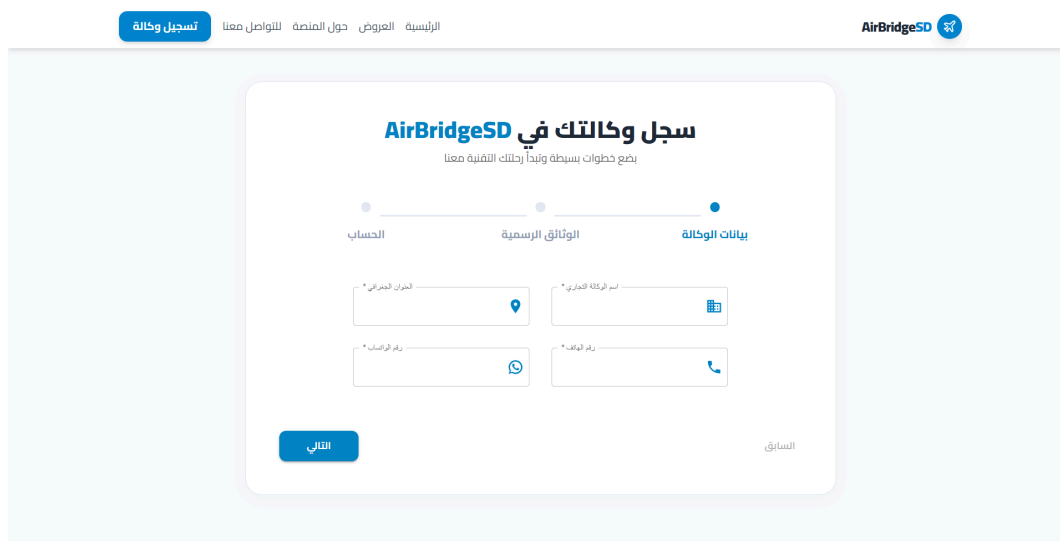


Screen 4.9 Privacy Policy Page



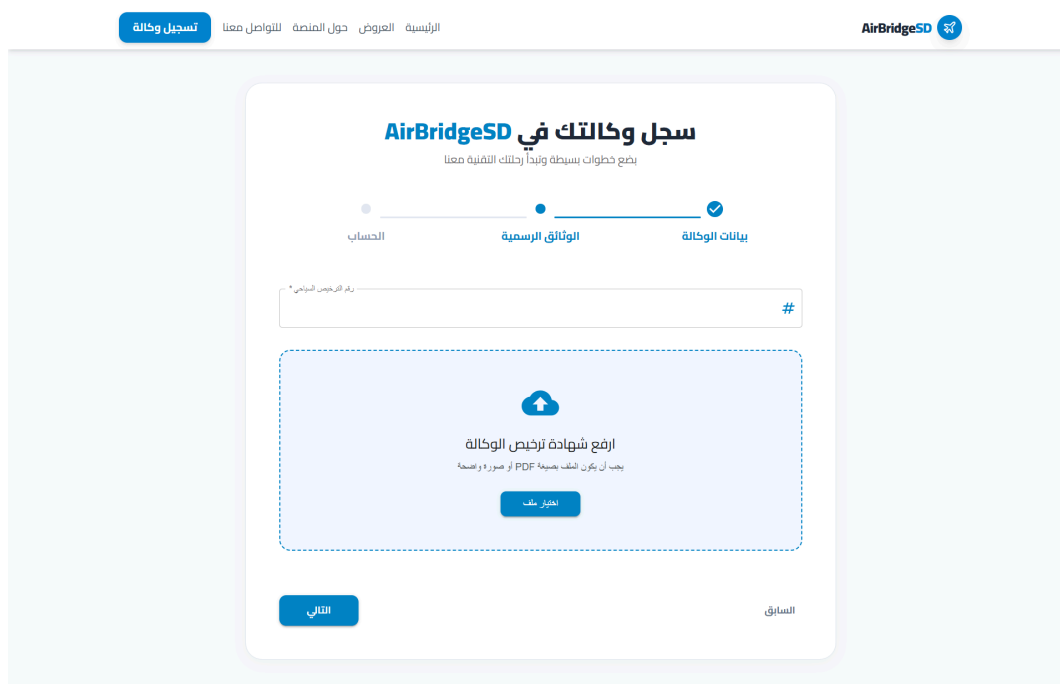
Screen 4.10 Terms and Conditions Page

4.3.2 Agency interfaces :



The screenshot shows the first step of the agency registration process on the AirBridgeSD website. The header includes a navigation bar with links for 'الرئيسية' (Home), 'العروض' (Offers), 'حول المنصة' (About the Platform), 'للتواصل معنا' (Contact Us), and a 'تسجيل وكالة' (Register Agency) button. The main content area is titled 'سجل وكالتك في AirBridgeSD' (Register your agency on AirBridgeSD) with a subtitle 'يضع خطوات بسيطة وتبدأ رحلتك التقنية معنا' (It sets simple steps and starts your technical journey with us). A progress bar at the top indicates three steps: 'الحساب' (Account), 'الوثائق الرسمية' (Official Documents), and 'بيانات الوكالة' (Agency Data), with the third step being the current active step. The form contains four input fields: 'الاسم التجاري' (Commercial Name), 'اسم الوكالة التجاري' (Commercial Agency Name), 'رقم الهاتف' (Phone Number), and 'رقم الفاكس' (Fax Number). Each field has a corresponding icon (location pin, ID card, telephone, and fax machine respectively). A blue 'التالي' (Next) button is at the bottom left, and a 'السابق' (Previous) link is at the bottom right.

Screen 4.11 New Agency Registration Page



The screenshot shows the second step of the agency registration process. The header and progress bar are identical to the previous screen, but the progress bar now shows the second step, 'الوثائق الرسمية' (Official Documents), as the current active step. The form area is titled 'ارفع شهادة ترخيص الوكالة' (Upload Agency License Certificate) with a subtitle 'يجب أن يكون الملف بصيغة PDF أو صورة واضحة' (The file must be in PDF format or a clear image). There is a large dashed blue box for file upload, containing a cloud upload icon and a blue 'اختر ملف' (Choose File) button. Above this box is a text input field for 'رقم الترخيص التجاري' (Commercial License Number) with a '#' symbol on the right. A blue 'التالي' (Next) button is at the bottom left, and a 'السابق' (Previous) link is at the bottom right.

Screen 4.12 New Agency Registration Page 2

سجل وكالتك في AirBridgeSD

يضع خطوات بسيطة وتبدأ رحلتك التقنية معنا

●

✓

✓

الحسابالوثائق الرسميةبيانات الوكالة

البريد الإلكتروني لعميل *



اسم المسؤول (مدير الحساب) *



تاريخ كلمة المرور *



كلمة المرور *



إرسال الطلب للمراجعة

السابق

Screen 4.13 New Agency Registration Page 3



بوابة الوكالة

مرحباً بك، يرجى تسجيل الدخول للمتابعة

البريد الإلكتروني

agency@example.com



كلمة المرور



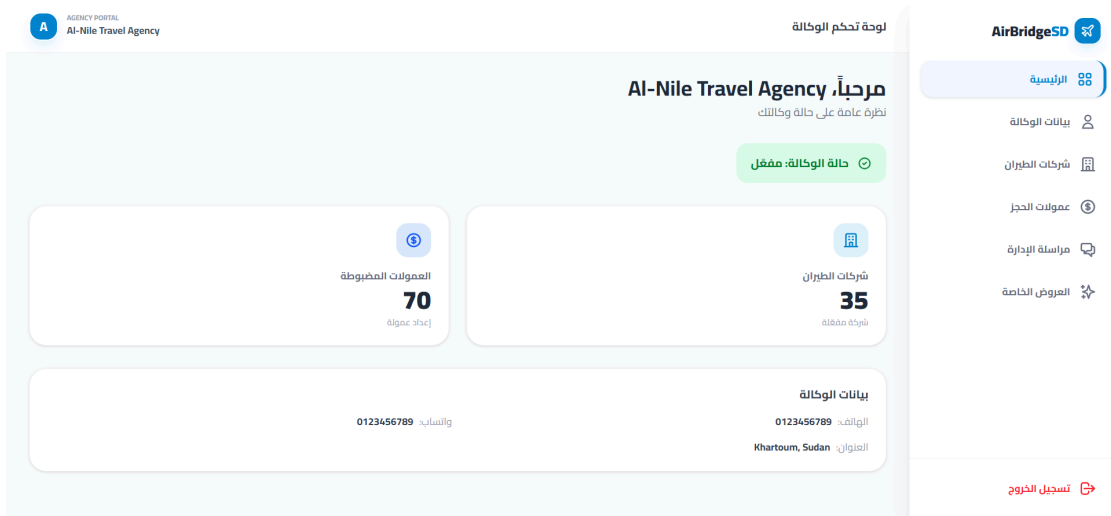
نسيت كلمة المرور؟

تسجيل الدخول للنظام

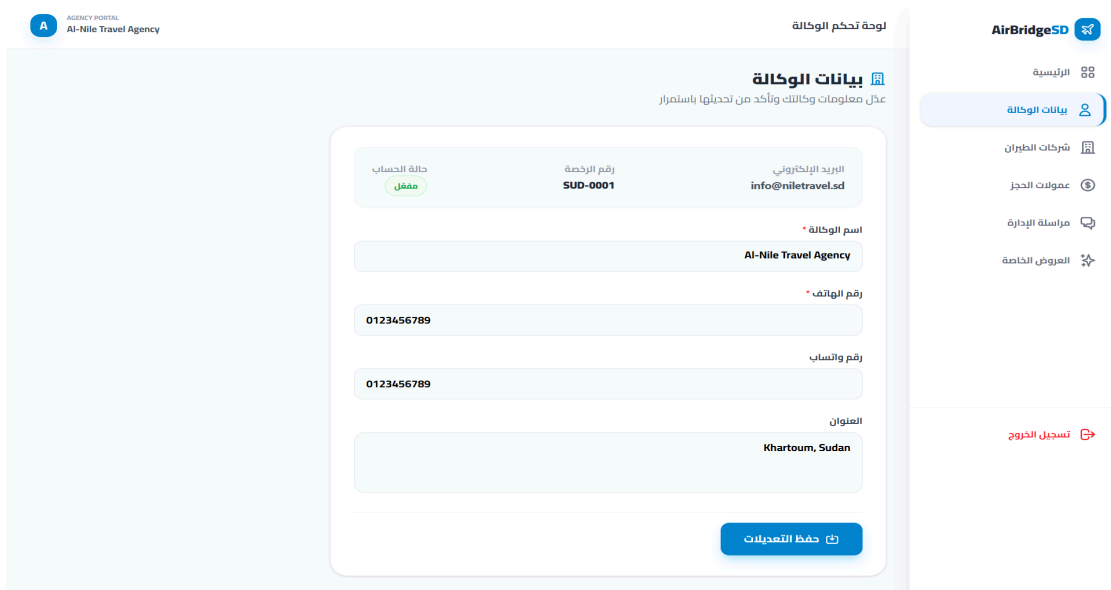
جميع الحقوق محفوظة © 2026 **ArtBridge50**

Screen 4.14 Agency login Page

4.3.2.1 Agency Dashboard

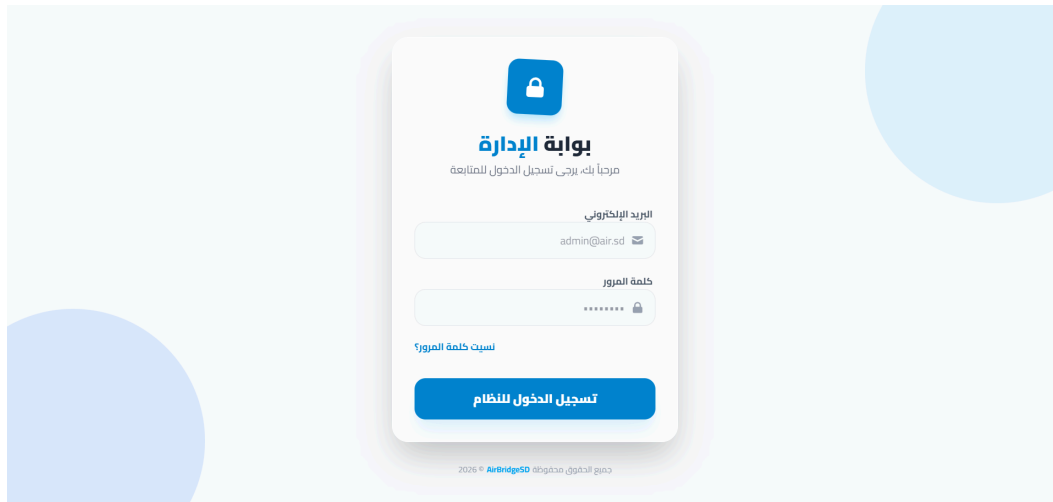


Screen 4.15 Agency dashboard Page

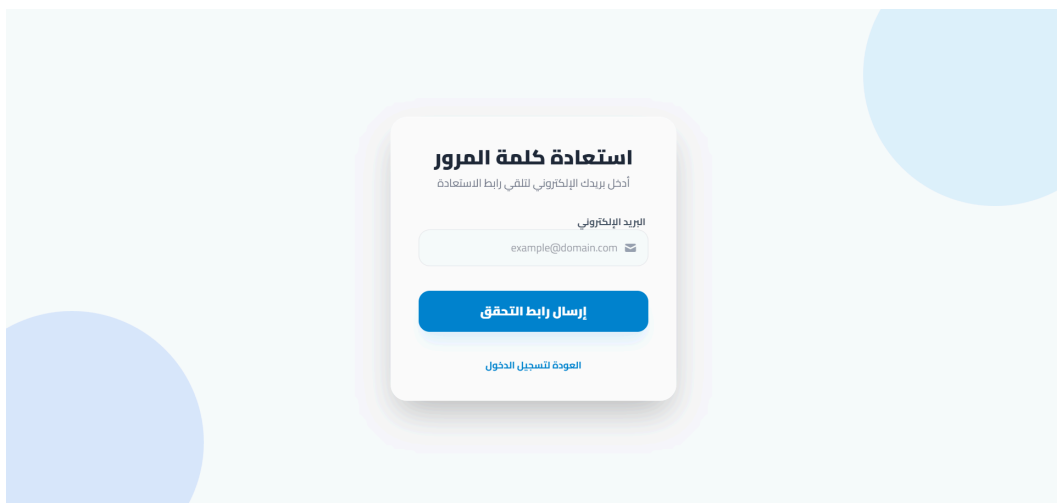


Screen 4.16 Agency data Page

4.3.3 Admin interfaces



Screen 4.22 Admin login page

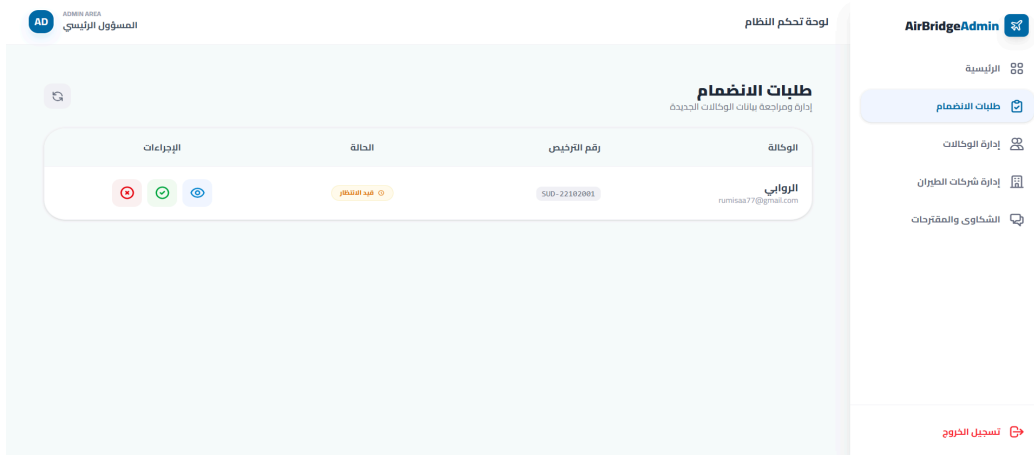


Screen 4.23 Forgot Password page

4.3.2.1 Admin dashboard



Screen 4.24 Admin dashboard Page



Screen 4.25 Pending agencies Page

ADMIN AREA

المسؤول الرئيسي

لوحة تحكم النظام

إدارة الوكالات

التحكم في حسابات الوكالات المسجلة والمقبولة في النظام

الوكالة	تاريخ الانضمام	الحالة	الإجراءات
الروابي numisaa77@gmail.com	٢٠٢١/٦/١	مقبول	  
Al-Safir Travel info@safirtravel.sd	٢٠٢١/٥/٢٩	مقبول	  
Al-Jazira Travel contact@aljaziratravel.sd	٢٠٢١/٥/٢٩	مقبول	  
Al-Musafer Travel info@musafer.sd	٢٠٢١/٥/٢٩	مقبول	  
Al-Nile Travel Agency info@niletravel.sd	٢٠٢١/٥/٢٩	مقبول	  
Dar Al-Salam Travel contact@daralsalam.sd	٢٠٢١/٥/٢٩	مقبول	  

AirBridgeAdmin

الرئيسية

طلبات الانضمام

إدارة الوكالات

إدارة شركات الطيران

الشكاوى والمقترحات

تسجيل الخروج

Screen 4.26 Agencies Management Page

ADMIN AREA

المسؤول الرئيسي

لوحة تحكم النظام

إيقاف الحساب

إجمالي الحجزات

0

فرد البناء

الحالة الحساب الحالية

نشط ويعمل

Al-Nile Travel Agency

مقرى الوكالة: 18

عودة

البريد الإلكتروني

info@niletravel.sd

تاريخ الانضمام

٢٠٢١/٥/٢٩

اسم المدير المسؤول

Nile Agency Manager

رقم التواصل

0123456789

الموقع الجغرافي

Khartoum, Sudan

المستندات الرسمية

رخصة السفر والسياحة

FILE

AirBridgeAdmin

الرئيسية

طلبات الانضمام

إدارة الوكالات

إدارة شركات الطيران

الشكاوى والمقترحات

تسجيل الخروج

Screen 4.27 Agency data page

ADMIN AREA

المسؤول الرئيسي

لوحة تحكم النظام

إضافة شركة

شركات الطيران

إدارة وتحديث بيانات شركات الطيران المسجلة

اسم الشركة	رمز IATA	الإجراءات
Air Algérie	AH	 
Air Arabia	G9	 
Air Arabia Abu Dhabi	YY	 
Air Arabia Maroc	30	 
Air France	AF	 
Air India	AI	 

AirBridgeAdmin

الرئيسية

طلبات الانضمام

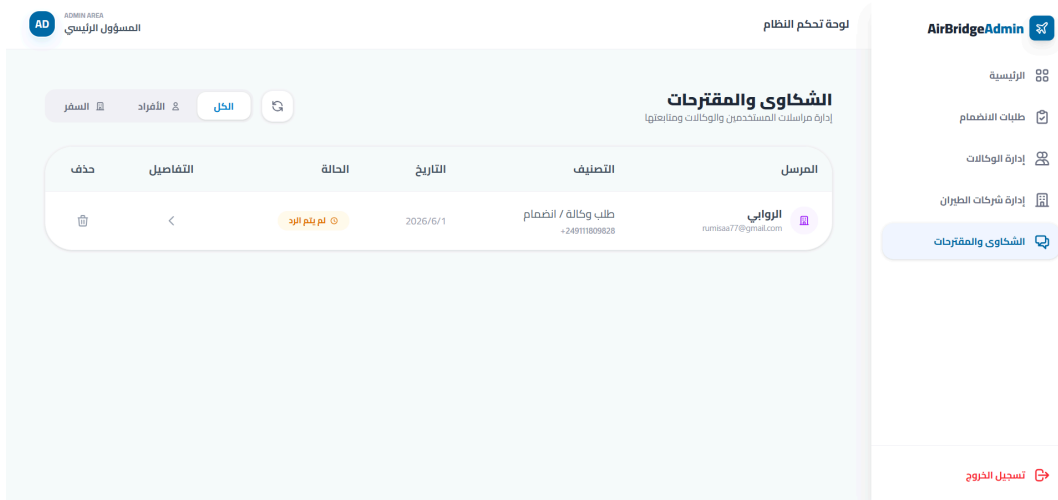
إدارة الوكالات

إدارة شركات الطيران

الشكاوى والمقترحات

تسجيل الخروج

Screen 4.28 Airline Management Page



Screen 4.29 Messages and Suggestions page



Screen 4.30 Messages details

CHAPTER 5

RESULTS AND RECOMMENDATIONS

5.1 Introduction:

This chapter presents the final evaluation of the AirBridgeSD platform after completing the development phase. The chapter is divided into four main sections. Section 5.2 discusses the testing strategy and testing results, including functional testing, API integration testing, responsive design testing, and code quality analysis. Section 5.3 presents the project results and explains how each system objective was achieved. Section 5.4 outlines the current limitations of the system and provides recommendations for future improvements. Finally, Section 5.5 concludes the chapter with a summary of the project outcomes and contributions.

5.2 System Testing:

5.2.1 Overview of Testing Strategy:

Software testing is an important stage in the Software Development Life Cycle (SDLC) because it helps ensure that the implemented system behaves according to the required functional and non-functional requirements. For the AirBridgeSD platform, multiple testing approaches were used to verify the correctness, reliability, and maintainability of the system.

The testing process focused on three main categories:

1. **Functional Testing:** to verify that the implemented features work according to the defined use cases and requirements.
2. **API Integration Testing:** to validate the communication between the React.js frontend, Laravel backend, and the external Duffel API.
3. **Code Quality Analysis:** to evaluate the maintainability, reliability, and security of the source code using SonarQube static analysis.

5.2.2 Functional Testing:

Functional testing was performed to verify that the implemented features satisfy the use cases and system requirements defined earlier in the project.

Each test case was derived directly from the related use case to maintain consistency between requirements and testing results.

The test cases were organized according to the three system actors: Customer, Agency, and Administrator. Each test case includes the tested feature, expected result, actual result, and status.

Table 5.1 Customer Functional Test Cases

TC ID	Feature Tested	Expected Result	Status
TC-C01	Search Flights	Flight results	✓ Pass
TC-C02	Invalid Search Input	Validation message displayed	✓ Pass
TC-C03	No Search Results	“No flights found” message shown	✓ Pass
TC-C04	View Agency Offers	Offers displayed correctly	✓ Pass
TC-C05	WhatsApp Contact	WhatsApp link opened correctly	✓ Pass
TC-C06	View Agency Profile	Agency profile details page view to customer.	□ failed
TC-C07	Submit Complaint	Complaint stored successfully	✓ Pass

Table 5.2 Agency Functional Test Cases

TC ID	Feature Tested	Expected Result	Status
TC-A01	Agency Registration	Account created successfully with pending status	✓ Pass
TC-A01	Agency Login	Dashboard accessed successfully	✓ Pass

TC-A02	Add Special Offer	Offer stored and displayed	✓ Pass
TC-A03	Edit Offer	Updated information displayed	✓ Pass
TC-A04	Delete Offer	Offer removed successfully	✓ Pass
TC-A05	Fixed Commission Setup	Fixed commission applied correctly	✓ Pass
TC-A06	Percentage Commission Setup	Percentage calculation correct	✓ Pass
TC-A07	Manage Partnered Airlines	Airline relationships updated	✓ Pass
TC-A8	Update Agency Profile	Profile updated successfully	✓ Pass
TC-A9	Agency Logout	Session terminated successfully	✓ Pass

Table 5.3 Administration Functional Test Cases

TC ID	Feature Tested	Expected Result	Status
TC-AD01	Admin Login	Dashboard accessed successfully	✓ Pass
TC-AD02	View Agencies	Agency list displayed correctly	✓ Pass
TC-AD03	Approve Agency	Agency activated successfully	✓ Pass
TC-AD04	Reject Agency	Agency rejected successfully	✓ Pass
TC-AD05	Disable Agency	Agency access disabled	✓ Pass
TC-AD06	Manage Airlines	Airline list updated successfully	✓ Pass
TC-AD07	View Complaints	Complaints displayed correctly	✓ Pass

TC-AD08	Reply to Complaint	Reply delivered successfully	✓ Pass
TC-AD09	Filter Complaints	Filtering applied correctly	✓ Pass
TC-AD10	Admin Logout	Session terminated successfully	✓ Pass

All test cases passed successfully, resulting in a **100% functional testing pass rate**. No major functional issues were identified during testing.

5.2.3 API Integration Testing

API integration testing was conducted using Postman to verify the correctness of the RESTful API endpoints implemented in the Laravel backend. The testing process focused on validating request and response handling, HTTP status codes, authentication behaviour, and error handling.

Each endpoint was tested under different scenarios, including valid requests, invalid input, and unauthorized access attempts.

Table 5.4 API Endpoint Test Results

Endpoint	Method	Expected Status	Actual Status	Result
/api/auth/register	POST	201 Created	201	✓ Pass
/api/auth/login	POST	200 OK	200	✓ Pass
/api/flights/search	GET	200 OK	200	✓ Pass

/api/offers	GET	200 OK	200	✓ Pass
/api/offers	POST	201 Created	201	✓ Pass
/api/offers/{id}	PUT	200 OK	200	✓ Pass
/api/offers/{id}	DELETE	200 OK	200	✓ Pass
/api/commission	POST	201 Created	201	✓ Pass
/api/complaints	POST	201 Created	201	✓ Pass
/api/admin/agencies	GET	200 OK	200	✓ Pass
/api/admin/agencies/{id}/approve	PATCH	200 OK	200	✓ Pass

All tested API endpoints returned the expected HTTP status codes and behaved correctly during testing. Authentication and authorization middleware also worked correctly by preventing unauthorized access to protected routes.

5.2.4 Code Quality Analysis (SonarQube)

Static code analysis was performed using SonarQube Community Edition to evaluate the quality of both the React.js frontend and Laravel backend codebases. The analysis focused on identifying bugs, vulnerabilities, duplicated code, and maintainability issues.

Table 5.5 SonarQube Analysis Summary

Metric	Frontend	Backend	Status
Bugs	5	9	✓ Pass
Vulnerabilities	0	0	✓ Pass

Code Smells	4	6	✓ Pass
Duplicated Lines	0.0%	3.4%	✓ Pass
Maintainability Rating	B	A	✓ Pass
Reliability Rating	B	B	Require A to pass
Security Rating	A	B	✓ Pass

The analysis confirmed that no critical bugs or security vulnerabilities were detected in the final codebase.

5.2.5 Responsive Design Testing

Responsive design testing was conducted to verify that the system interface works correctly across multiple screen sizes and devices. Testing was performed using browser developer tools and physical mobile devices.

Table 5.7 Responsive Design Test Results

Viewport	Device Type	Status
375px	Mobile	✓ Pass
768px	Tablet	✓ Pass
1280px	Laptop	✓ Pass
1920px	Desktop	✓ Pass

5.3Results

This section presents the final outcomes of the AirBridgeSD project according to the objectives defined in Chapter 1.

5.3.1 Objective 1 : Unified Flight and Offer Aggregation

Result: Achieved.

The platform successfully integrates international flight data from the Duffel API with locally Flights and managed agency offers into a single unified interface accessible to customers.

5.3.2 Objective 2 : Multi-Criteria Search and Filtering

Result : Achieved.

The system provides dynamic search and filtering capabilities that allow users to browse flights and Special offers using multiple criteria such as destination, airline, agency, and price.

5.3.3 Objective 3 : Automated WhatsApp Booking Request Routing

Result: Achieved.

The platform automatically generates and routes pre-filled WhatsApp booking requests containing flight and agency details to the selected travel agency.

5.3.4 Objective 4 : Dynamic Commission Calculation Engine

Result: Achieved.

The implemented commission engine supports both fixed and percentage-based pricing models and dynamically calculates the final ticket prices displayed to customers.

5.3.5 Objective 5 : Agency Self-Service Management Dashboard

Result: Achieved.

The Agency Dashboard enables agencies to independently

manage profiles, airline partnerships, commissions, Special offers, and customer interactions through a secure interface.

5.3.6 Objective 6 : Centralised Administration and Governance Panel

Result : Achieved.

The Admin Panel provides centralized management features including agency approval, airline management, complaint handling, and platform monitoring.

5.3.7 Objective 7 : Secure Agency Onboarding and Authentication

Result: Achieved.

The system implements a secure agency registration and approval workflow with authentication controls that restrict access to unauthorized or suspended agencies.

5.4 Recommendations for Future Development

The following limitations represent intentional project scope boundaries and contextual design decisions rather than unresolved implementation failures. These decisions were made to maintain feasibility within the academic scope while aligning the platform with observed operational practices in the Sudanese travel sector:

R1 - Payment Gateway Integration:

Integrate online payment solutions to support complete booking transactions inside the platform.

R2 - Diversified Travel Services:

Expand system capabilities to support additional local operations, such as visa application tracking, Hajj and Umrah package management, and domestic hotel reservations.

R3 - Native Mobile Application Development:

Develop Android and iOS mobile applications to improve accessibility and user engagement.

R4 - Automated Agency Verification:

Deploy a secure document upload and verification pipeline for registering travel agencies. This feature ensures that only licensed operators can post flight data and promotional packages, enhancing platform trust.

R5 - English Language:

Add English localization Language support.

R6 - Multi-Provider Flight Integration:

Integrate additional flight providers to improve data availability and route coverage.

R7 - Statistical Analytics Panel:

Deploy a data-driven dashboard for system administrators and travel agencies to monitor high-traffic flight routes, peak seasonal patterns, and financial revenue trends.

R8 - Extended Customer Request and History Management:

Future versions may introduce complete request tracking and historical interaction records.

5.5 Conclusion

The project provides a centralized web platform that combines international flight data and local Sudanese travel agency offers within a single system. Through its hybrid architecture using React.js, Laravel, and external API integration, the platform demonstrates a practical and scalable solution for improving flight discovery and agency communication within the Sudanese travel market.

The identified Recommendations represent opportunities for future enhancement rather than critical system failures. Overall, the project achieved its intended objectives and delivered a functional, maintainable, and deployable platform.

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